

PoP Quiz

1. In the projection systems, what properties could be preserved or distorted?
2. Name types of projection surfaces?
3. What is the standard parallel in the projections concept?
4. Name one popularly used projection system.

Course Evaluation (CE 3420 and CE 6610)

- Two announced Quiz – Best of each : CE3420 - 25 % (CE 6610 - 15%)
- End Semester: CE3420 - 45% (CE 6610 - 40%)
- Assignments/Practical component: Google Earth Engine and Quantum GIS (QGIS) : CE3420 - 20% (CE 6610 - 5%)
- Pop Quiz (Numbers not defined) + Proxy for attendance : CE3420 - 10% (CE 6610 - 5%)
- Segments 5-6 (CE 6610) : Dr. Satish Regonda (35%)

What will you learn?

The concepts and techniques of
spatial and **non-spatial** geoprocessing tools
operated on vector geospatial data

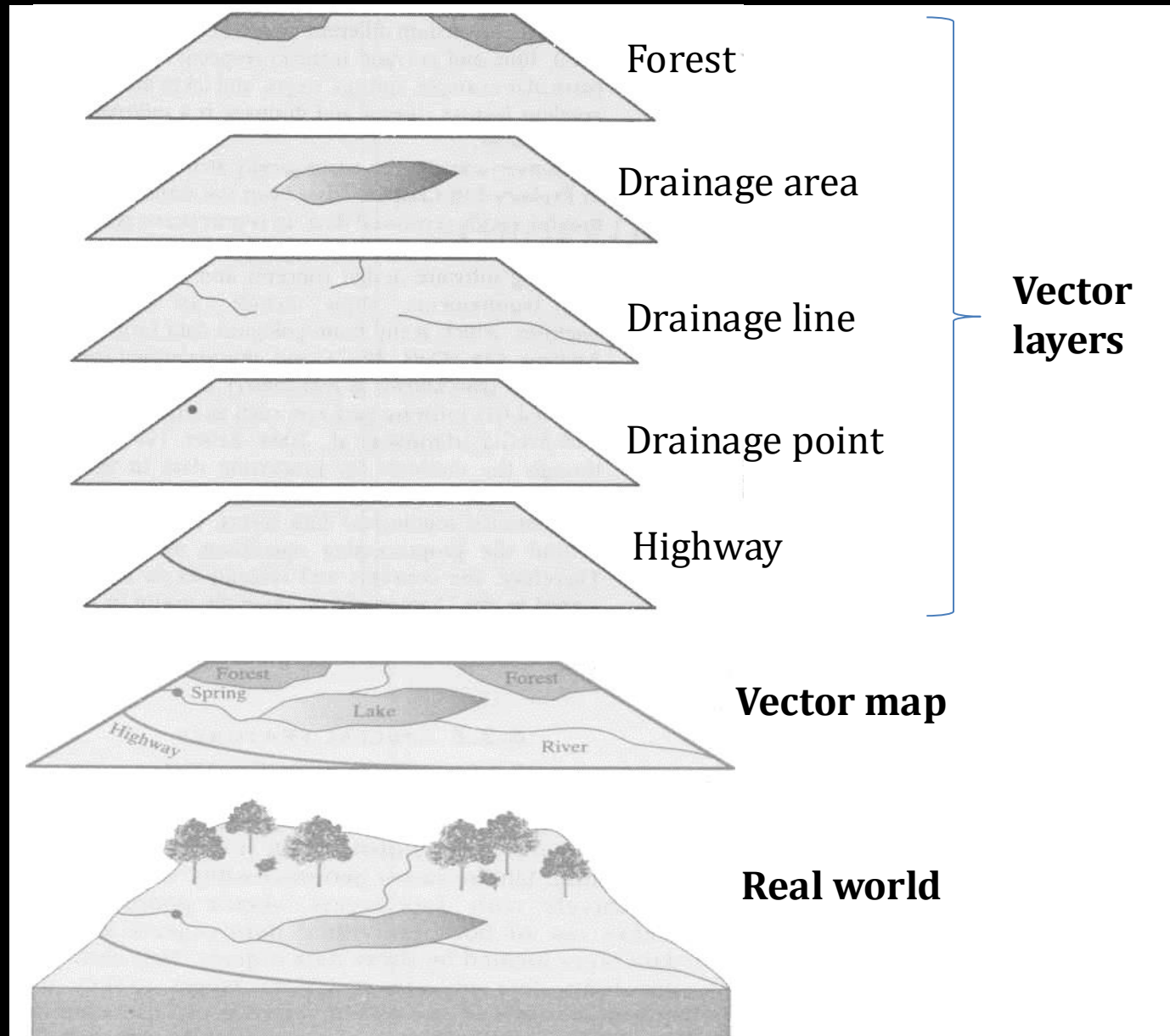
Characteristics of Vector Geoprocessing

Vector data:

- Depict individual spatial features as discrete entities using 3 basic graphical elements: **points, lines, and polygons**.
- In the database, these entities are identified as **feature classes**, each pertaining to a particular theme (e.g.: transportation, drainage, vegetation, land parcels, administrative boundaries *etc.*).
- Used for applications focusing on the characteristics of individual classes of spatial features in a particular geographic area of interest.
i.e. one vector feature class → one theme in the real world

Vector data layer :

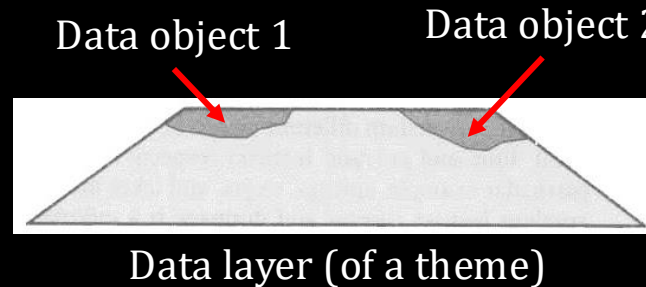
- In a digital geospatial database, feature classes are logically organized as **layers or themes**.
- Features relevant to a particular theme are identified separately according to their respective types of graphical elements.
- Example:
Springs, rivers, and lakes → independent feature classes
belonging to the theme 'drainage'.
But,
 springs → point feature;
 rivers → line feature;
 lakes → polygon feature.



The vector method of representing real world features

Special features of vector geoprocessing

- Vector geoprocessing makes use of **both** individual **data objects** and the **data layer** formed by these data objects.

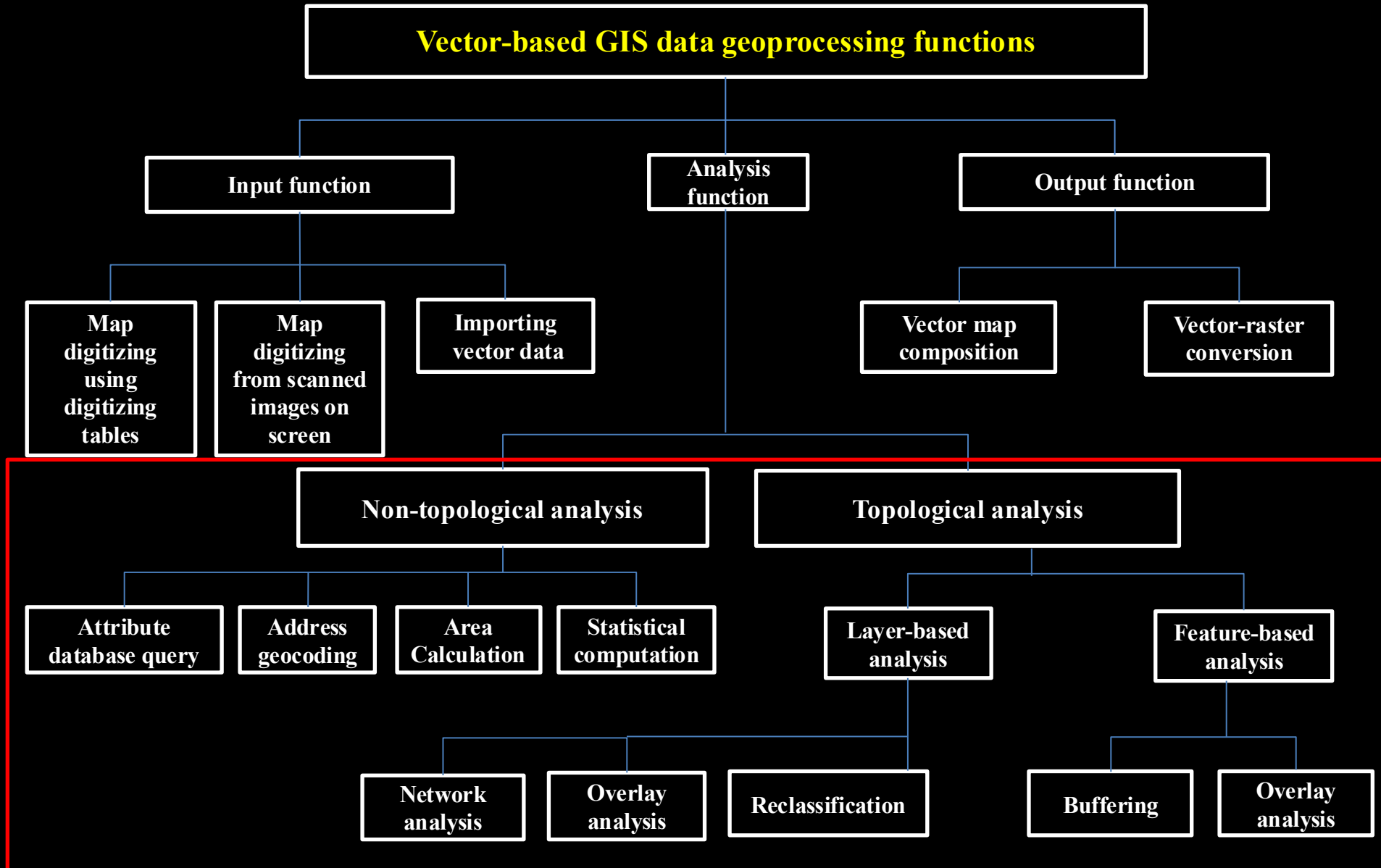


Examples of Vector-based processing based on:	Layer-based Output	Object-based Output
Input by Layers	Site suitability analysis <i>(by overlay of several input layers)</i>	Query to select individual features according to their attribute/location characteristics <i>(E.g.: <u>Schools</u> within 1km of <u>Rivers</u> → spot only certain school objects by overlay of the two layers)</i>
Input by Objects	Creation of surfaces <i>(by interpolating only a few sample points)</i>	Calculation of area, perimeter, and centroid of polygons <i>(each object will have unique results)</i>

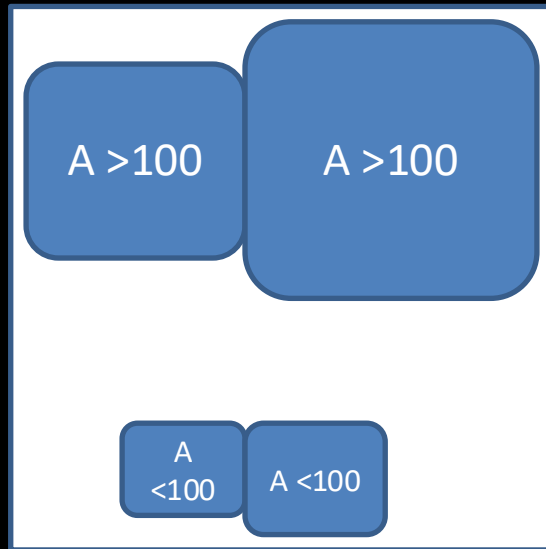
- Hence, compared to raster geoprocessing, Vector geoprocessing → requires a **larger, more complex set of** spatial and non-spatial tools.

- Relies heavily on **topological relationships** among data objects.
(as spatial problem solving depends on spatial relationships among objects)
- Often makes use of **both graphical and attribute data** in a tightly coupled manner.
- Possible to access:
 - graphical or locational data by specifying attribute values, and
 - attribute data by means of graphics or locations.
- Can be carried out using spatial and non-spatial tools, either *independently or in conjunction* with one another.
E.g.: 1. State population > 10 Lakh → Entirely non-spatial (use only attribute data)
2. Schools with strength > 200 lying within 1km of river →
combination of spatial and non-spatial tools.
- Vector data are normally recorded and processed using ground coordinates in double precision, *creating impression* of a better accuracy than raster data.
Yet, precision ≠ accuracy !!

Classification of Vector Geoprocessing Functions



Reclassification and Aggregation



Reclassified

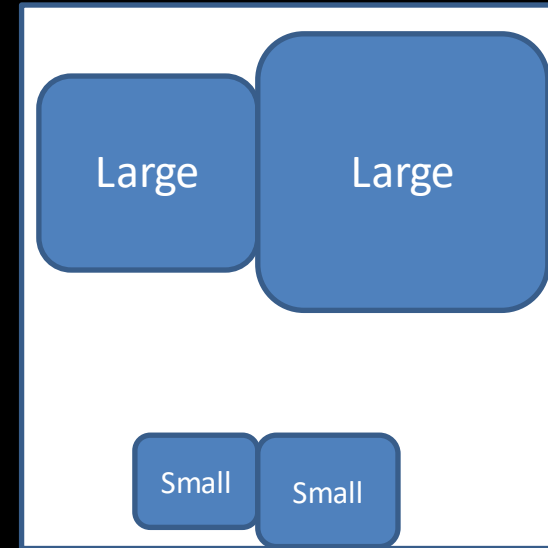


Based on data
attribute query
(non- topological)

A > 100 ?

1 (True) → Large

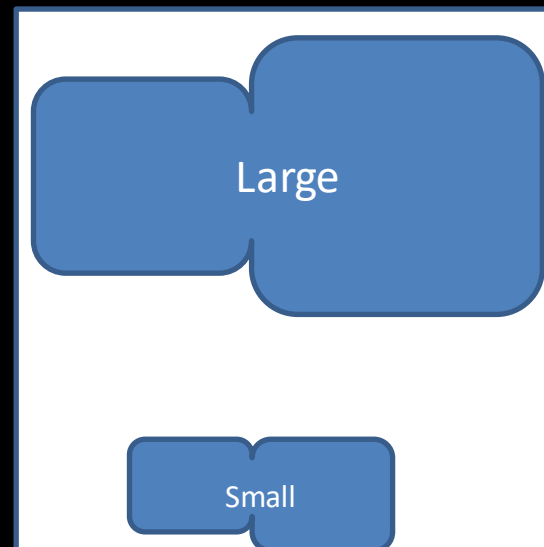
0 (False) → Small



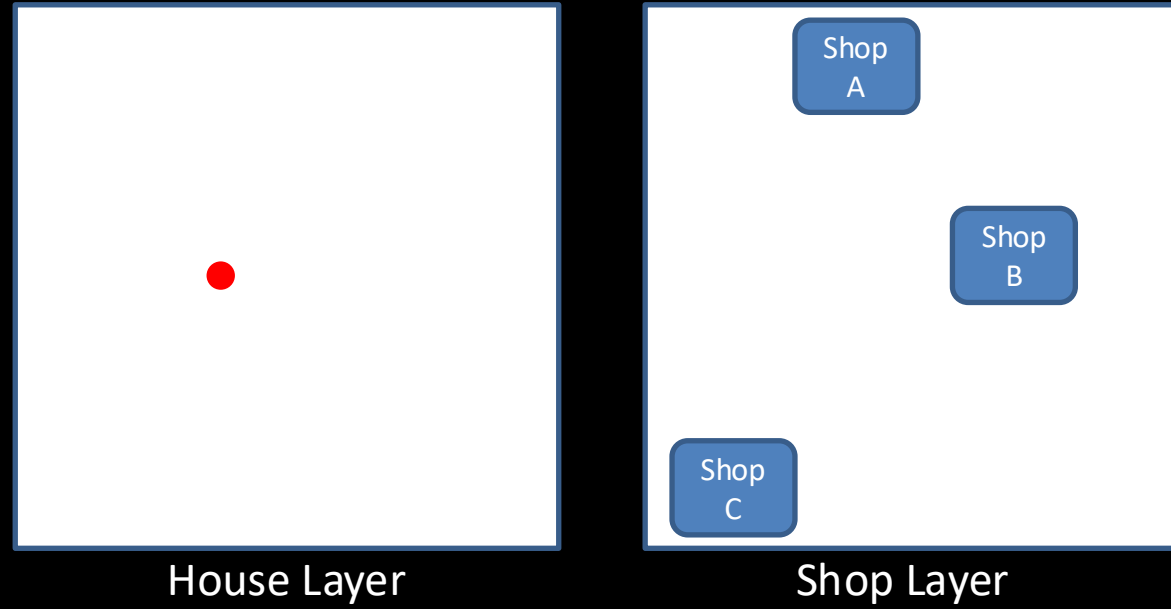
Aggregated



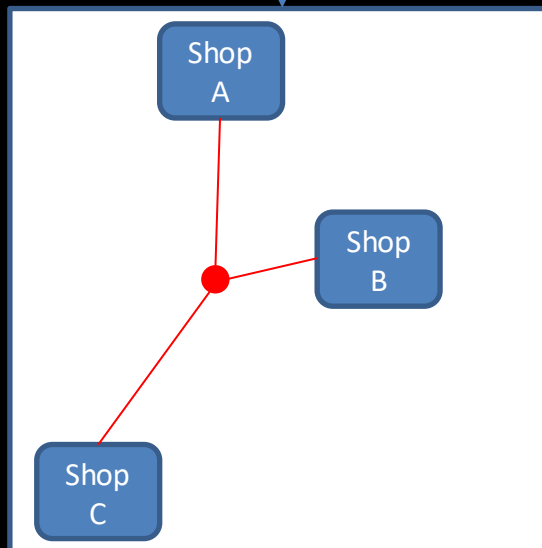
Combined
based on
topological
relation
(adjacency)



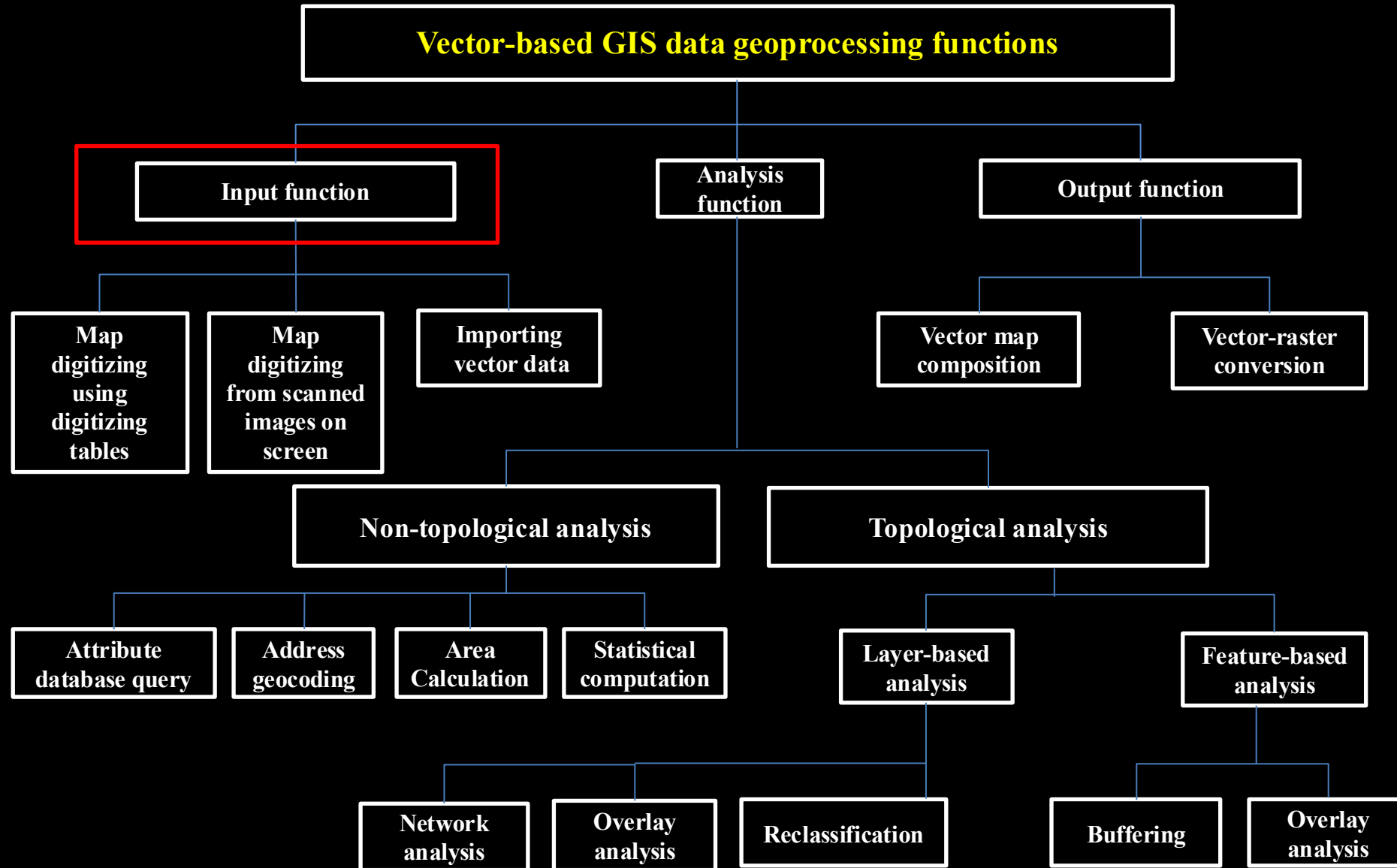
Network Analysis



Overlay
of layers
+
Distance
measurement
(geometric
property)



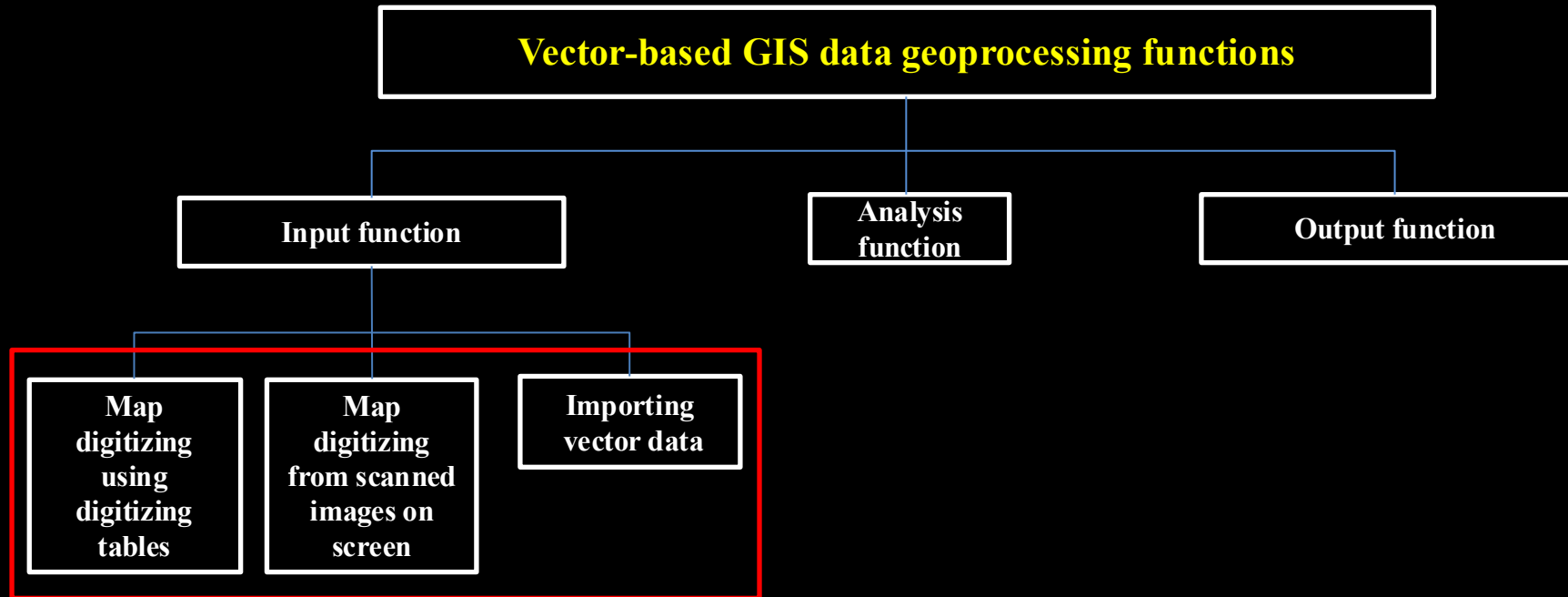
Classification of Vector Geoprocessing Functions



Vector Geoprocessing Input Functions

- In GIS, digital data conversion is made up of two components:
 - **Graphical data conversion:** To create digital map layers.
 - **Attribute data conversion:** To create tabular data files associated with the graphical elements on a layer.
- Digital Data Conversion Process:
 - **Acquisition:** Digitizing existing maps, purchasing from Govt. agencies or commercial data suppliers, (or) by collecting new data using field surveying, GPS, and photogrammetric and remote sensing methods.
 - **Editing:** Cleaning the acquired digital data to achieve acceptable quality with respect to the analysis objectives.
 - **Formatting/Translating:** Converting the edited data into the specific physical database format compatible to GIS (e.g.: .shp)
 - **Linking:** Relating graphical data with associated attribute data.

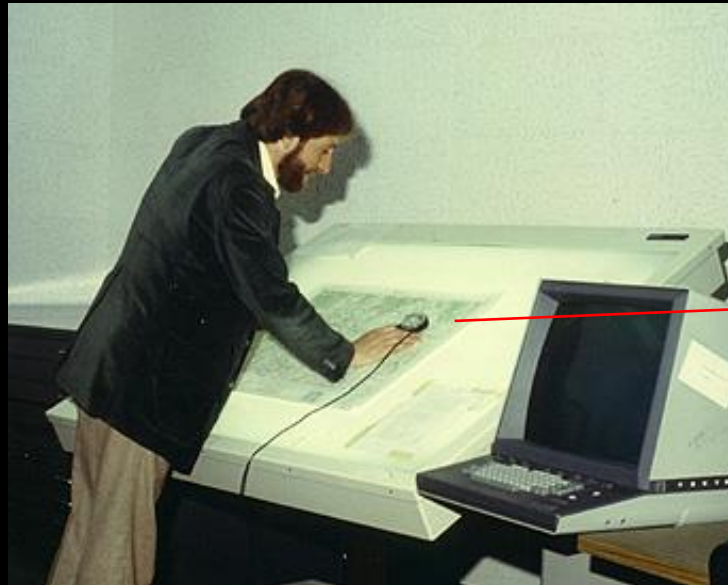
Classification of Vector Geoprocessing Functions



Vector Geoprocessing Input Functions

1. Map digitizing using digitizing tables:

- Conventional map digitizing device → **digitizing table** or **digitizer**.
- Useful for GIS applications requiring project-specific data (i.e., data unlikely to be available from external sources).
- Diminishing with the increased availability of digital data from various means.



Digitizing Table



Digitizing Puck

1. Map digitizing using digitizing tables- Flowchart

Digitizing template:

- It contains → tic points (reference/control points), map neat lines and graphical elements that are common to all layers that are to digitized/created.
- Eliminates the need to digitize common features on different layers.
- Ensures perfect registration of these features when layers are overlaid.

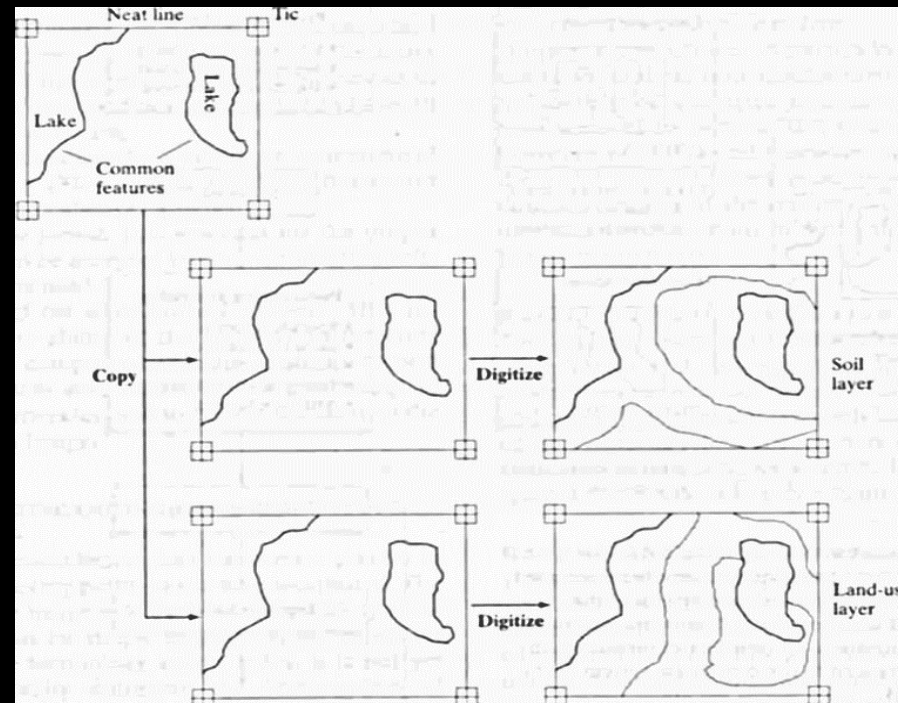
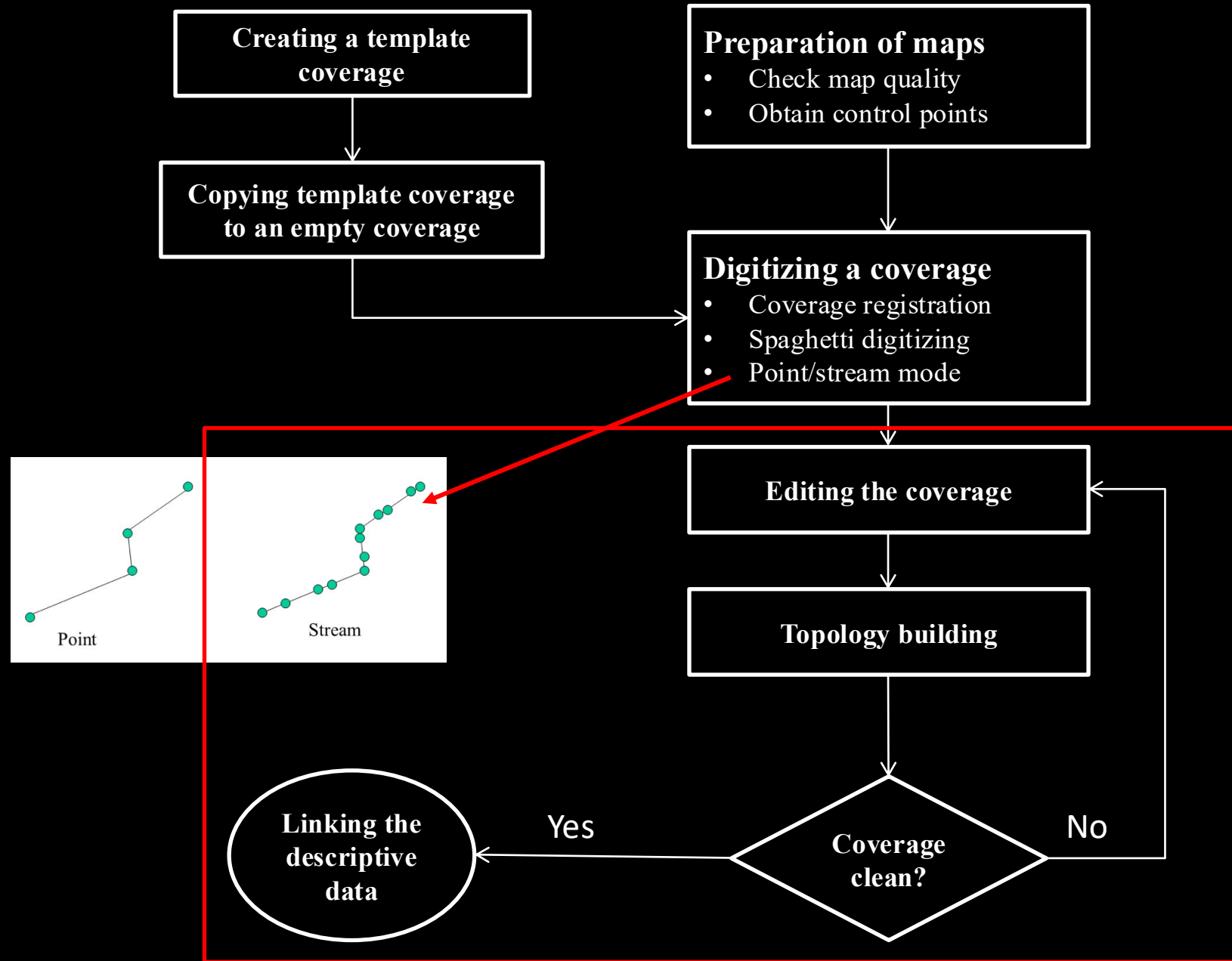


Illustration of a digitizing Template

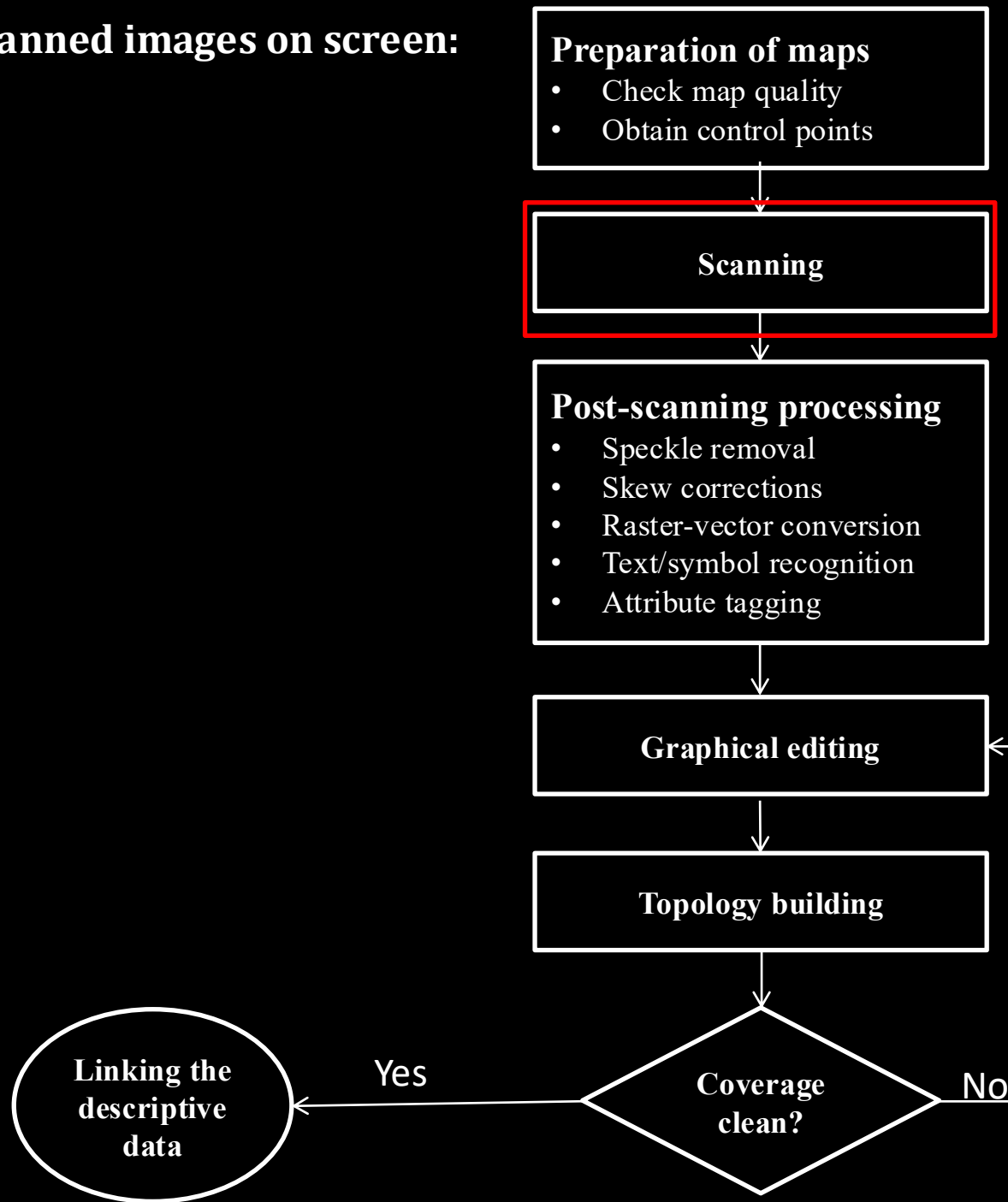
1. Map digitizing using digitizing tables- Flowchart



2. Map digitizing from scanned images on screen:

- This process is called **screen digitizing** or **heads-up digitizing** to distinguish it from conventional table digitizing.
- Frequently used method in the recent times, due to improvement in hardware design, software capabilities, and data-compression techniques.
- This approach of digital data conversion is capable of **converting a large number of maps** in a relatively short time frame and at a cost comparable to or lower than conventional table digitizing.

2. Map digitizing from scanned images on screen:

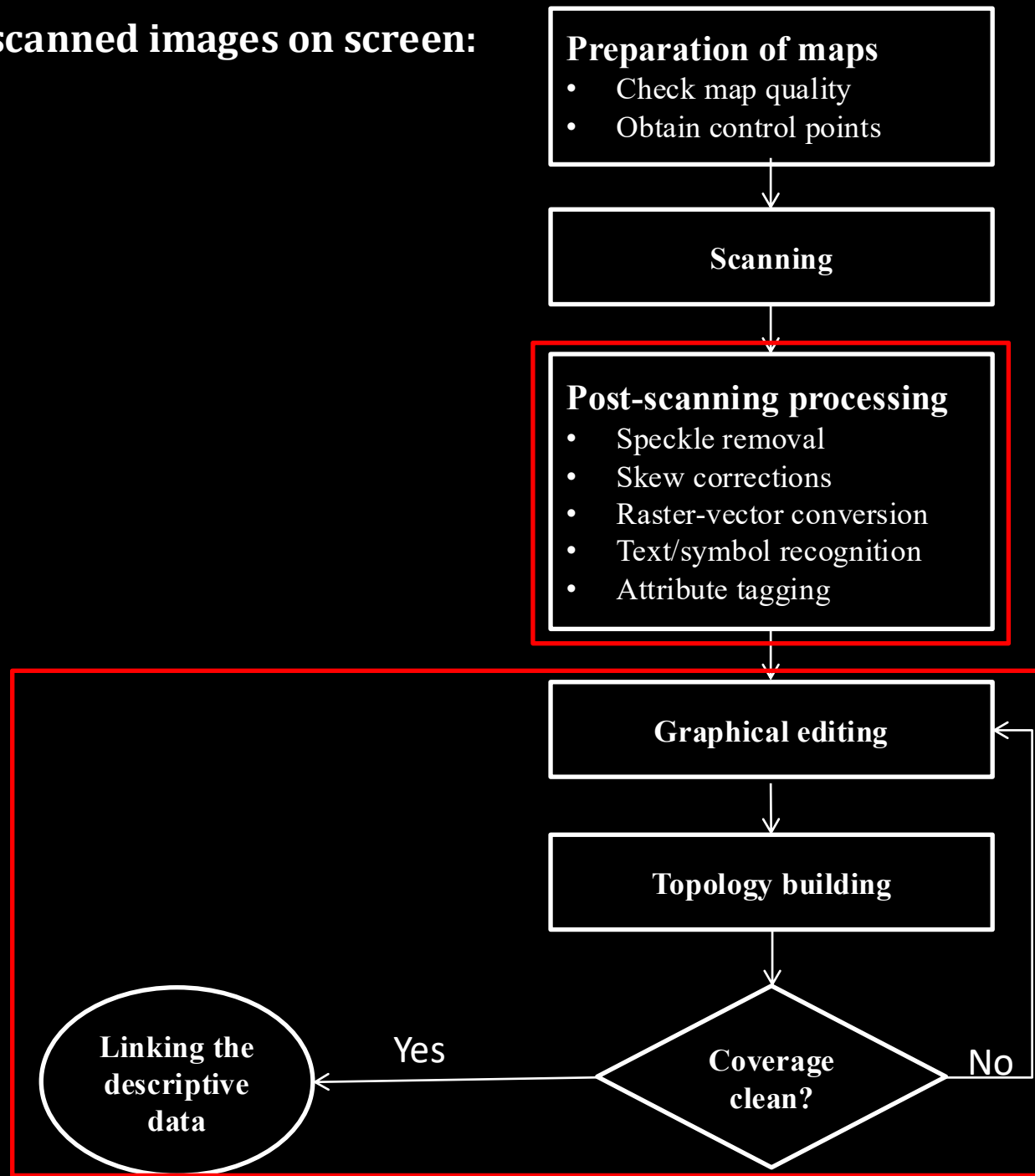


2. Map digitizing from scanned images on screen:

Scanned Maps- Technical specifications:

- **Resolution:** Defines the sharpness of the image; in *dots per inch (dpi)*. Usually, 300-600 dpi is preferred for GIS applications.
- **Scanner width and length:** A minimum scan width of 36 inches (91.4 cm) is required. Document length is not always a problem.
- **Output file format:** It is essential for the output formats to be accepted by the vectorization soft-ware to be used.
- **Quality of the accompanying software:** This includes the ability of the software to integrate with the computer hardware platform, to perform batch and wizard-driven processing, to remove speckles, and to correct the skew of the resulting image.

2. Map digitizing from scanned images on screen:



2. Map digitizing from scanned images on screen:

Contd..

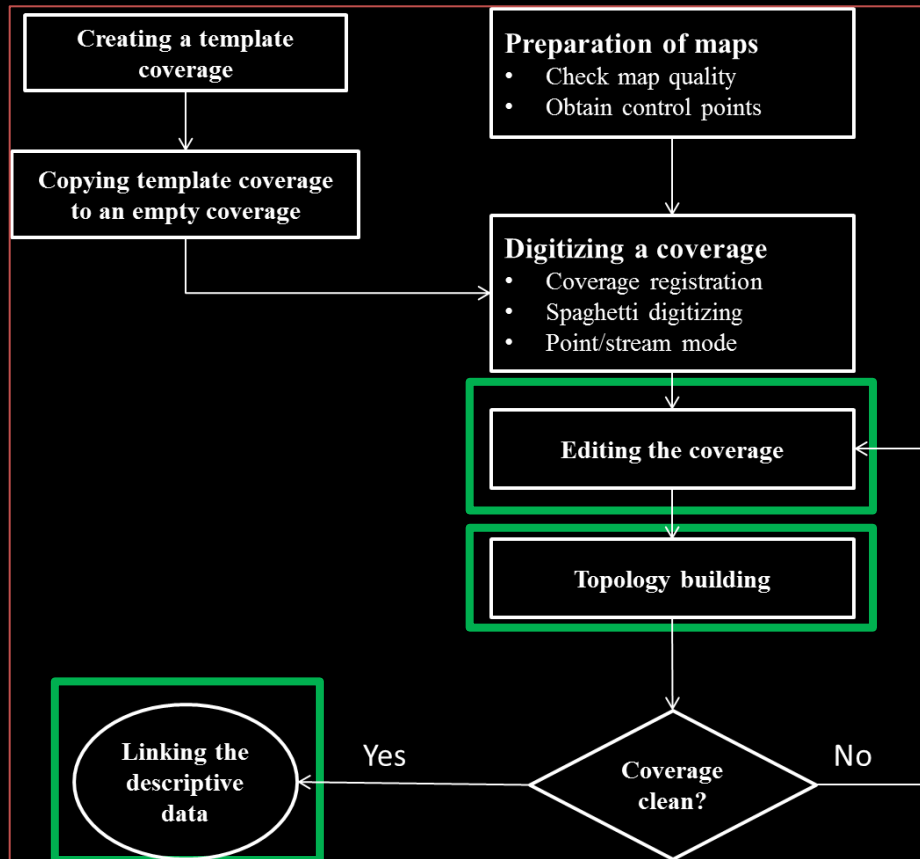
Postscanning data processing:

- Scanning is a **non-selective** data conversion process in the sense that *every point on the map is captured*.
- The resulting raster image is of limited use for GIS applications other than for simple displays on the computer screen.
- Hence, postscanning data processing, using either computer-assisted or manual methods, is necessary.
- Amount of work required for postscanning data processing depends on a variety of factors:
the quality of the original maps, the complexity of the map contents, as well as the functionality of the vectorization software .
- In general, postscanning processing is carried out as a sequence of processes:
 - (1) raster-vector conversion,
 - (2) raster text conversion,
 - (3) raster symbol conversion,
 - (4) graphical data editing, and
 - (5) attribute data tagging.

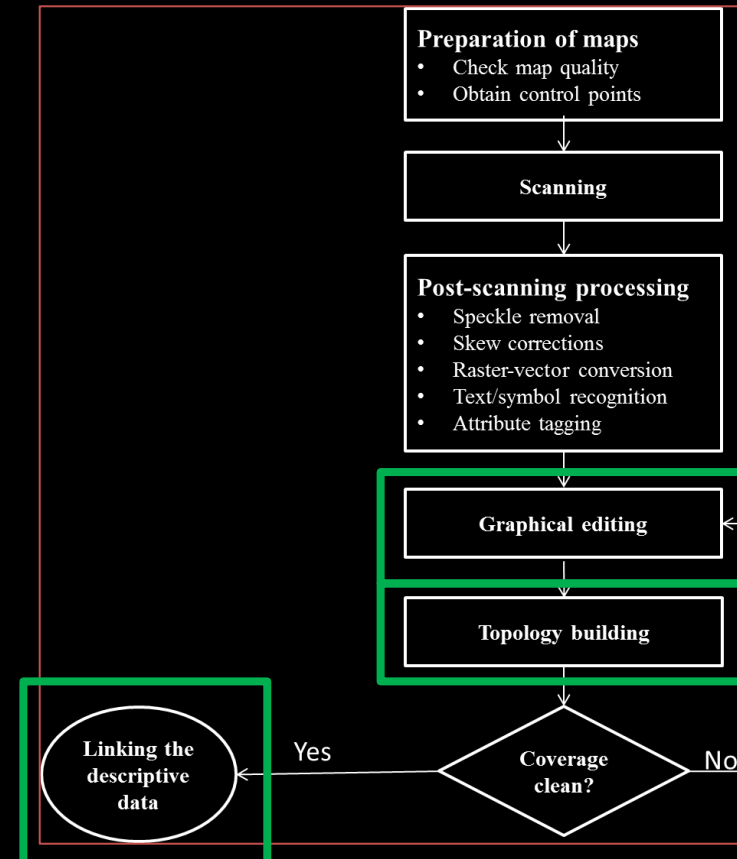
3. Importing vector data

- Imported data from different sources?
 - Can have multiple map projection or coordinate systems !!
- Hence, the imported data must be geometrically transformed to a common coordinate system. For this, we use the geoprocessing function **‘coordinate transformation’**.
- Current GIS software packages support a large set of standard projections and coordinate systems, with pre-defined transformation parameters (e.g.: ellipsoid, central meridian, standard parallel etc.)

Vector Geoprocessing Input Functions



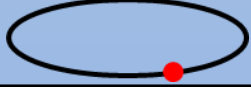
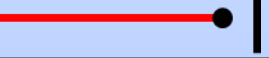
**1. Map digitizing using digitizing tables-
Flowchart**



**2. Map digitizing from
scanned images on screen:**

Graphical Data Editing

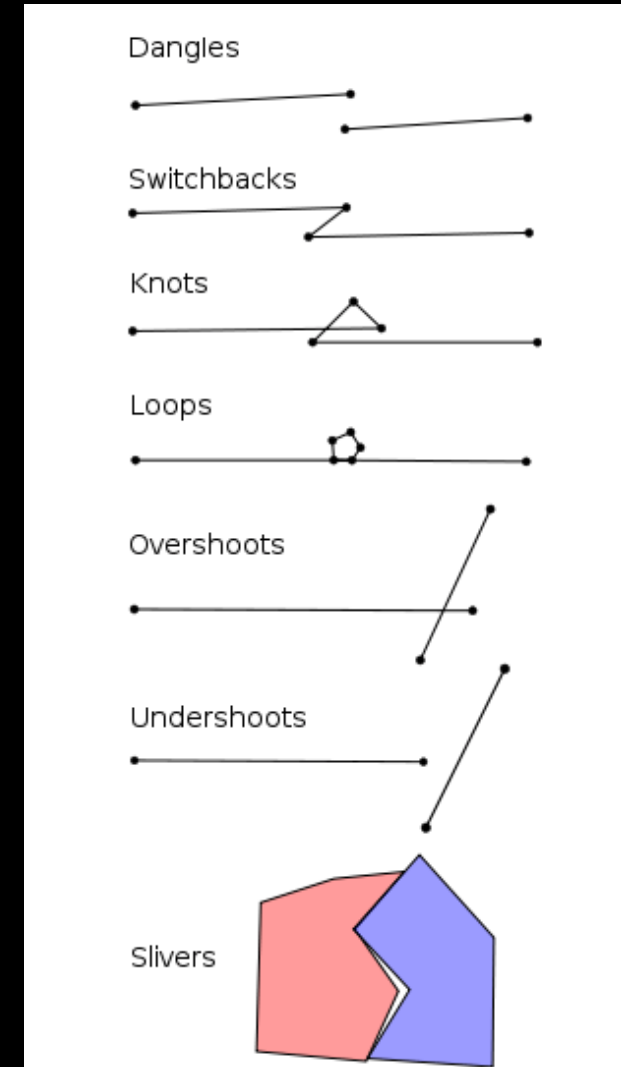
- Graphical data editing is a **post-digitizing process** to ensure the **integrity of the data** before they can be used in a geospatial database.
- Integrity of the data in the context of a topologically structured layer means that they are **free from errors**.

Feature	Errors	Examples
Node	Dangling node	
	Pseudo node	
	Pseudo node	
Arc	Dangling arc	
	Overshoot	
	Undershoot	
	Intersection(no node)	
Label	Missing label	
	Duplicate/multiple labels	

Graphical Data Editing

Hence, an error-free graphical data implies that:

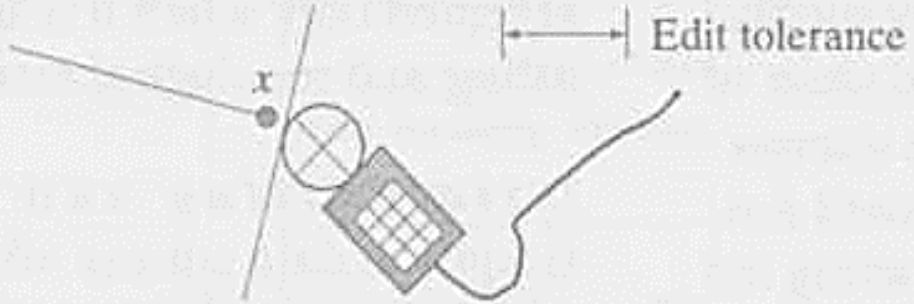
- lines intersect where they are expected to intersect (i.e., no undershoot or overshoot);
 - nodes are created at all points where lines intersect;
 - all polygons are closed;
 - each polygon contains a label point;
 - the topology of the layer is built.
- Graphical data editing is a **core function** of GIS.
 - It applies to any vector data input obtained (i.e. newly created or imported data).
 - It is made up of four basic phases of work:
 - i) Setting the editing environment,
 - ii) Topology building,
 - iii) Data editing and error corrections, and
 - iv) Joining adjacent layers.



Source:

<http://www.gislounge.com/digitizing-errors-in-gis/>

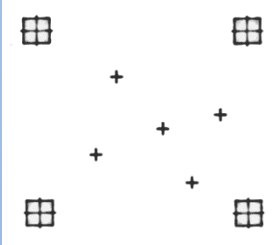
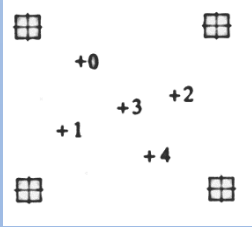
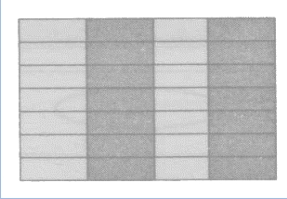
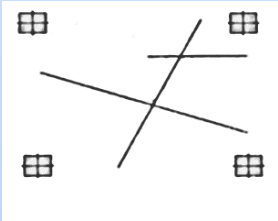
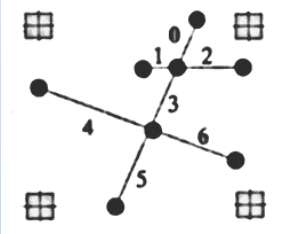
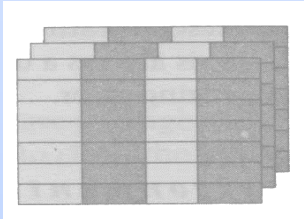
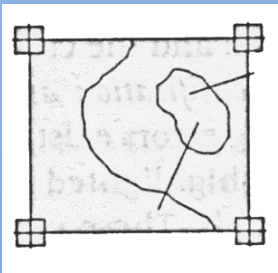
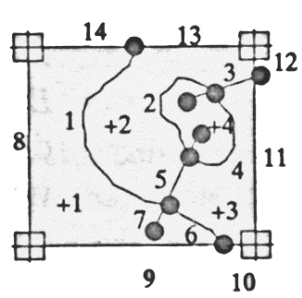
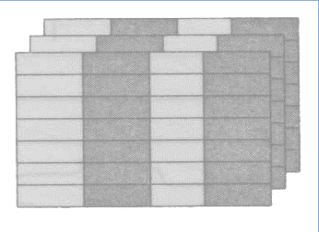
Graphical data editing- Phase 1: Setting the Editing Environment

Tolerance	Illustration
(a) Edit tolerance Cursor is able to select point x when it is clicked within the edit tolerance	
(b) Weed tolerance Digitized arcs smaller than weed tolerance are filtered	
(c) Grain tolerance Digitized points too close to one another are removed	

Graphical data editing- Phase 2: Topology Building

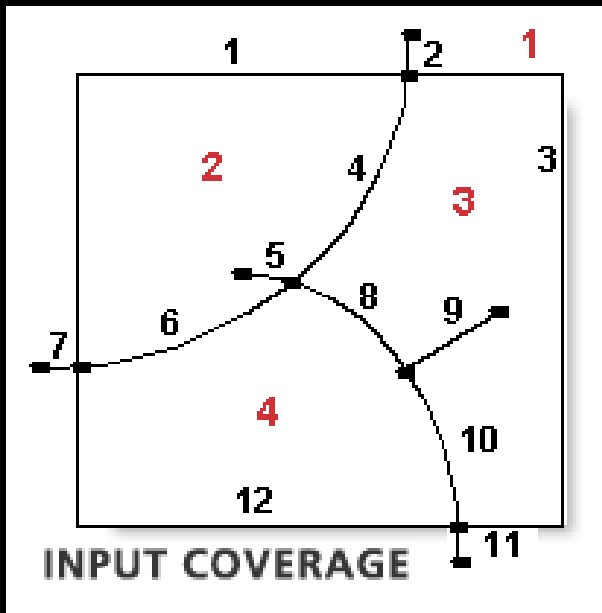
- This is probably the most important process in graphical data editing.
- It actually serves two interrelated purposes:
 - (1) Building the topological structure and relationships for the graphical elements on a layer.
 - Topology structure and relationships are necessary for performing spatial analyses on the various graphical elements in a layer.
 - Point, line, and polygon topology are created by assigning an internal identifier to each graphical element and creating the associated attribute tables containing feature information.

Graphical data editing- Phase 2: Topology Building

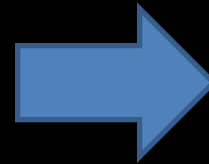
Coverage type	Before process	After process	Attribute table created	Topological checking
Point		 - Internal ID for points - Point topology	 - Point attribute table (PAT) - Other tables	
Line		 - Create nodes - Internal ID for nodes/arcs - Arc topology	 - Arc attribute table (AAT) - Other tables	- Dangling nodes - Dangling arcs
Polygon		 - Create nodes - ID for nodes/arcs - ID for polygon - Node topology - Arc topology	 - Polygon Attribute Table(PAT) - Other tables	- Dangling nodes - Dangling arcs - Polygon labels

Graphical data editing- Phase 2: Topology Building

Example of Polygon Attribute Table:



• Node
2 Arc Internal Number
2 Polygon Internal Number



Polygons to the left and right of each arc

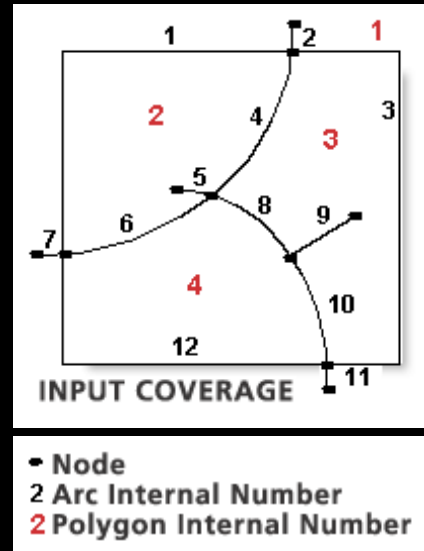
Arc number	Left poly	Right poly	Length
1	1	2	5.0
2*	1	1	0.2
3	1	3	8.8
4	2	3	1.2
5*	2	2	0.2
6	2	4	1.3
7*	1	1	0.2
8	4	3	1.9
9*	3	3	1.5
10	4	3	1.9
11*	1	1	0.2
12	4	1	3.1

Graphical data editing- Phase 2: Topology Building

- This is probably the most important process in graphical data editing.
- It actually serves two interrelated purposes:
 - (1) **Building the topological structure and relationships for the graphical elements on a layer.**
 - Topology structure and relationships are necessary for performing spatial analyses on the various graphical elements in a layer.
 - Point, line, and polygon topology are created by assigning an internal identifier to each graphical element and creating the associated attribute tables containing feature information.
 - (2) **Error identification and automated corrections**
 - Topology building highlights the digitizing errors in the vector file.
 - These errors can be **sometimes corrected automatically** by setting **user-defined tolerances** in the topology building process.
 - For example: CLEAN command in ArcGIS
 - INPUT: Specify tolerance values (node-to-node, and min. arc length)
[fuzzy tolerance] [dangling tolerance]
 - RUN: i) Builds topological relation and creates attribute table
ii) Compares the attribute table values
iii) Removes redundant features.

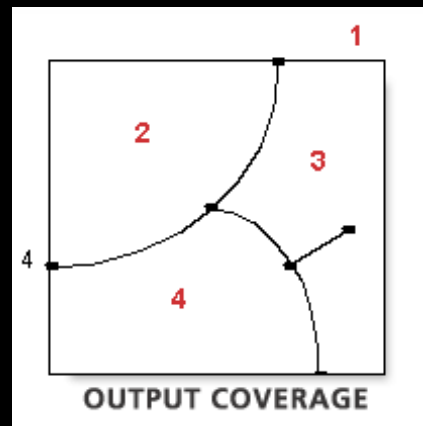
Graphical data editing- Phase 2: Topology Building

Illustration of CLEAN command:



Polygons to the left and right of each arc

Arc number	Left poly	Right poly	Length
1	1	2	5.0
2*	1	1	0.2
3	1	3	8.8
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7*	1	1	0.2
8	4	3	1.9
9*	3	3	1.5
10	4	3	1.9
11*	1	1	0.2
12	4	1	3.1



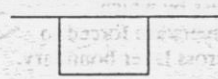
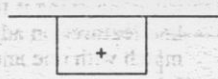
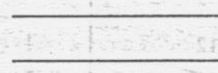
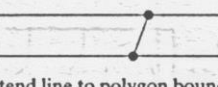
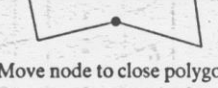
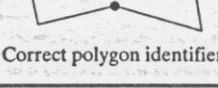
If dangling tolerance = 1.0,
then arcs 2, 5, 7, 11 will be removed

Graphical data editing- Phase 3: Data Editing and Error Corrections

- Automated tools building topological relations can remove some but not all errors.
- So, GIS allows graphical data editing procedures enabling the operator to select, delete, copy, and add graphical elements, as well as to change their properties, on a vector layer.
- Data Editing is an error-prone process, when complex map layers are processed.

(also new errors may be added in the process)

- It is also a **time-consuming** process because it must be performed layer by layer.

Type of errors	Before correction	After correction
Missing label		 Add polygon label
Missing arc		 Add arc
Overshoot		 Select and delete overshoot
Undershoot		 Extend line to polygon boundary
Dangling node		 Move node to close polygon
Wrong label ID		 Correct polygon identifier

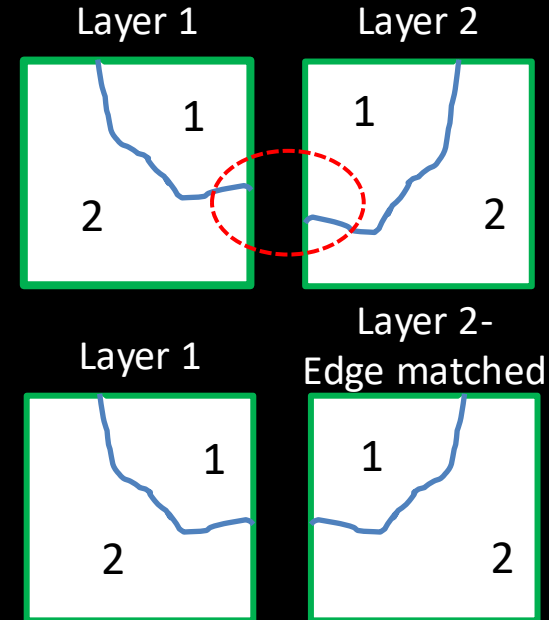
Examples of graphical editing.

Graphical data editing- Phase 4: Joining Adjacent Layers

- Generally, GIS applications require spatial data from more than one map.
- Hence, we must join adjacent layers to create a continuous geospatial database.
- This process involves three steps:

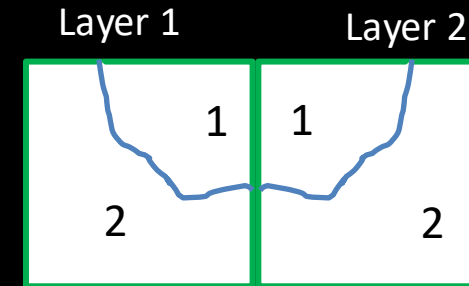
i) Edge matching:

To ensure that linear features across layer boundaries match with one another



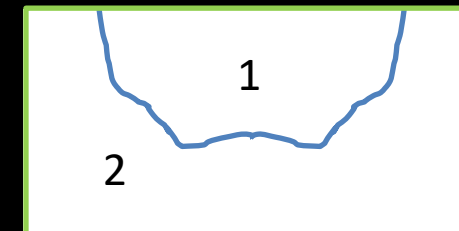
ii) Map joining:

To physically join the adjacent layers to a single layer

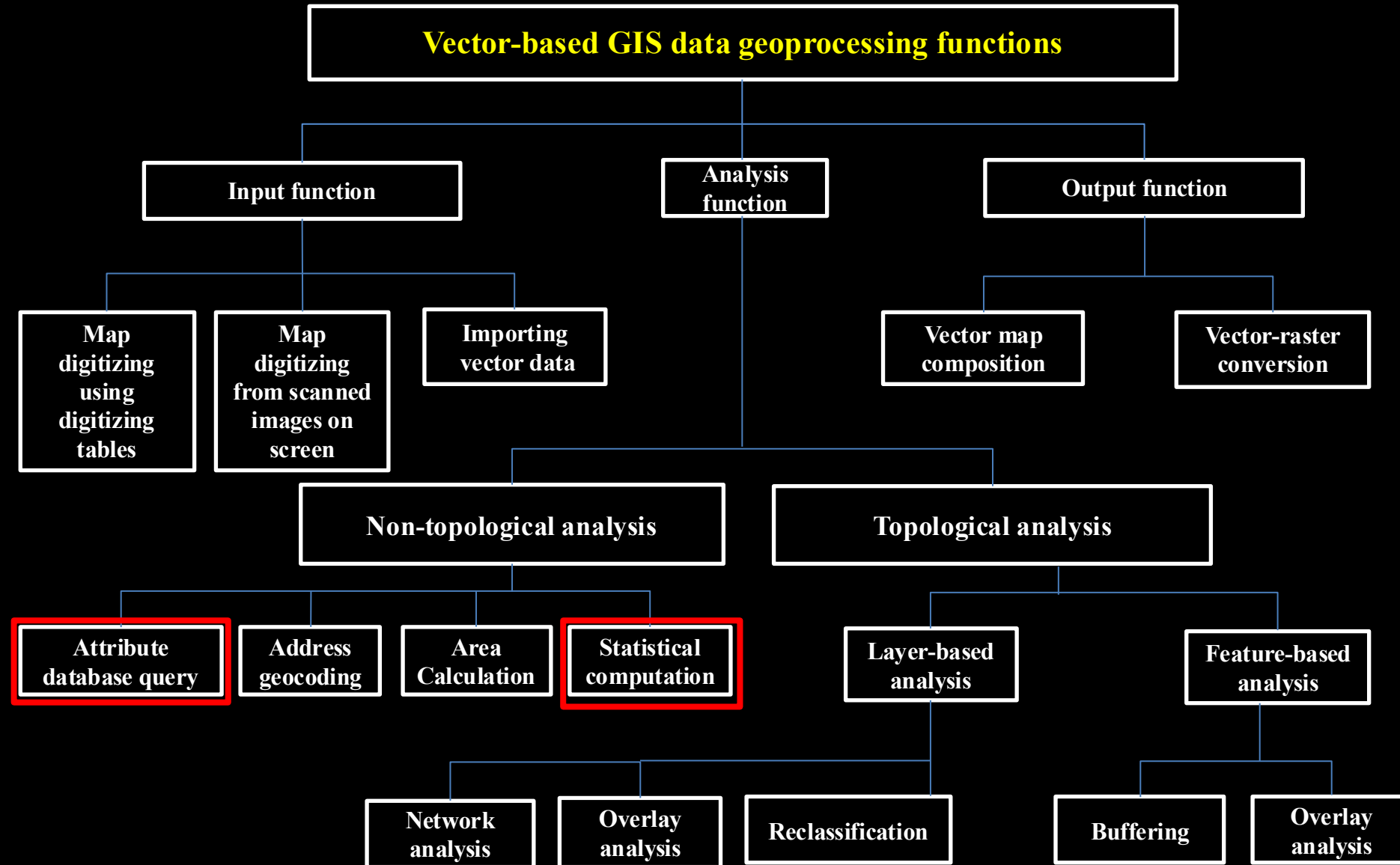


iii) Dissolving redundant polygon boundaries:

To merge adjacent polygons (from merged layers).
The topology of the resulting layer is rebuilt.



Classification of Vector Geoprocessing Functions



Attribute Database Query

- It is an essential function of vector geoprocessing.
- The storage of descriptive data in form of attribute tables in object-relational data models allows the data to be retrieved, updated, and analyzed using query commands.
- For geospatial data stored in external DBMS, SQL is the default query language.
- SQL is implemented using three key operations-
 - **SELECT**: To extract data items in specified rows of a table.
 - **PROJECT**: To extract data items in specified columns of a table.
 - **JOIN** (also called RELATIONAL JOIN): To merge two tables based on the values in the common columns of the tables.

Attribute Database Query

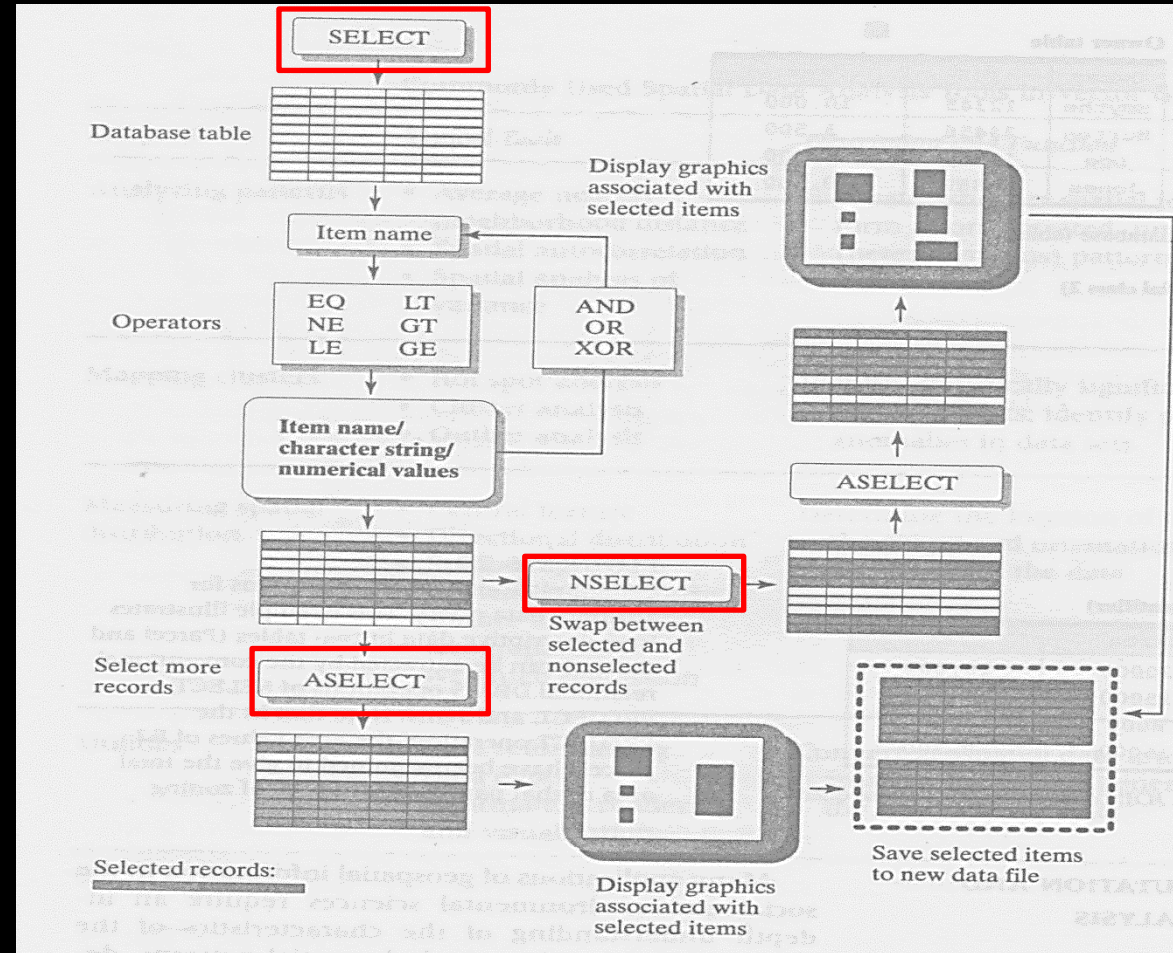
- It can be used interactively or as part of an application program (i.e., embedded within codes of a computer program), making it extremely flexible to use.
- It can use various relational (GT, LT, EQ, GE, and LE), arithmetic (=, +, -, *, /), and Boolean (AND, OR, NOT, exclusive OR) operators.
- The syntax of a typical SQL query is:

SELECT <attribute_name> FROM <table> WHERE <condition_statement>

E.g.: *SELECT state FROM India_Political_Map WHERE state_popln GT 1,00,00,000*

- SQL query results in a table, which can be displayed on screen/exported to a data file.

Example of Attribute Database Query



ASELECT (ADDSELECT): Additional Select
NSELECT (NONSELECT): Swap selection

Attribute Database Query

- SQL is implemented using three key operations-
 - **SELECT**: To extract data items in specified rows of a table.
 - **PROJECT**: To extract data items in specified columns of a table.
 - **JOIN** (also called RELATIONAL JOIN): To merge two tables based on the values in the common columns of the tables.

Parcel table

PID	Zoning	Area
12345	C1	20000
23456	R2	15000
12332	R2	8000
88889	I2	35000

Owner table

Owner	PID	Tax
Smythe	12345	10,000
Burton	23456	3,500
Doe	12332	5,000
Jones	88889	8,300

(a) The original database tables

SELECT parcel where zoning = R2 (residential class 2)

Zoning	PID	Area
R2	23456	15000
R2	12332	8000

PROJECT parcel over zoning using PID

Zoning	Area
C1	20000
R2	23000
I2	35000

JOIN parcel and owner over PID (parcel identifier)

PID	Owner	Zoning	Area	Tax
12345	Smythe	C1	20000	10,000
23456	Burton	R2	15000	3,500
12332	Doe	R2	8000	5,000
88889	Jones	I2	35000	8,300

(b) The SELECT, PROJECT, and JOIN operations in database query

Statistical Computation

- In essence, statistical computation is an extension of attribute database query.
- Conventional statistical measures→ **mean, maximum, minimum, range, standard deviation, and frequency.**
- These provide essential information about descriptive data in attribute tables, and statistical computation of these measures is a standard function of vector geoprocessing.
- However, as these statistical measures are **aspatial** and **deterministic in nature**, their usefulness for spatial analysis is rather limited.
- Classical statistical theories assume that the individual observations are independent of one another. But, in reality, the closer observations tend to be more similar than those farther apart. Remember **TOBLER'S FIRST LAW OF GEOGRAPHY??**

Statistical Computation

- However, many applications of spatial information (especially in social and environmental sciences) require an **in-depth understanding** of the spatial characteristics of the observation data.
- These include **spatial patterns**, **dependency** and **association**, **trends**, and **randomness** and **variability** that may be hidden in a spatial data set.
- These can not be assessed using classical statistics, resulting in a new branch of statistical science called ***Spatial Statistics or Statistical Spatial Data Analysis***.
- Spatial Statistics incorporate **spatial relationships into the model building process**. Spatial relationships include:
 - the physical distance between observations,
 - the existence of a hierarchy of administrative areas within which data are collected and analyzed, and
 - topological characteristics such as adjacency, containment, and connectivity of places and spatial objects.

Commonly Used Spatial Data Analysis Tools in Vector Geoprocessing

Category	Typical Tools	Description and Function
Analyzing patterns	• Average nearest neighborhood distance • Spatial autocorrelation • Spatial analysis of variance	Evaluate whether spatial features and their attribute values form a concentrated, uniform (homogeneous), or random (heterogeneous) pattern across a given geographic space.
Mapping clusters	• Hot spot analysis • Cluster analysis • Outlier analysis	Identify statistically significant values and locations of hot and cold spots; identify possible measurement errors and anomalies in data sets.
Measuring spatial distribution	• Central feature • Directional distribution • Spatial regression modeling and trend surface analysis • Geographically weighted regression	Determine the location of the center of gravity of the data, the shape and orientation of the data, and the degree of dispersion of the data.
Utilities	• Data reformatting • Data export • Information rendering and visualization	Convert geospatial data into a form suitable for statistical spatial analysis and generate information products from the results of analysis.

Spatial Calculation and Quantification

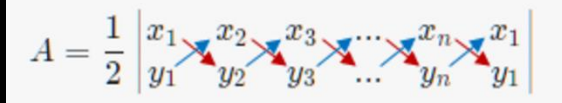
- Spatial calculation of **position**, **area**, **perimeter**, and **distance** provides important information required by numerous applications of geospatial information.
- In conventional map analysis and interpretation, the calculation of these attribute values was an extremely tedious and error-prone task.
- In vector geoprocessing, spatial calculation is **totally automated**.
- With the aid of built-in tools in GIS, users of geospatial information are now able to determine the positions, areas, perimeters, and distances of thousands of spatial features **in seconds** practically **free of measurement errors**.
- **Area of Polygon**: Based on the coordinates of the points defining the boundary of the polygon using the following equation:

$$A = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - y_i x_{i+1})$$

where

A is the area,

x and y are abscissa and ordinate values respectively of the nodes forming the polygon numbered in counterclockwise order.


$$A = \frac{1}{2} \left| \begin{array}{ccccccc} x_1 & x_2 & x_3 & \dots & x_n & x_1 \\ y_1 & y_2 & y_3 & \dots & y_n & y_1 \end{array} \right|$$

Spatial Calculation and Quantification

- **Centre of gravity/centroid of a polygon (x_c, y_c):** It is often required for label or name placement and polygon-to-polygon nearest-neighbour computation. It is calculated similarly by using node coordinates of the polygon:

$$x_c = \frac{1}{6A} \sum_{i=1}^n (x_i + x_{i+1})(x_i y_{i+1} - y_i x_{i+1})$$
$$y_c = \frac{1}{6A} \sum_{i=1}^n (y_i + y_{i+1})(x_i y_{i+1} - y_i x_{i+1})$$

- **Distances:** The calculation of the distance between a point and a line or a polygon is a little more complex, and so is the calculation between two lines, two polygons, and a line and a polygon.
- Thus, it is necessary to first determine **the nearest pair of points** in the spatial objects concerned and then apply the following equation to calculate the distance between these pair of points.
- The distance D between two points p_1 and p_2 is calculated using the Pythagorean theorem, expressed by the following equation:

$$D_{(p_1, p_2)} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Spatial Calculation and Quantification

- **Perimeters:** The perimeter or edge of a polygon defined by points p_1 through p_n (with coordinates (x_i, y_j) for i from 1 to n and $p_{n+1} = p_1$) is

$$E = \sum_{i=1}^n D(p_i, p_{i+1})$$

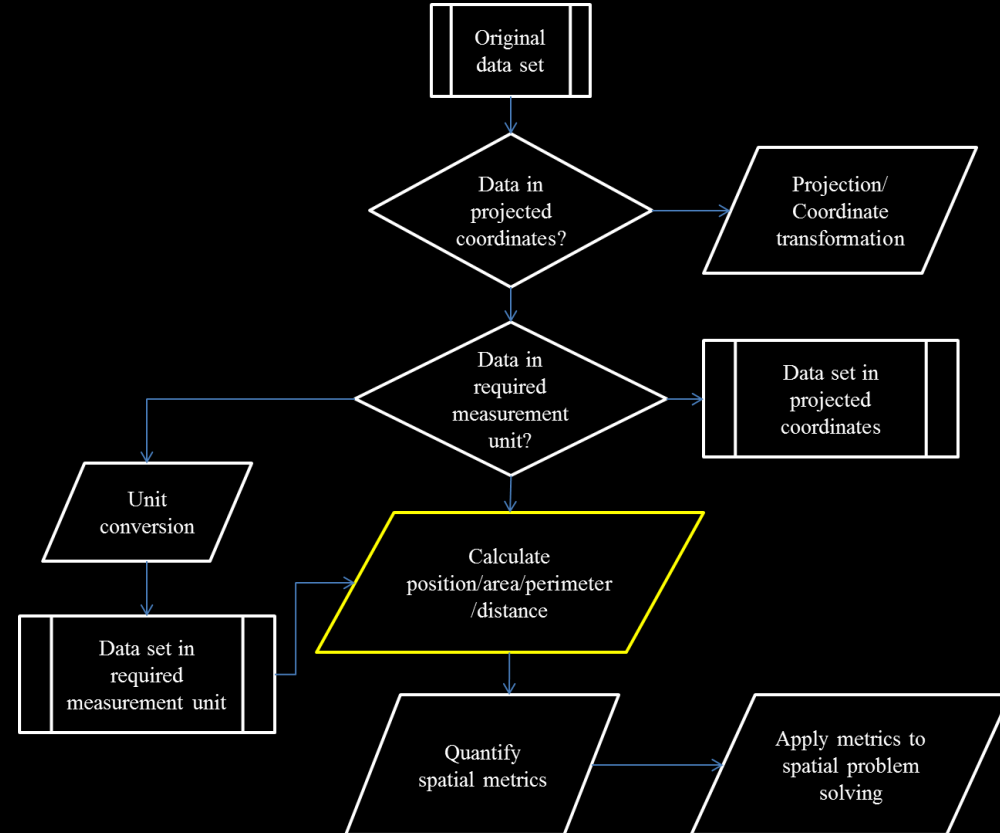
- This equation is also used to calculate the **length of a polyline**, defined in computer graphics as a continuous line object composed of one or more line segments, if $p_{n+1} \neq p_1$ (i.e., the line object does not close on itself to form a polygon).
- The power of digital computers allows the length of a polyline to be **more precisely** determined by using line interpolation algorithms based on the principles of vector functions in calculus.

Spatial Calculation and Quantification

- In vector geoprocessing, spatial calculation is often performed as an **intermediate step** to obtain new attribute data to support specific spatial analysis objectives.
- Spatial calculation involves not only the use of mathematical formulas or algorithms but also often requires **pre-processing** of the source data.

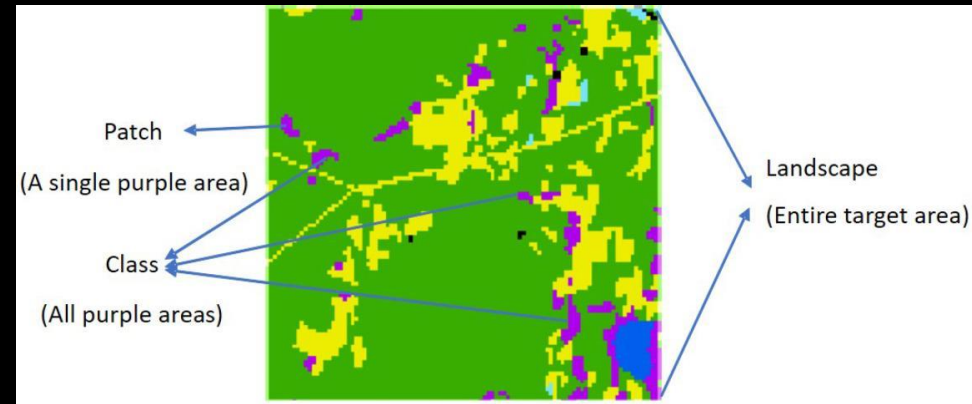
For example:

Projection and coordinate transformation and measurement unit conversion, to make sure that the results are compatible with the user's application needs.



Spatial Calculation and Quantification

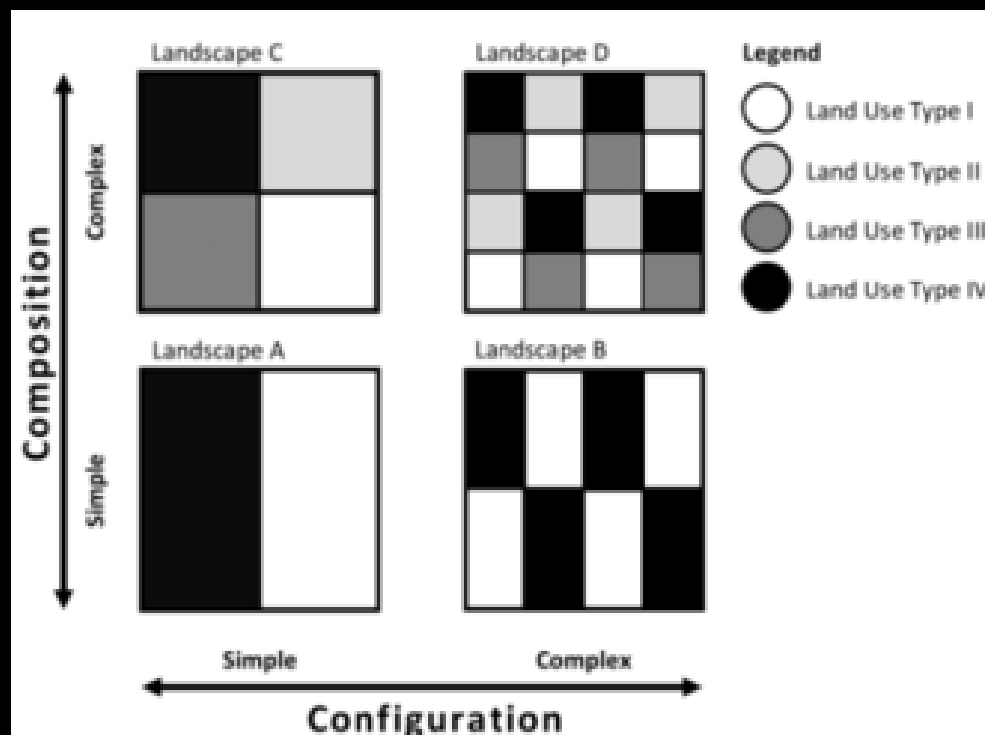
- Numerous spatial-analytical techniques have been developed to take advantage of the power of GIS in the spatial calculation of area, perimeter, and distance.
- One of these is the methods of **landscape metrics**. Simply put, landscape metrics are **numerical values or indices** that quantify the **spatial and non-spatial characteristics** of patch, class, and landscape features.
- A **"patch"** is defined as each individual occurrence of a particular land-cover or land-use type in the landscape.
- A **"class"** refers to all the occurrences of a particular land-cover or land-use type in the landscape.
- A **"landscape"** is the entire geographic space of interest. Metrics can be computed and reported in terms of individual patches, by class, or for the landscape as a whole.



Try
LecoS - Landscape Ecology Statistics tool in QGIS

Spatial Calculation and Quantification

- By using a combination of area, perimeter, and distance values and associated categorical attributes of individual patches, several landscape metrics can be computed to describe the landscape composition and configuration.
- **Landscape composition** → variety and abundance of patch types within the landscape without considering space (e.g., patch areas, density, texture etc.).
- **Landscape configuration** → on the other hand, is the spatial characteristics, arrangement, and association of patches within a class or landscape (e.g. isolation/proximity, connectivity etc.).



Spatial Calculation and Quantification

Proximity Analysis:

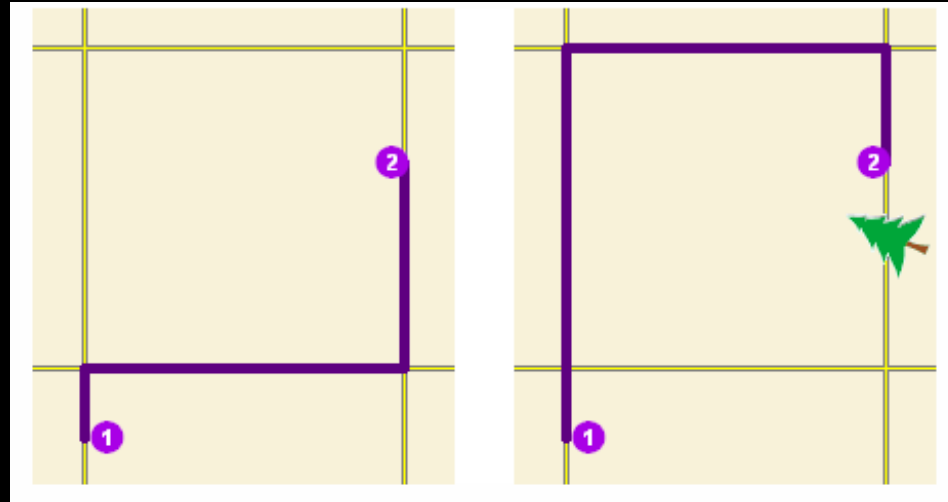
- An important spatial quantification is **proximity**, which is conceptualized using spatial measurement of point-to-point distances.
- In vector- based proximity measurement, the results are represented by a **proximity map of equally distant zones** around a location or a set of locations, rather than a table of distance values between pairs of locations.
- In practice, the concept of **proximity can be extended from physical measurement of distance to the calculation of movement times**.

For example: Proximity → in terms of distance / time / cost / energy consumed etc.

- Here, the amount of resistance, or cost, required to traverse a path in a network, or to move from one element in the network to another is called **Impedance**.
- Example:
 - Forest fires → Travel times depends on various impedances → wind speed, direction and other weather conditions, presence of nearby water bodies, etc.
 - Emergency response vehicles → Travel times depend on the impedances → traffic density and speed, number of necessary stops, etc.

Spatial Calculation and Quantification

- Another key term in proximity analysis is **Barriers**. These are topographic features in the measured geographic space that modify distance, and hence travel times.

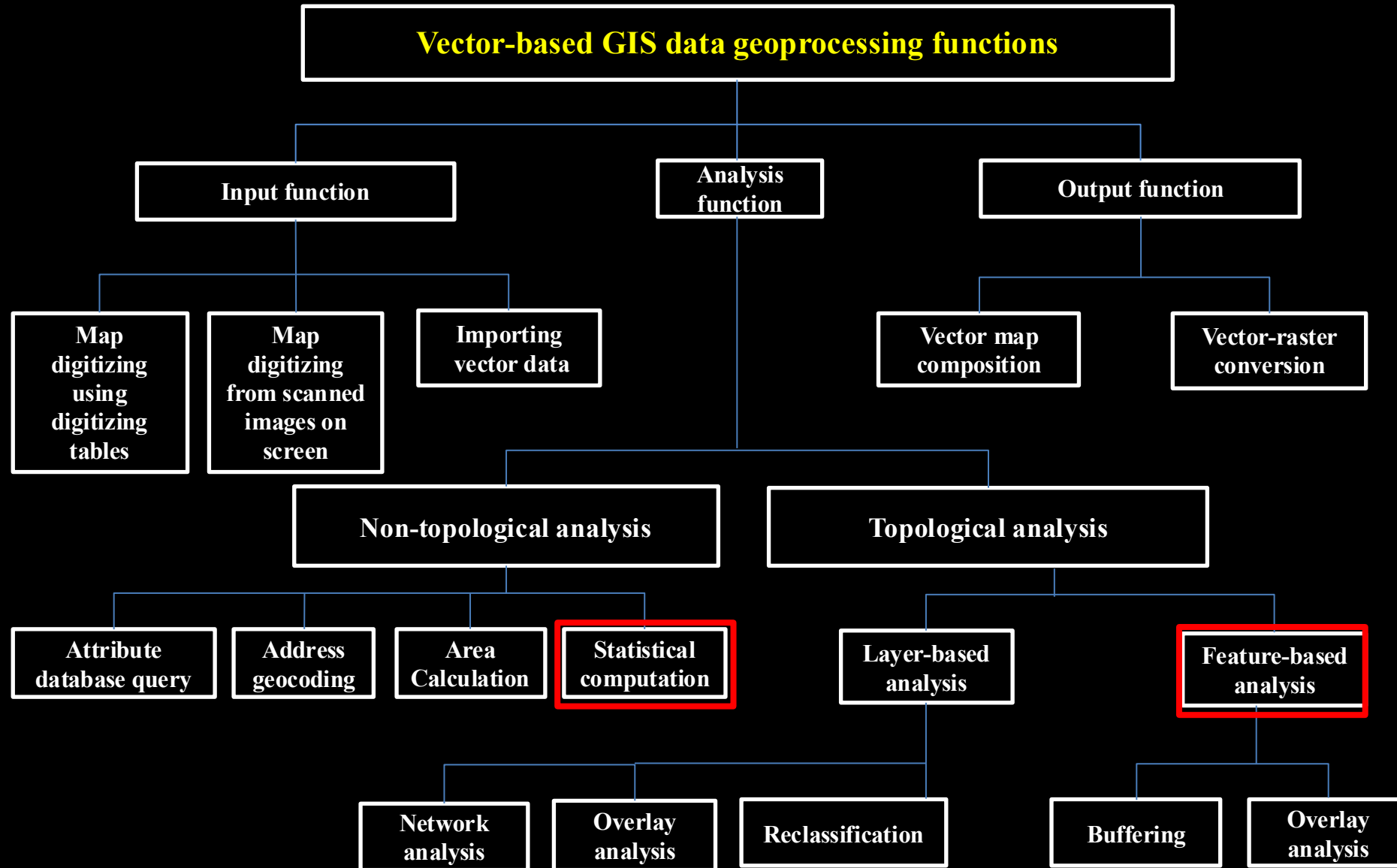


Without barriers

Without barrier

- Barriers can be either:
 - Absolute barriers** → completely restrict movement unless an alternate path exists, or
 - Relative barriers** → passable only at a cost that may be equated with an increase in the physical distance and/or effort used. E.g.: Toll booths.

Classification of Vector Geoprocessing Functions

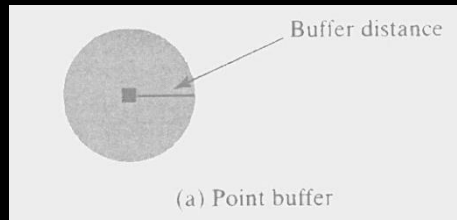


Feature- based Topological Functions

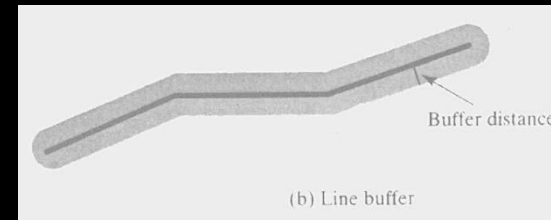
- Topological geoprocessing includes a variety of processes that rely on the use of topological relationships (**adjacency**, **containment**, and **connectivity**) for spatial problem solving.
- All methods of topological geoprocessing are constructed using one or more of these relationships.
- Similar to non-topological methods, topological methods can be used individually as **stand-alone** spatial problem-solving tools. They can also be used in **conjunction** with other topological and non-topological methods in more complex GIS applications.
- The methods of topological geoprocessing are **more sophisticated** than their non-topological counterparts, both conceptually and technically.
- Common examples of feature-based topological functions are:
 - Buffering
 - Overlay Analysis

Buffering

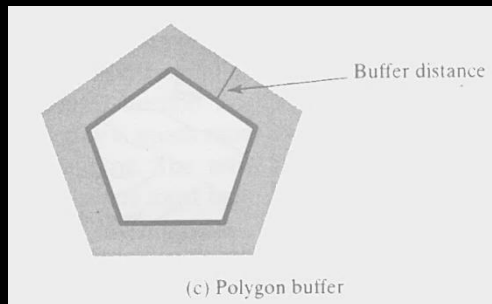
- A buffer is a **zone with a specified width** surrounding a spatial feature.
- The process of creating a buffer (buffering) is an essential function in vector geoprocessing.
- Since the basic graphical elements of vector geospatial data are **points**, **lines**, and **polygons**, buffers can be generated around each of these elements.
- In each case, the buffer generated takes the **shape from the feature**.



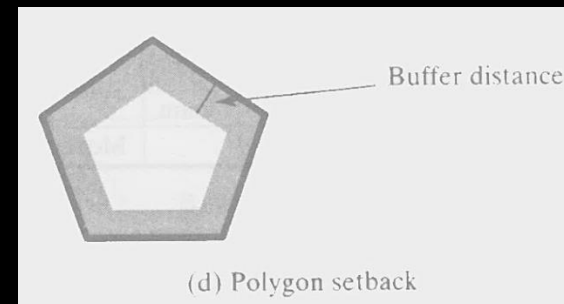
Point Buffer: A circle with a specified radius (as buffer distance) drawn around the point.



Line buffer: A band with a specified distance created on **both sides** of the line conforming to the line's curve.



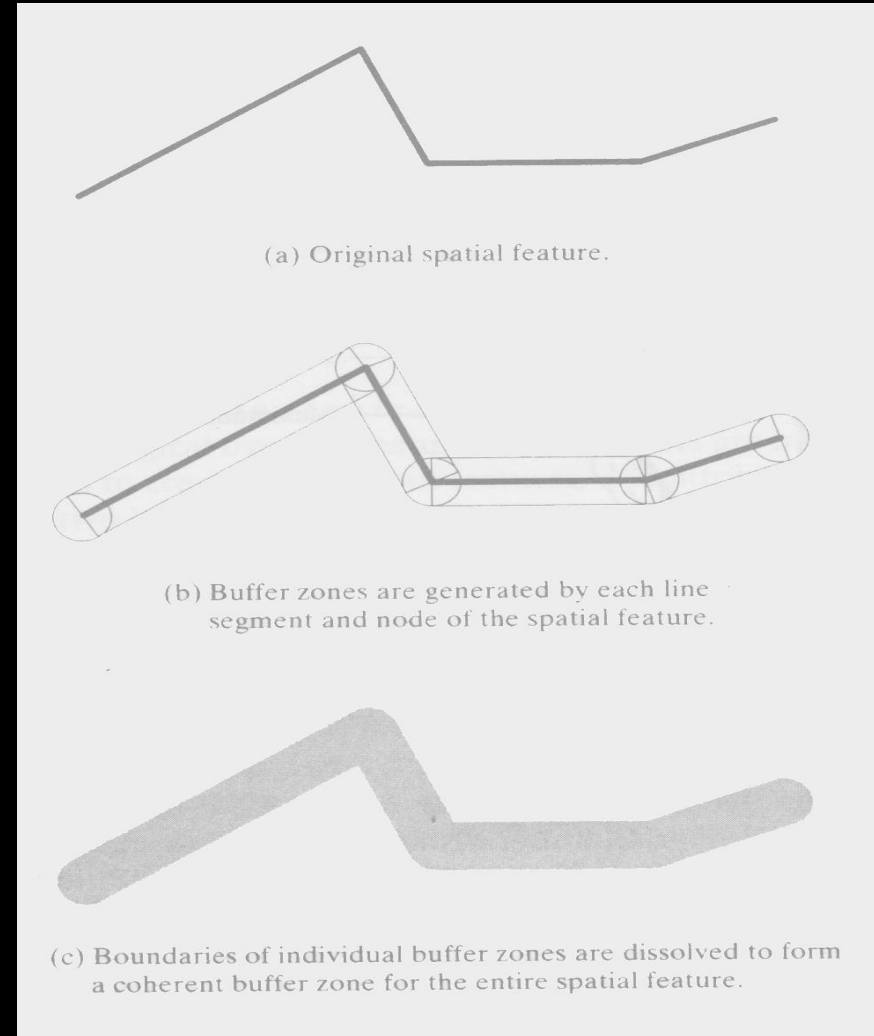
Polygon buffer: A belt of specified buffer distance from the edge of the polygon surrounding the polygon and conforming to its shape.



When the buffer zone is constructed inward from the polygon boundary, it is called **polygon setback**.

Buffering

- Buffer zones are **polygons**, no matter whether they are generated from a point, line, or polygon feature.
- In geoprocessing, the buffer distance is a **user-defined** parameter that is used by the system to generate the buffer zones.
- For highly convoluted line features, such as a meandering river or complex polygon boundaries, buffer zones generated tend to overlap.
- However, boundaries of overlapping buffer zones are **automatically dissolved** to produce a single coherent buffer zone for the entire spatial feature.



Buffering

- The objective of buffering is to **delineate the area of interest** for a neighborhood analysis.
- It is often used in conjunction with other geoprocessing functions to **search spatial features** that fall inside or outside this area of interest and analyze their spatial patterns and interrelationships.
- Common **applications** of buffering are:
 - identification of properties **within a certain distance** of a zoning change application in urban development,
 - delineation of **areas around** natural features where human activities are restricted in natural resource management, and
 - determination of areas affected by location, movement, or spread of hazardous materials in environmental management.

Featured-based Overlay Analysis

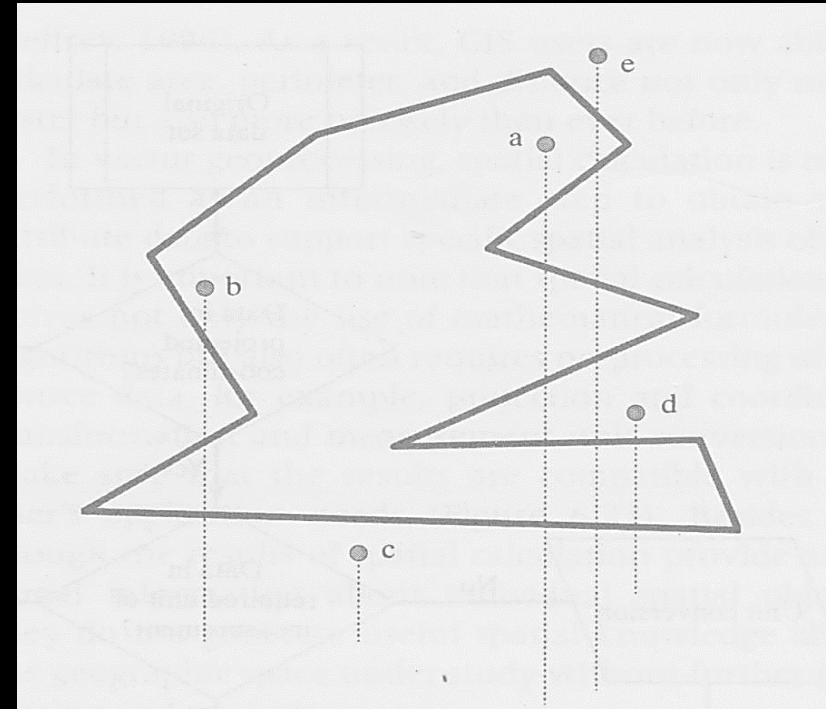
- The aim of this function → to find the **relationships** between individual features of one layer and those of another layer.
- One of the most commonly used feature-based overlay functions is **point-in-polygon** matching.
- It determines whether a given point feature falls **inside a polygon**.
- The most straightforward test to determine the position of a point relative to a polygon is the **plumb-line algorithm**.

In this method, a point is:

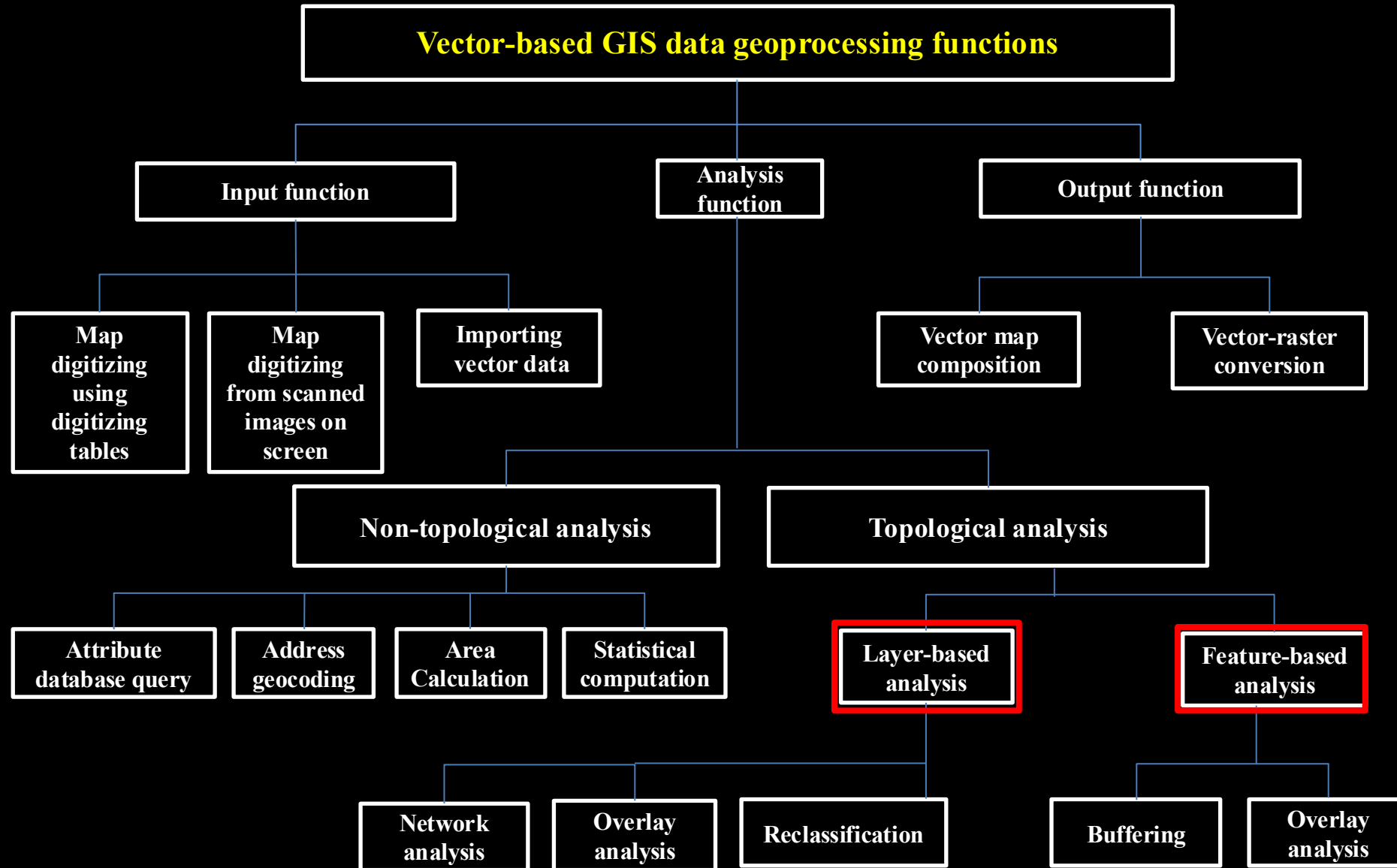
- within a polygon if its plumb line has an odd number of intersections with polygon boundaries, or
- outside the polygon if the plumb line has zero/even number of intersections with polygon boundaries.

E.g.:

Points	No. of intersections	Conclusion
a, b	5, 3 respectively	Within the polygon
c, d, e	0, 2, 6 respectively	Outside the polygon



Classification of Vector Geoprocessing Functions



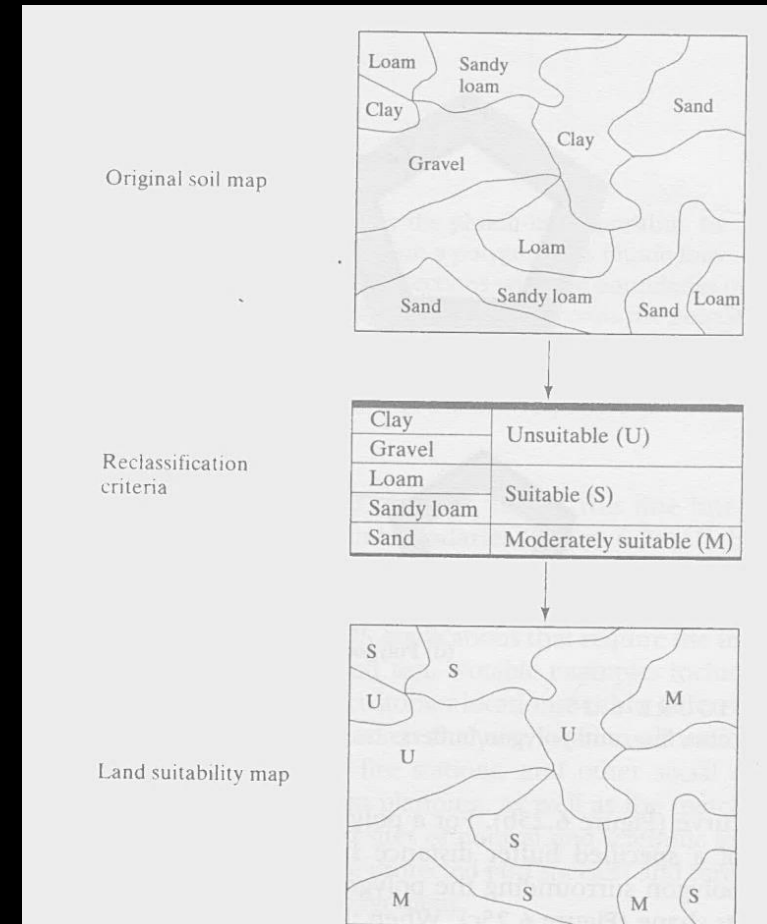
Layer-based topological functions:

- These functions work on the **entire layer** rather than on individual spatial features.
- Map layers that are used as input to this particular class of geoprocessing functions must be **topologically structured**.
- The output layers from these functions are topologically structured on completion of the function.
- Layer-based topological functions, as a rule, are **computationally intensive** in nature.
- Common examples of layer-based topological functions are:
 - Reclassification and Aggregation
 - Overlay Analysis

Reclassification and Aggregation

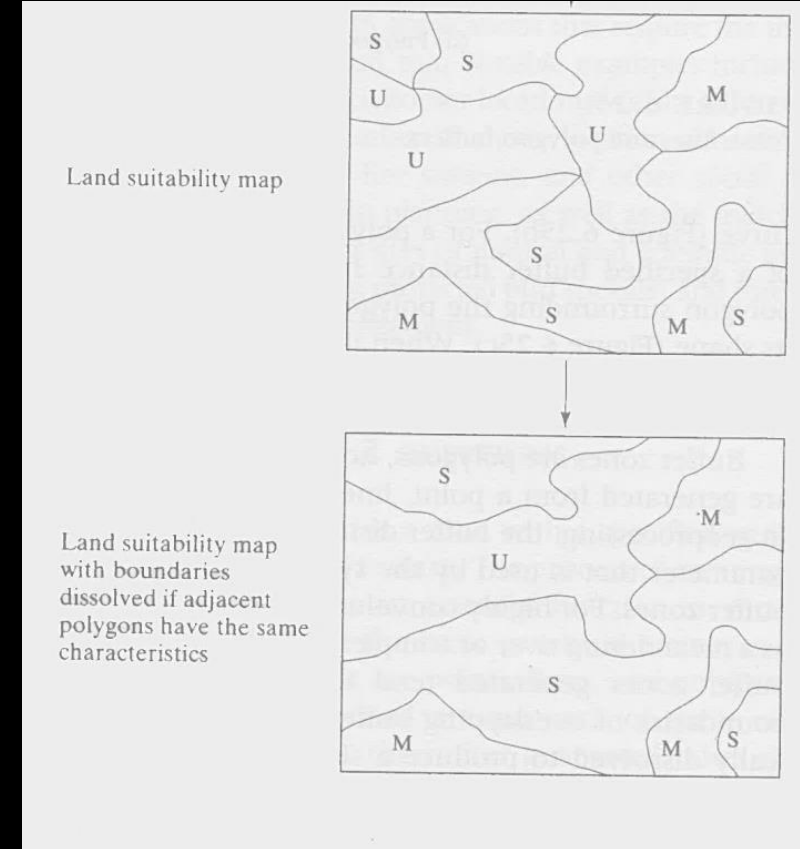
- It is a **database simplification process** that aims at reducing the number of categories of attribute data on a layer.
- This process is made up of **two steps** that involve:
 - i) Reclassification: a **non-topological process** using attribute data, and
 - ii) Aggregation: a **topological process** using graphical data.

- Step 1: Reclassification→
 - Using non-topological methods of attribute data query
 - **select attribute data** of a particular value (or range of values) by an arithmetic operator (=) or a relational operator (EQ) and
 - **then assign a new value** to them according to the new classification scheme.



Reclassification and Aggregation

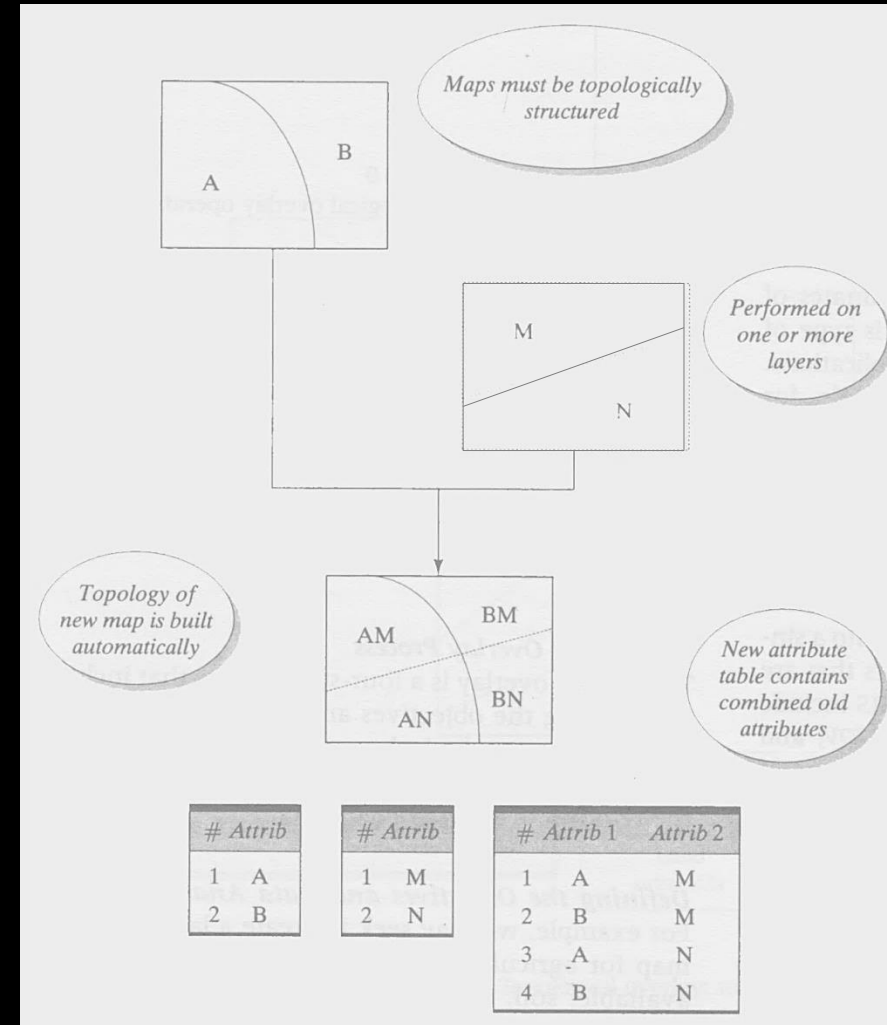
- Step 2: Aggregation→
 - After reclassification, some polygons will have the **same attribute values** as their neighbors.
 - The **common boundaries** between polygons with identical attribute values become **redundant** and have to be removed in order to maintain database integrity.
 - So, the second step is a topological process using the **adjacency relationship** to determine neighbouring polygons with similar attribute values and remove the redundant boundaries in between.



As the topology of the original layer has been changed after reclassification, it must be **rebuilt** to generate new attribute tables and establish the topological relationships among the graphical elements in the new layer

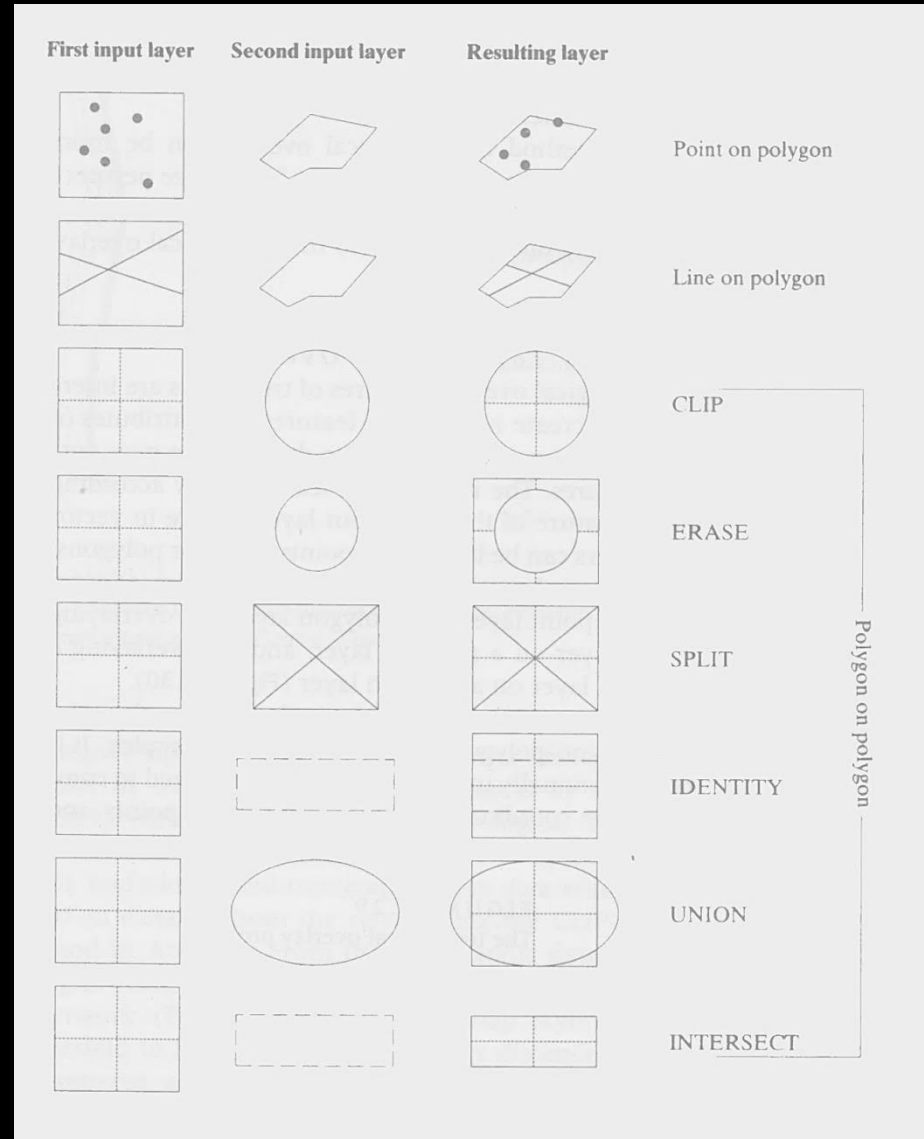
Layer-based overlay analysis

- “Topological overlay” → implies that
 - the layers must have been **topologically structured**, and
 - after the overlay, the topology of the combined layer (created from the overlay operation) have to be updated.
- The resulting attribute table for the new layer consists of **both old attributes and the new attributes** formed by logical or mathematical operation on the old ones.
- The method of topological overlay can be most easily understood by explaining it from three perspectives:
 - (1) **the types of overlay**,
 - (2) **the topological overlay operators**,
 - (3) **the topological overlay process**.



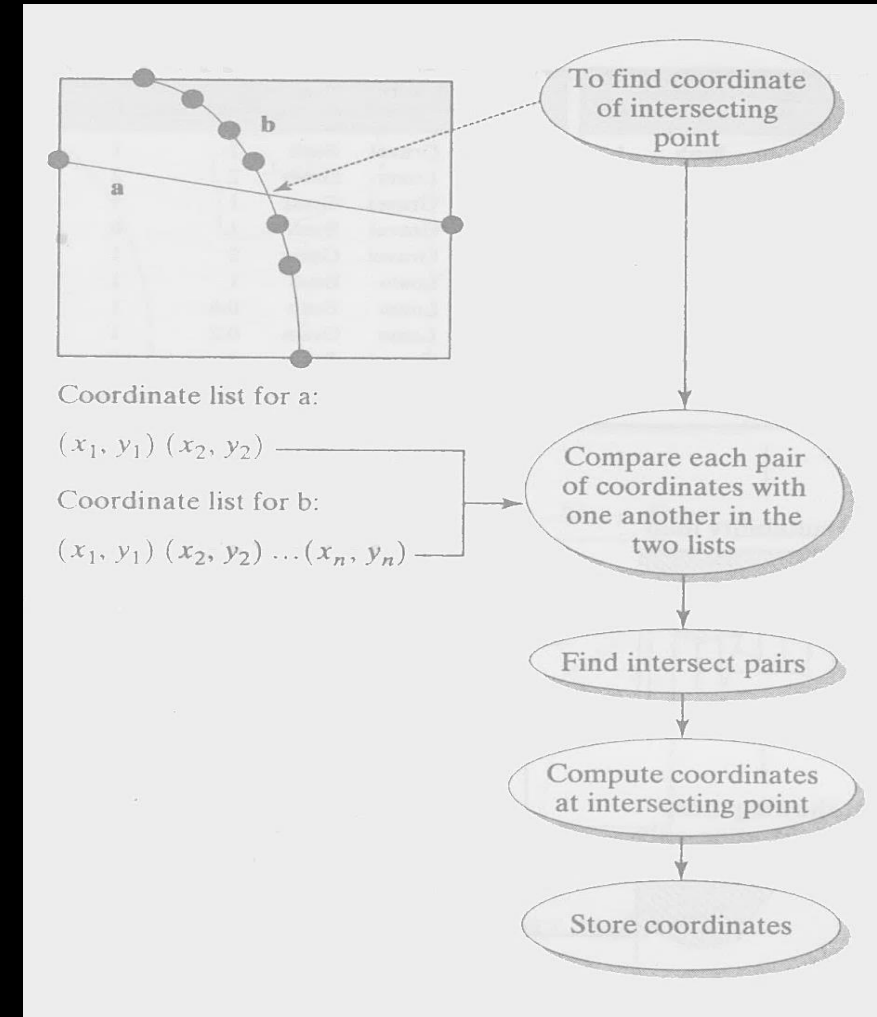
Layer-based overlay analysis

Types of Topological Overlay and Overlay Operations



Layer-based overlay analysis

- Of three topological overlay types, the polygon-on-polygon overlay is **complex**.
- The need to compute the coordinates of the intersecting points and then to compare them with the list of coordinates of the two intersecting lines (See Fig.)
- Hence, it is **computationally intensive**.
- This type of overlay is particularly useful for **modeling applications**.
- For example: By overlaying soil types with land-use parcels, it is possible to **associate soil types with land use**, and hence determine which types of land use have occupied fertile soil and where they are located.



Layer-based overlay analysis

It involves a four-step process:

- Defining the objectives and data analysis criteria,
- Applying topological overlay operators,
- Post-overlay processing and interpreting the results, and
- Removal of sliver polygons.

Example:

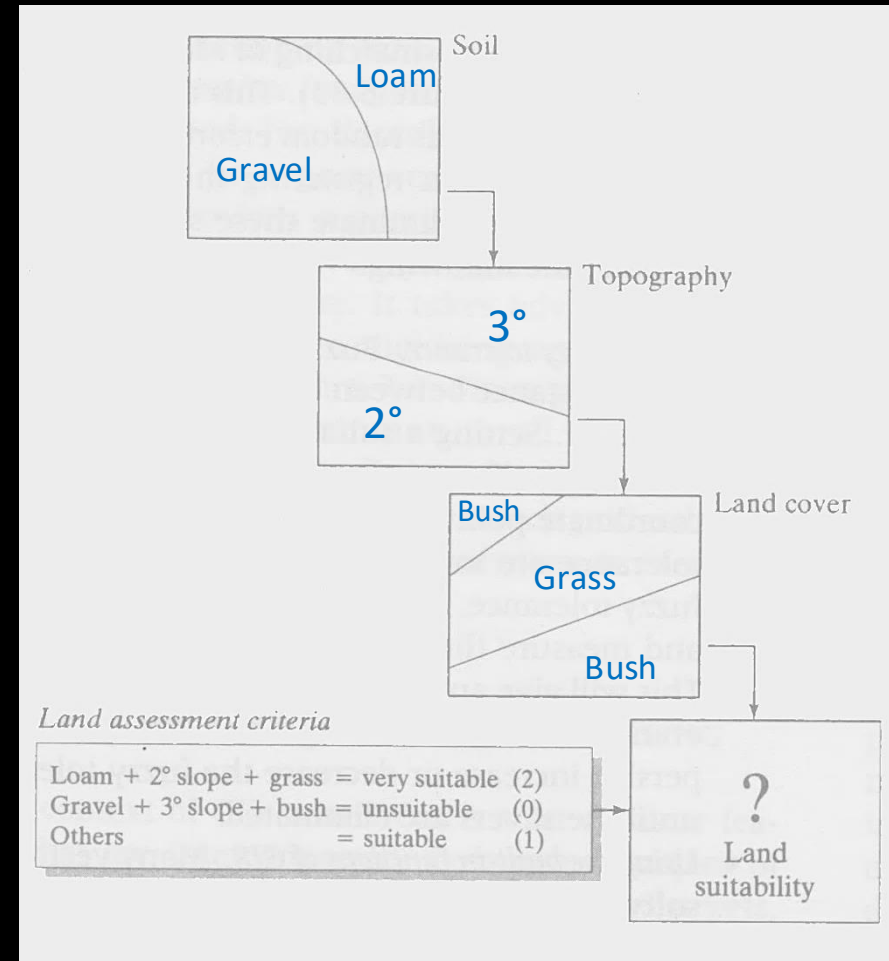
i) Defining the objectives and data analysis criteria:

Objective: Land suitability map for agricultural development.

Input layers: Soil, Topography (slope angles), and Land cover.

Data analysis criteria:

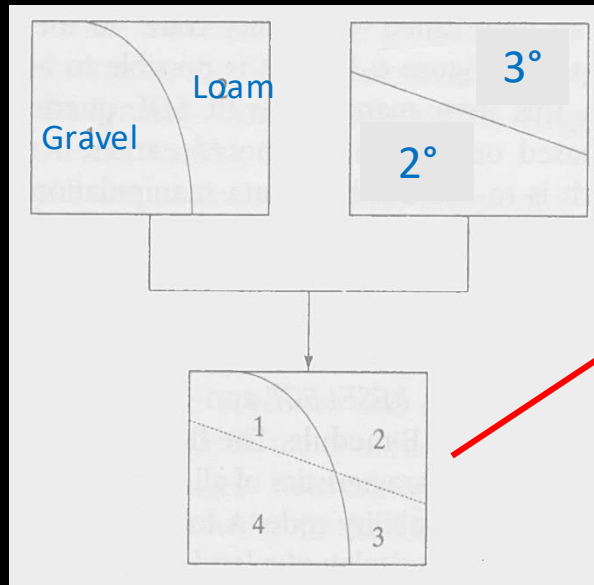
- Loam + 2° slope + grass → very suitable (code = 2)
- Gravel + 3° slope + bush → unsuitable (code = 0)
- Others → suitable (code = 1)



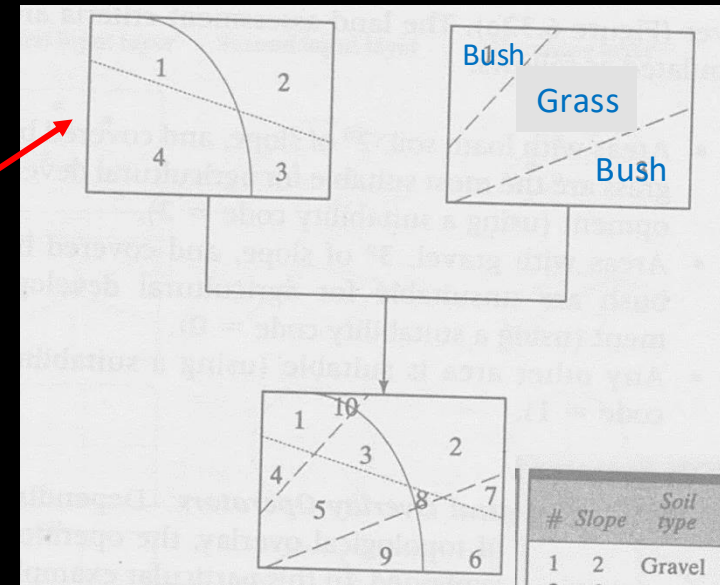
Layer-based overlay analysis

ii) Applying topological overlay operators:

- Based on the objective of topological overlay, operators can be applied as follows:
- In this example, only 'UNION' operator is needed. It is applied twice, as shown:
 - to overlay soil (S) and topography (T) → creating a new layer S-T and its attribute table.
 - to overlay S-T layer with the land-cover (L) layer → creating a new layer S-T-L and its attribute table.



#	Slope	Soil type	Area
1	2	Gravel	2
2	2	Loam	3
3	3	Loam	1
4	3	Gravel	4

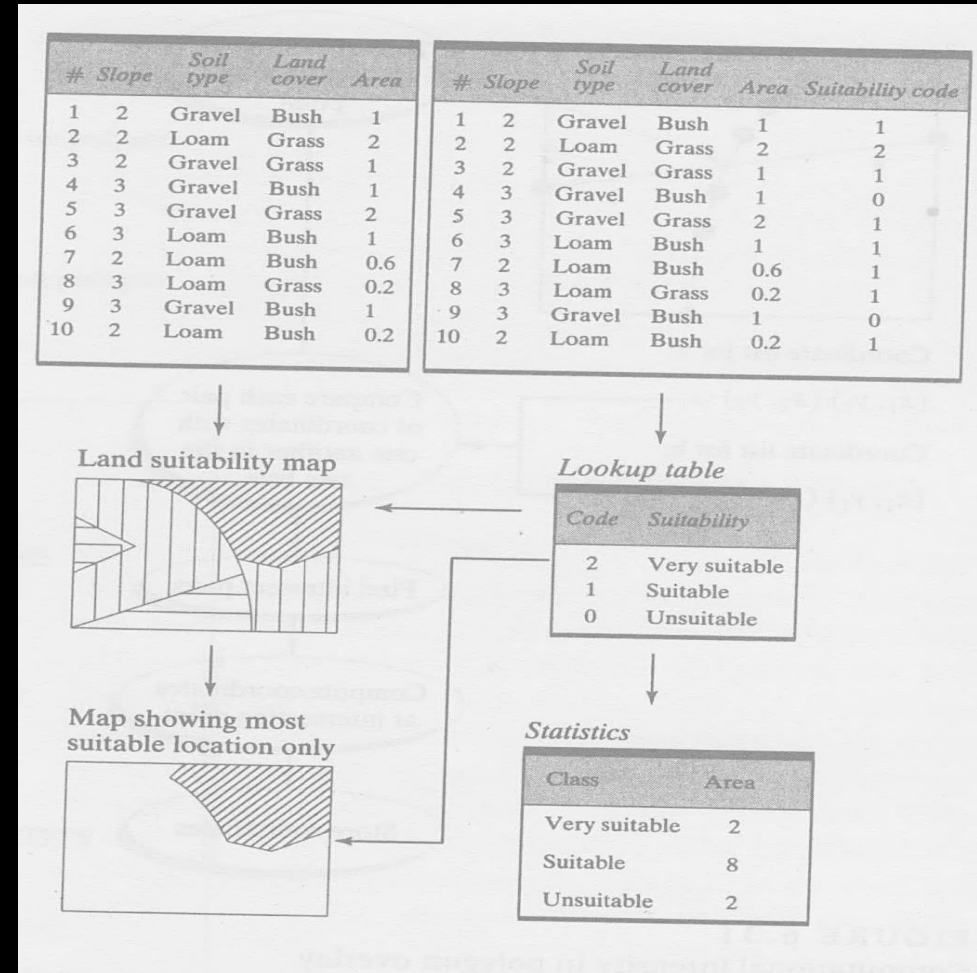


#	Slope	Soil type	Land cover	Area
1	2	Gravel	Bush	1
2	2	Loam	Grass	2
3	2	Gravel	Grass	1
4	3	Gravel	Bush	1
5	3	Gravel	Grass	2
6	3	Loam	Bush	1
7	2	Loam	Bush	0.6
8	3	Loam	Grass	0.2
9	3	Gravel	Bush	1
10	2	Loam	Bush	0.2

Layer-based overlay analysis

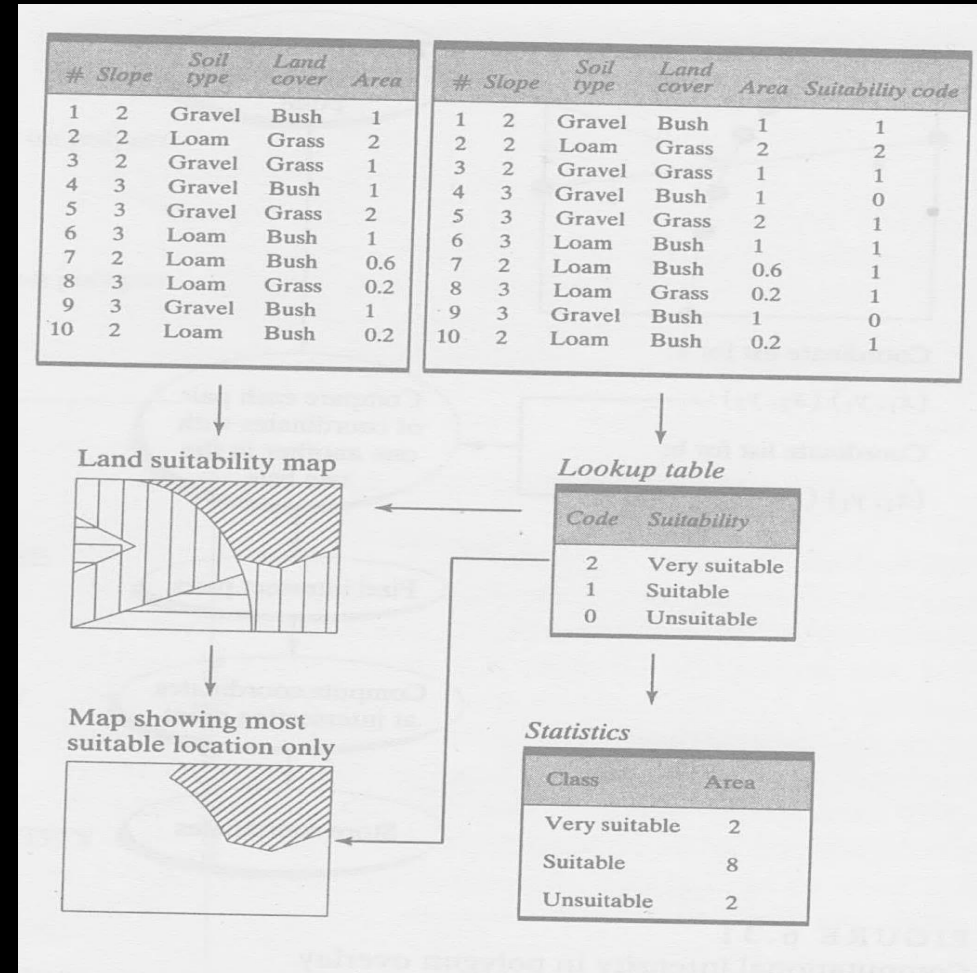
iii) Post-overlay processing and interpreting the results:

- This step requires the use of database management tools to change the structure of the attribute table by adding an item called "suitability code" to the S-T-L feature table.
- It is possible to add the code to this item manually with SQL queries and joins, based on the criteria noted earlier.
- Another approach is to use attribute data manipulation commands to select from the attribute file of the S-T-L layer the type of soil, topography, and land cover that meet the stipulated criteria and then add the correct suitability code.



Layer-based overlay analysis

- The final attribute table will incorporate characteristics of all three input layers and an item of suitability code.
- A lookup table can be used to control the display of a land suitability map.
- It is also possible to extract only those polygons that are most suitable for agricultural development (suitability code = 2).
- Therefore, the end product of topological overlay is a land suitability map from which the most suitable locations may be selected.
- Using the function of statistical computation, a table showing the statistics of different categories of land-use suitability may also be generated.



Layer-based overlay analysis

Illustrations of Various Overlay Analysis

Inputs:

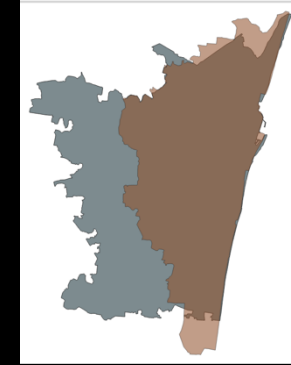
Layers



A



B



Map of A and B
(B made transparent for viewing)

Attribute Table →

Info_A	
1	A

Info_B	
1	B

Layer-based overlay analysis

Illustrations of Various Overlay Analysis

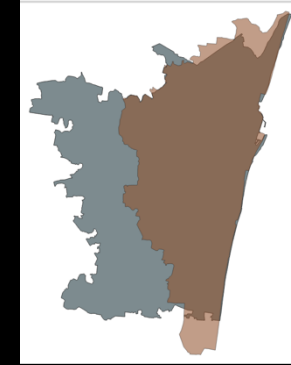
Inputs:



A



B

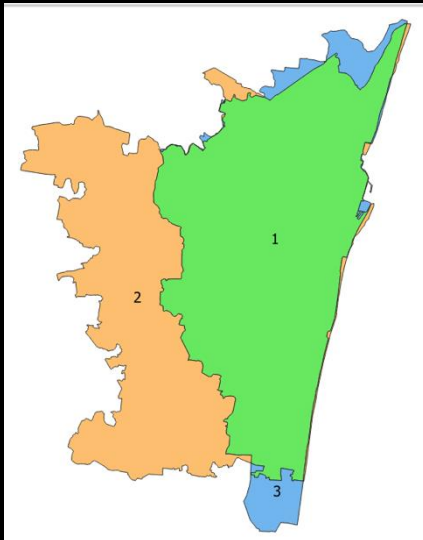


Map of A and B

(B made transparent for viewing)

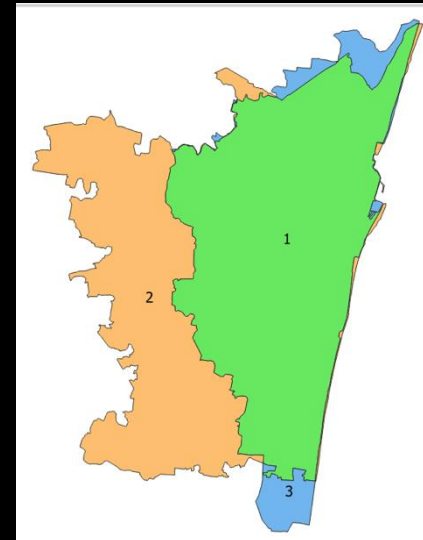
Union (QGIS):

*Input layer - A;
Overlay layer- B*



Union1 :: Features Total: 3, Filtered: 3,...			
	Info_A	Info_B	ID
1	A	B	1
2	A		2
3		B	3

*Input layer- B;
Overlay layer- A*



Union2 :: Features Total: 3, Filtered: ...			
	Info_B	Info_A	ID
1	B	A	1
2		A	2
3	B		3

Layer-based overlay analysis

Illustrations of Various Overlay Analysis

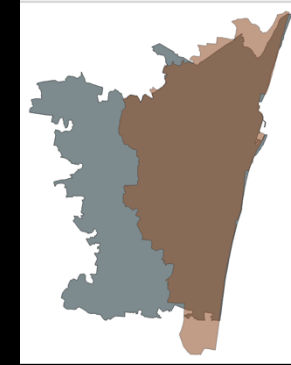
Inputs:



A



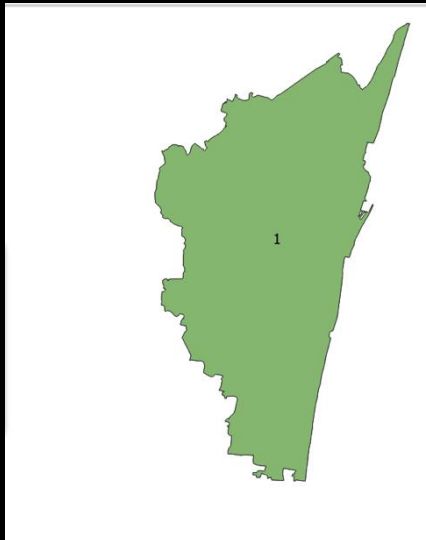
B



Map of A and B

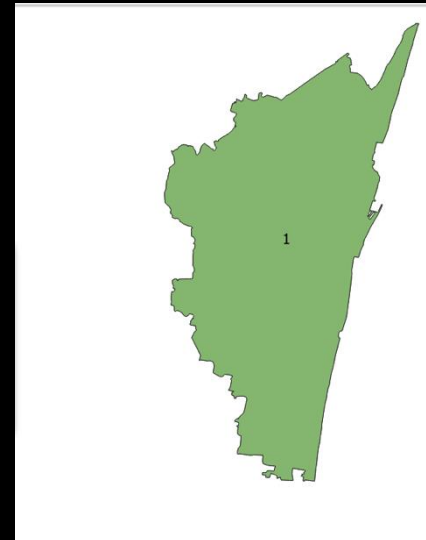
(B made transparent for viewing)

Intersection (QGIS): *Input layer - A;*
Overlay layer- B



	Info_A	Info_B	ID
1	A	B	1

Input layer- B;
Overlay layer- A



	Info_B	Info_A	ID
1	B	A	1

Layer-based overlay analysis

Illustrations of Various Overlay Analysis

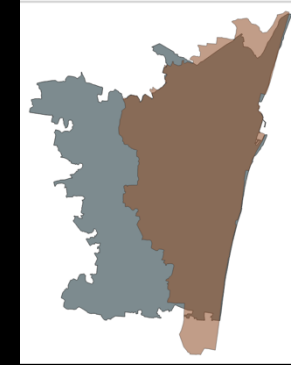
Inputs:



A



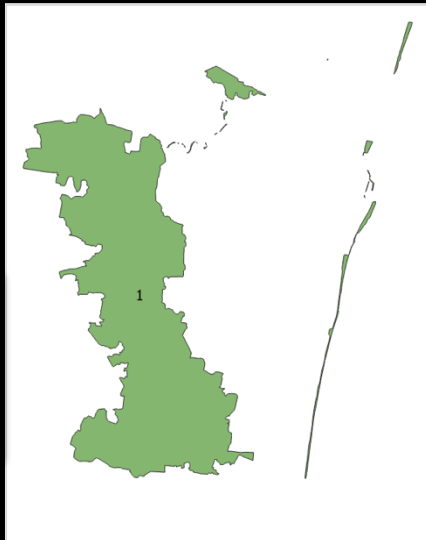
B



Map of A and B

(B made transparent for viewing)

Difference (QGIS): *Input layer - A;*
Overlay layer- B



Info_A	ID
1 A	1

Input layer- B;
Overlay layer- A

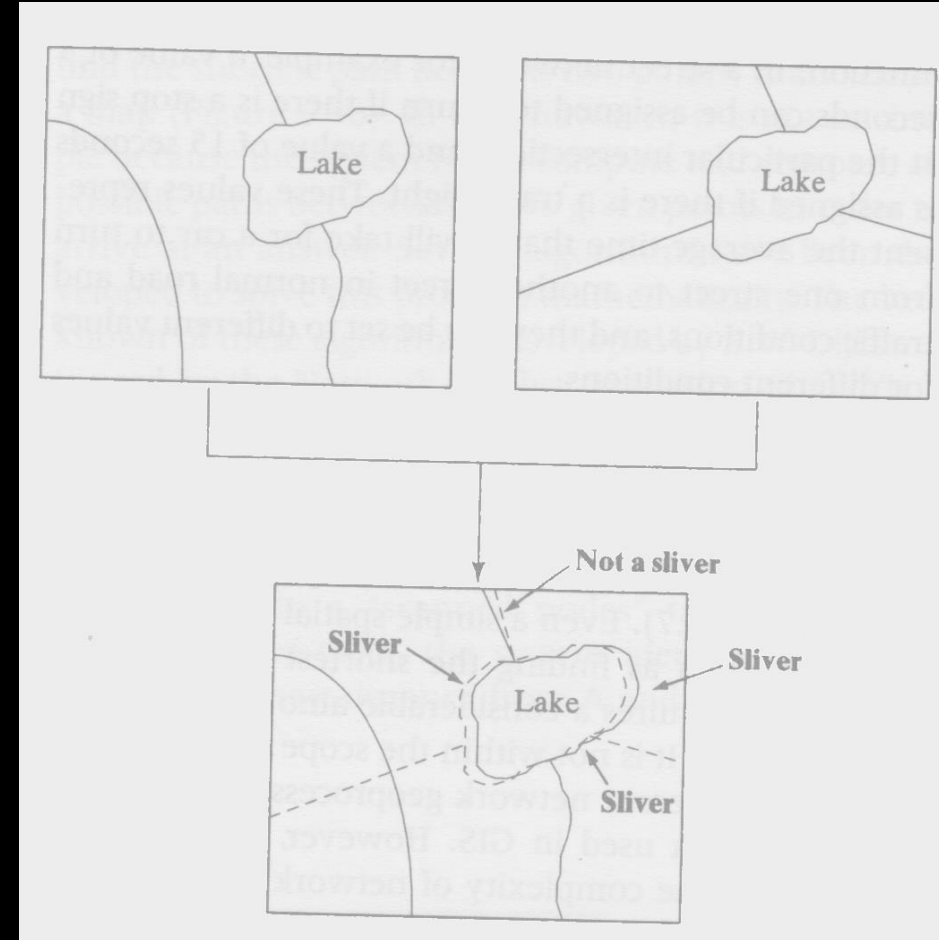


Info_B	ID
1 B	1

Layer-based overlay analysis

iv) Removal of Sliver Polygons:

- Sliver polygons, also called **spurious polygons**, are very small polygons that are generated in topological overlay as a result of the mismatching of identical features on different layers.
- This can be due to a variety of factors:
 - random errors in line digitizing and
 - imperfection in registering the layers.
- There are different ways to eliminate these sliver polygons:
 - Using fuzzy tolerance
 - Using the built-in functions of GIS
 - Using a template in digitizing



Layer-based overlay analysis

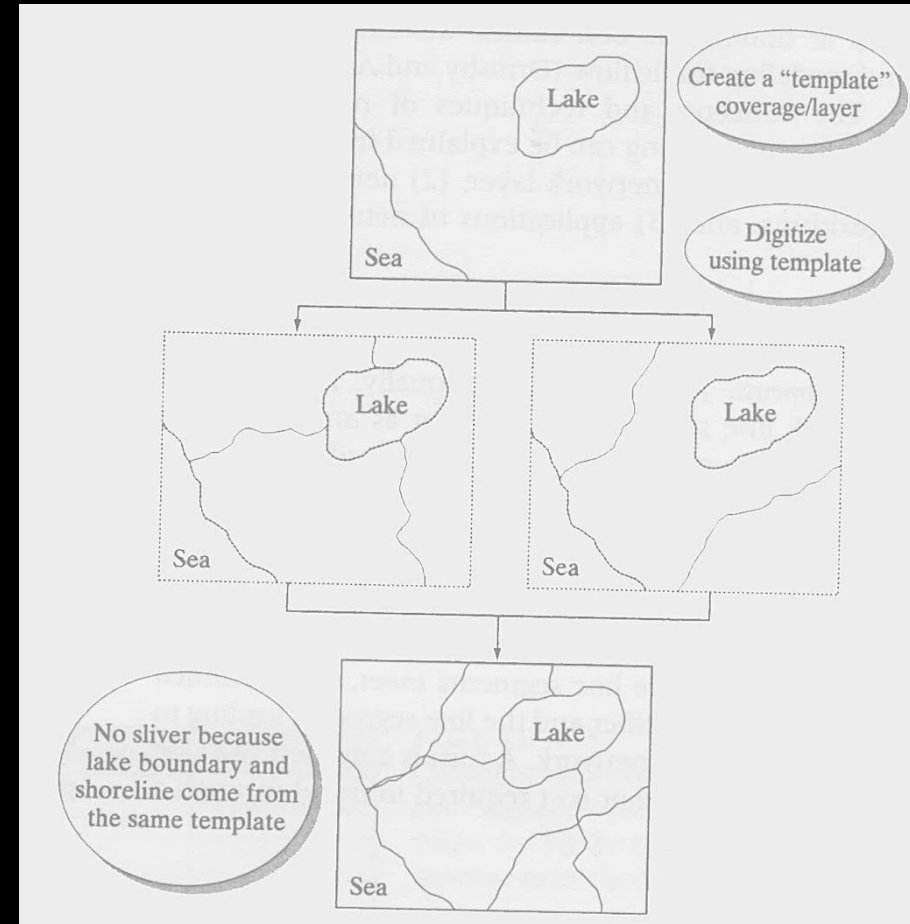
iv) Removal of Sliver Polygons:

- **Using fuzzy tolerance:**
 - It is the minimum distance between coordinates on the output layer.
 - Setting a **suitable fuzzy tolerance** will cause the sliver polygon to disappear when coordinate points that are closer than the fuzzy tolerance are merged.
 - To determine the correct fuzzy tolerance, make a **hard copy** of the overlay and measure the amount of sliver with a scale. This will give an initial estimate of the fuzzy tolerance to use in "cleaning" the overlay. If slivers persist, increase or decrease the fuzzy tolerance until the slivers are eliminated.
- **Using the built-in functions of GIS:**
 - Many vector GIS software packages have a built-in function to eliminate the sliver polygons resulting from topological overlay.
 - In ArcGIS, for example, the command ELIMINATE, which merges selected polygons by dropping the longest shared border between them, can be used to remove slivers based on one or more criteria.
 - These criteria can be the **area** (sliver polygons being very small), **the shape** (sliver polygons being very long and narrow), and **the number of arcs** (slivers having only two bounding arcs).

Layer-based overlay analysis

iv) Removal of Sliver Polygons:

- *Using a template in digitizing:*
 - In contrast to the two methods just described, which treat sliver polygons as a postoverlay process, this approach provides a preventive measure to ensure that **no sliver polygon** will occur in topological overlay.
 - It takes advantage of the method of using a digitizing template.
 - Since polygons (e.g., lakes) that are common to all layers in the overlay process are derived from the same digitizing template, they will match perfectly and the occurrence of sliver polygons is therefore avoided.



What will you learn?

The concepts and techniques of **spatial**
and **non-spatial** geoprocessing tools
that operate on raster geospatial data

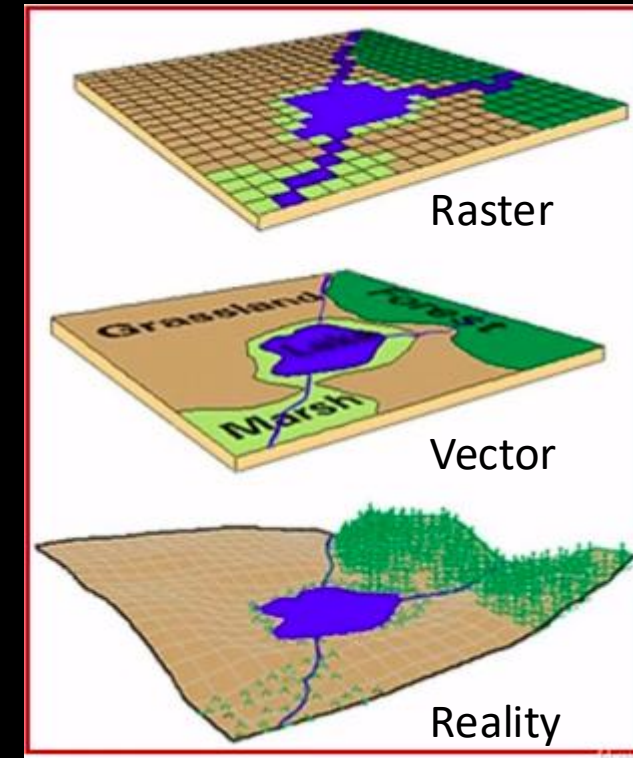
Introduction

- The term *geoprocessing* is now commonly used in GIS literature for "**geographic data processing**."
- A GIS typically contains hundreds of geoprocessing tools designed for all aspects of implementing geospatial technology and using geographic information, including:
 - data digitizing,
 - database creation and maintenance,
 - spatial data analysis and modeling, and
 - generation of cartographic and statistical information products.
- Geoprocessing is **central to the use of GIS** because it turns geospatial data into useful spatial information and intelligence.

Special Features of Raster Data

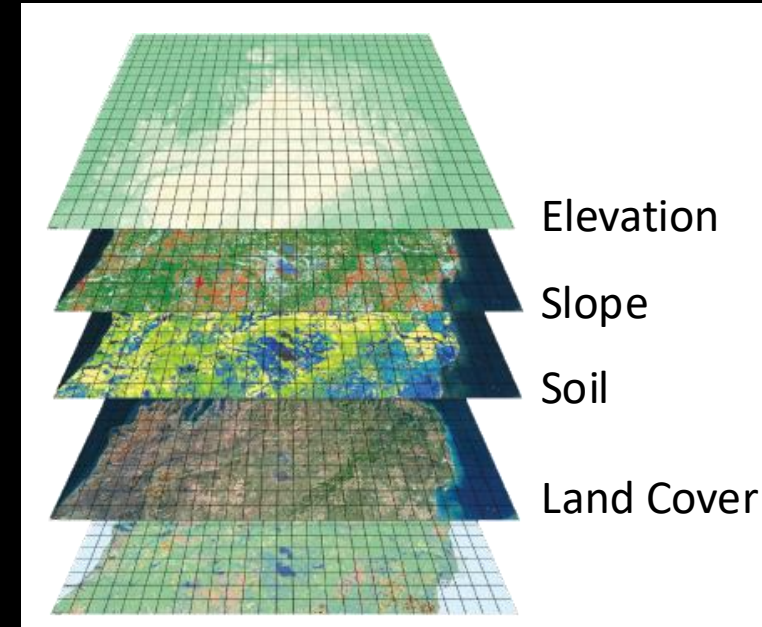
- In the raster data model,
 - the geographic space of interest is divided into **regular cells** of specific dimensions, and
 - the attribute or value of each cell is represented by a **digital alphanumeric code**.
- Within the geographic space,
the locations of raster cells are not explicitly recorded but are inferred from their positions as defined by **row and column** numbers in the data set.

If the ground coordinates of two or more cells are known, the locations of all the cells in the entire raster data set can be computed.
- This raster model of geospatial data representation allows non-spatial attribute values to be **spatially indexed by location**.



Special Features of Raster Data

- Logically, raster data are organized into **layers**, which are also called variously as **bands**, **themes**, or **overlays**.
- For applications in GIS, these raster layers must be **georeferenced** to a common map coordinate system.
- Depending on the application objectives, a raster geospatial database often contains multiple layers of data.



- Each of the layers depicts a specific characteristic of Earth's surface, such as topography, soil type, hydrology, vegetation cover, or land use. In each cell of a particular layer, there is **one and only one thematic attribute or value**.
- Raster layers are best used to represent **continuous** spatial phenomena such as elevation, temperature, and pollution concentration.
- Discrete spatial features such as roads, rivers and lakes, land parcels, as well as land-use and land-cover types may be represented.

File Formats of Raster Data

- In computer, raster data can be stored in several ways that are formally referred to as **file formats**.
- Different file formats have different characteristics, depending on factors such as
 - the sources of the data,
 - the methods of compression, and
 - the design specifications of different raster GIS software packages.
- Raster file formats can be classified into the following categories:
 - Generic Raster File Formats- ASCII file format and binary file format
 - Raster Data Interchange Format- TIFF and Geo TIFF
 - Raster Data Compression Formats- GIF, JPEG, JPG etc.
 - Remote Sensing Image Formats- BIL, BSQ, and BSP
 - Proprietary Raster Geospatial Data Formats- GRID format by ArcGIS

Classification of Raster Geoprocessing Analysis (Giordano et al. 1994)

Based on type of functions:

- Input functions
- Analysis functions
- Output functions

Based on type of operations:

- Logical operations
- Arithmetic operations
- Overlay operations
- Geometric Property operations
- Geometric Transformation operations
- Geometric Derivation operations

Based on extent of raster cells considered:

- Local operations
- Neighborhood operations
- Extended neighborhood operations
- Regional operations

Types of Raster Geoprocessing Function (Giordano et al. 1994)

Classified into three categories:

- **Input functions:**

To **prepare and structure raster data** for use in GIS, such as restructuring, compilation, and editing.

- **Analysis domain:**

To **derive spatial relationships** implicit in the source data, such as logical operations, arithmetic operations, overlay operations, and geometric operations.

- **Output functions:**

To present the **results of the analysis functions** in a suitable form for communication and spatial problem solving, such as maps, graphs, and statistical reports.

Classification of Raster GIS Analysis Operations

Six categories by Giordano et al. (1994)

1. Logical operations

Use logical operators (i.e., "AND," "OR," and "XOR") to create a new layer from two input layers.

2. Arithmetic operations

Use arithmetic operators (i.e., addition, subtraction, multiplication, division, and assignment) to create a new layer from two input layers or by transforming the cell values in an input layer.

3. Overlay operations

Merge attribute values from two or more layers, using logical operators or arithmetic operators.

Classification of Raster GIS Analysis Operations

4. Geometric property operations

Compute indices that describe the geometric properties pertaining to spatial features on a layer, such as shape, size, angle, and topological relationships.

5. Geometric transformation operations

Modify the spatial properties of features on a layer by applying linear transformations, such as scale change and translation, and nonlinear transformations, such as warping in image rectification.

6. Geometric derivation operations

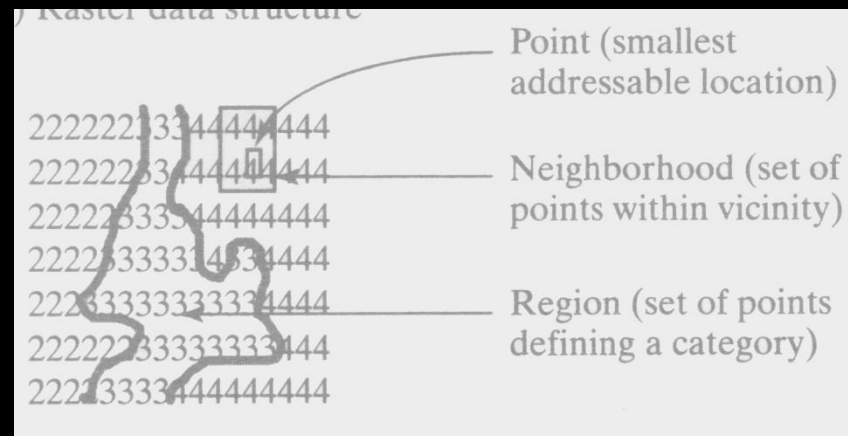
Create new features from existing features on a layer, using techniques such as triangulation, filtering, and surface interpolation

Classification of Raster Analysis based on the extent of the cells considered

Another common method is based on the ways of **using raster cells** in the operations.

- On a raster layer, the smallest addressable unit is a **point**, which is represented by a **cell**.
- Each point can be addressed as a part of a **neighborhood** of surrounding values.
- When all neighboring points having the same attribute value are grouped together, a **region** is identified.

In raster geoprocessing, some processes use the values of individual cells only, and others rely on neighborhood relationships or regional associations.



Classification of Raster Analysis based on the extent of the cells considered

- | | |
|-------------------------------------|---|
| 1. Local operations | Creating an output layer on which the value of each cell is a function of the cell at the same location on the input layer. |
| 2. Neighborhood operations | Creating an output layer on which the value of each cell is a function of the cells neighboring the cell at the same location on the input layer. |
| 3. Extended neighborhood operations | Creating an output layer on which the value of each cell is a function of the cells neighboring and beyond the immediate neighborhood of the cell at the same location on the input layer. |
| 4. Regional operations | Creating an output layer by identifying cells that intersect with or fall within each region on the input layer. |

Raster Geoprocessing Techniques

	Local Operations	Neighborhood Operations	Extended Neighborhood Operations	Regional Operations
Logical Operations	• Reclassification			
Arithmetic Operations	• Reclassification	• Aggregation • Filtering	• Statistical analysis	
Overlay Operations	• Logical • Arithmetic			• Category-wide overlay
Geometric Property Operations		• Slope and aspects	• Distance, proximity, and connectivity	• Area • Perimeter • Shape
Geometric Transformation Operations			• Rotation • Translation • Scaling	
Geometric Derivation Operations			• Buffering • Viewshed analysis	• Identification and reclassification

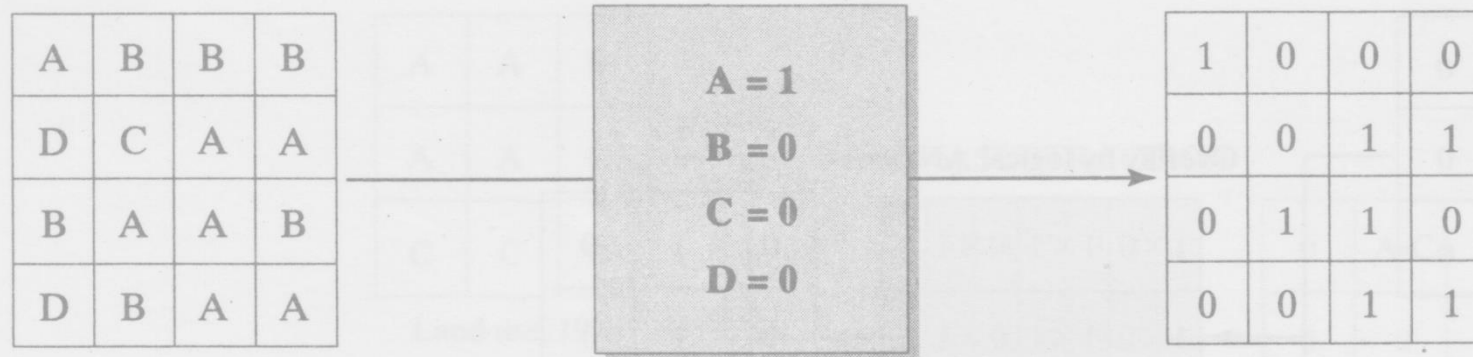
Local operations

Reclassification(recoding):

To create a new raster layer by **changing the attribute values of the cells** of the input layer.

Using one of the following forms through either logical or arithmetic operators:

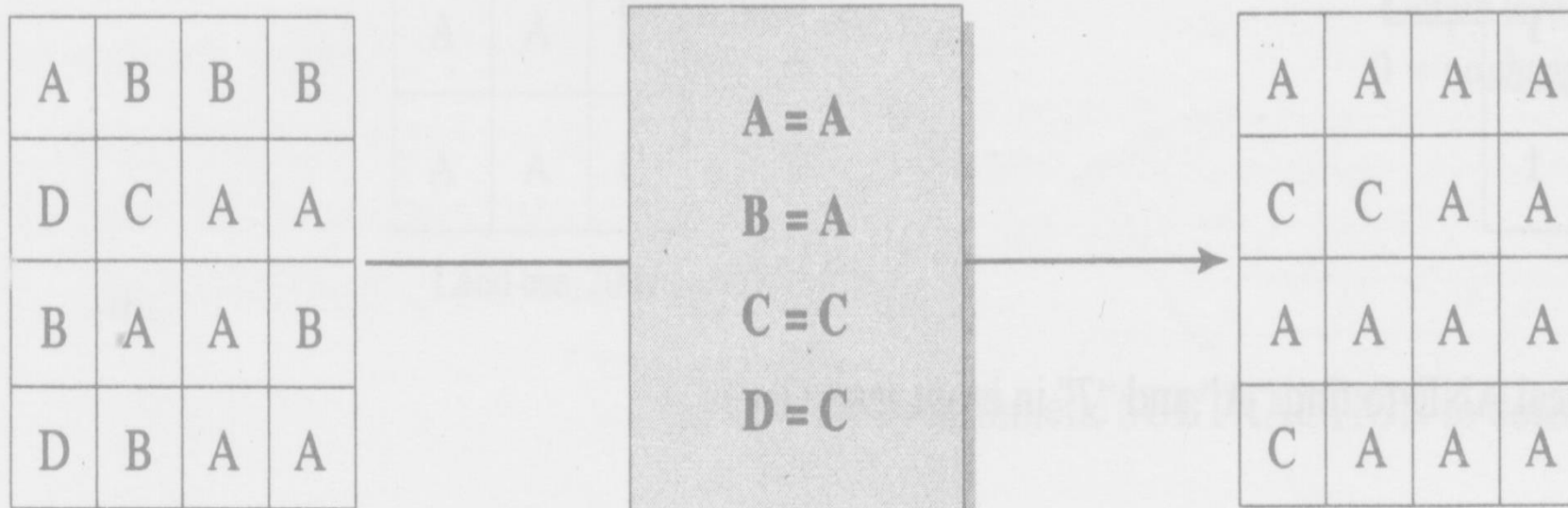
Assigning a new value to each value on the input map layer with the purpose of developing a binary (0 and 1) mask for use in subsequent GIS analysis, a process known as **binary masking**.



(a) Binary masking

Local operations

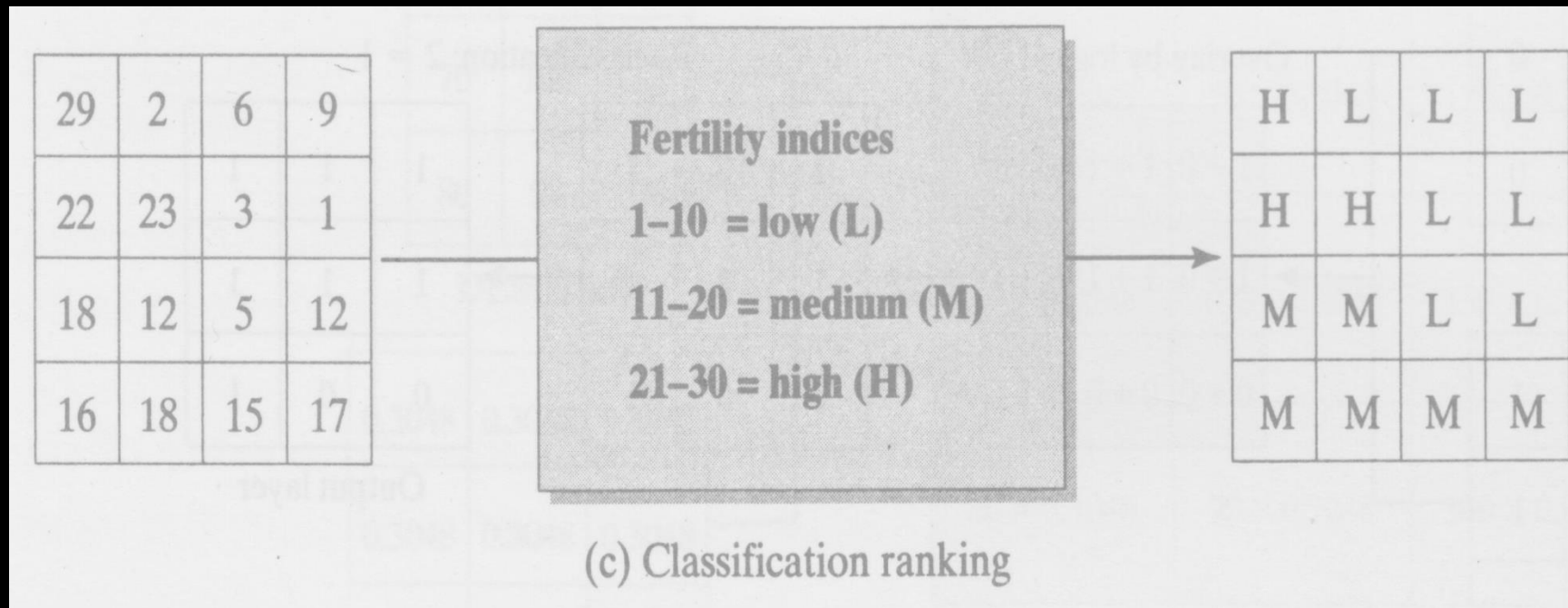
Assigning new values to classes or ranges of old values with the purpose of reducing the number of classes in the original input layer or to group values into categories in a new classification scheme.



(b) Classification reduction

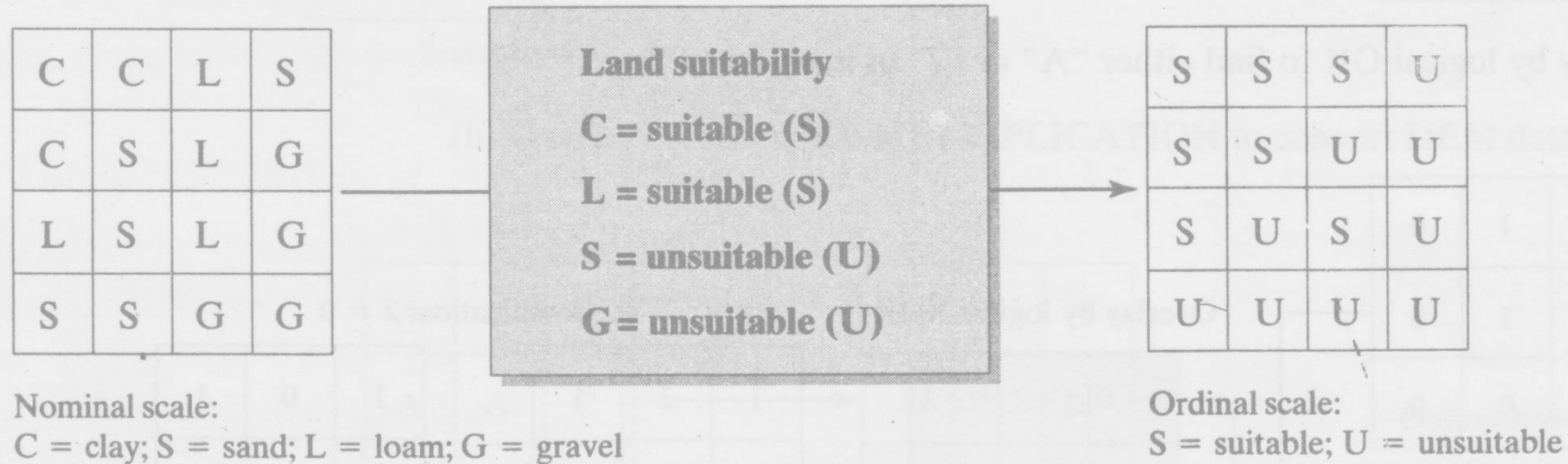
Local operations

Assigning ranks to unique values or categories of values found on the input layer.



Local operations

Assigning ranks or weighting to a qualitative (nominal scale) map layer to generate a quantitative map layer (ordinal, interval, or ratio scales).



(d) Changing measurement scales

Raster Geoprocessing Techniques

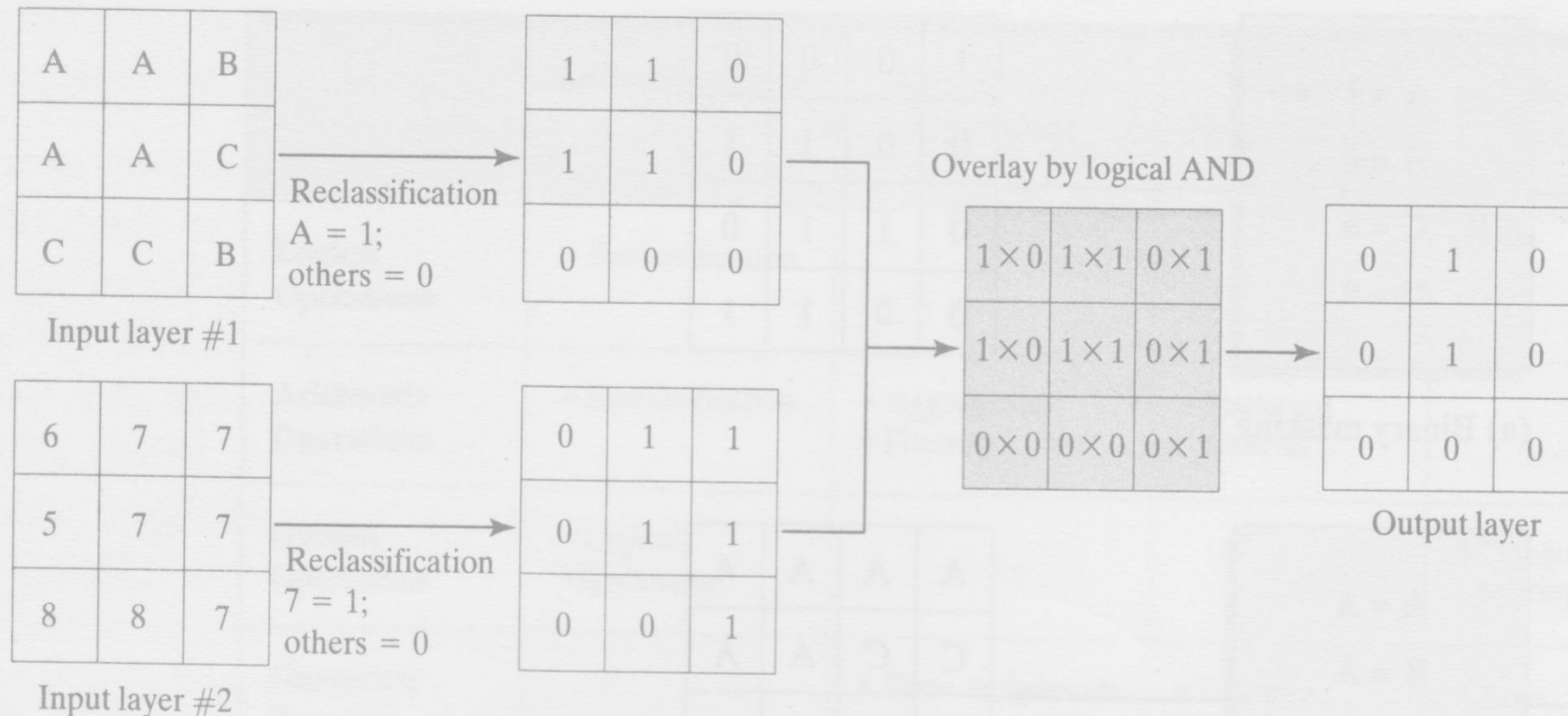
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Overlay analysis

- Overlay analysis uses either **logical or arithmetic** operators.
- Logical overlay methods use logical **AND**, **OR**, and **XOR** to generate a new layer from two input layers.
- Mathematically, the logical AND operation **multiplies** individual values of corresponding cells in the input layers, and the logical OR and XOR operations **add** individual values of corresponding cells in the input layers.
- The difference between logical OR and XOR is dependent on the way of producing the resulting binary layer.
- XOR is also the same way as the *logical* "either/or." The output is "true" if either, but not both, of the inputs are "true."

Examples of logical overlaying operations

a) Overlay by logical AND to find 'A' and "7" in input raster layer

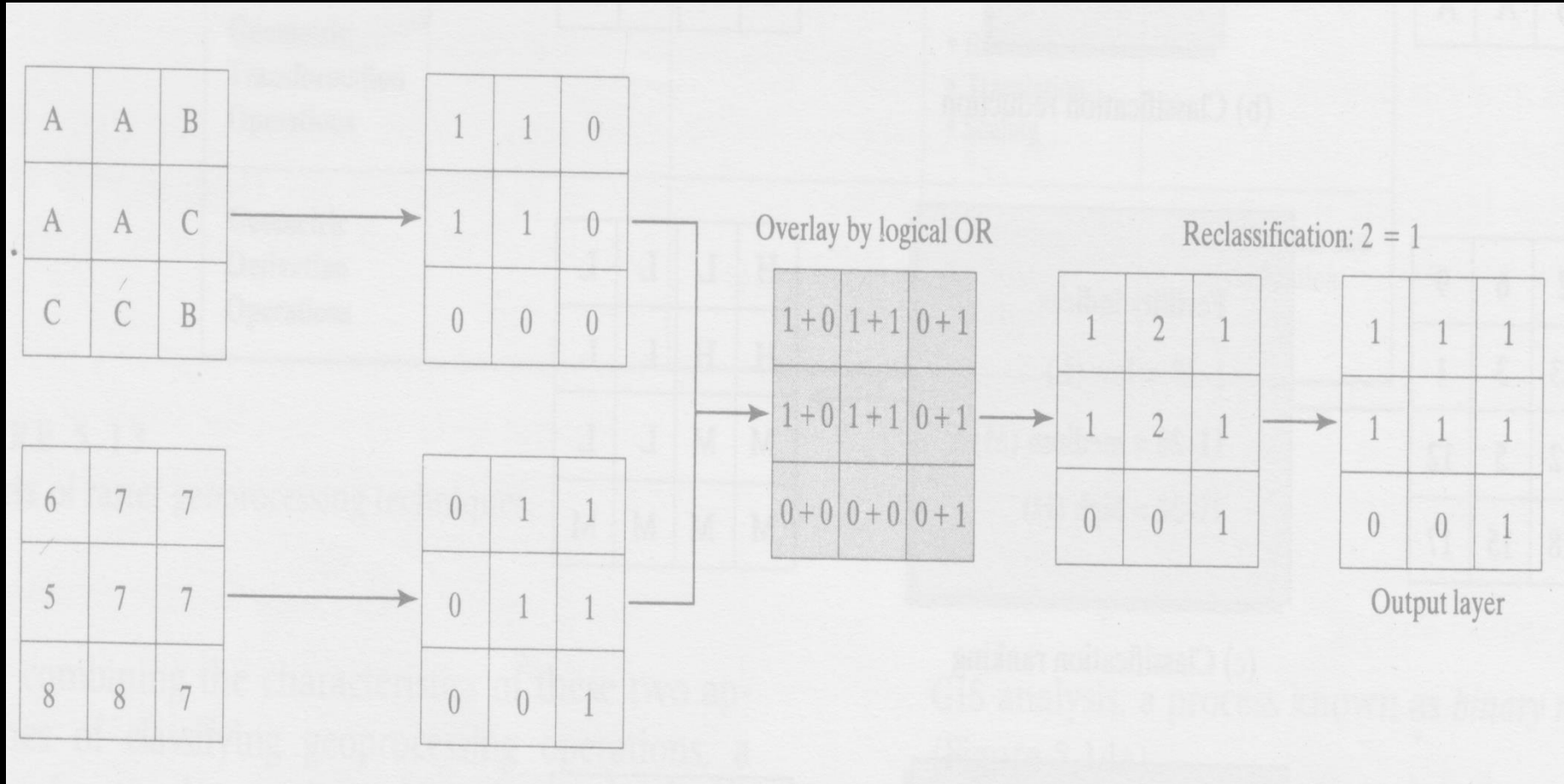


(a) Overlay by logical AND to find "A" and "7" in input raster layer

It is thought of as "only both".

Examples of logical overlaying operations

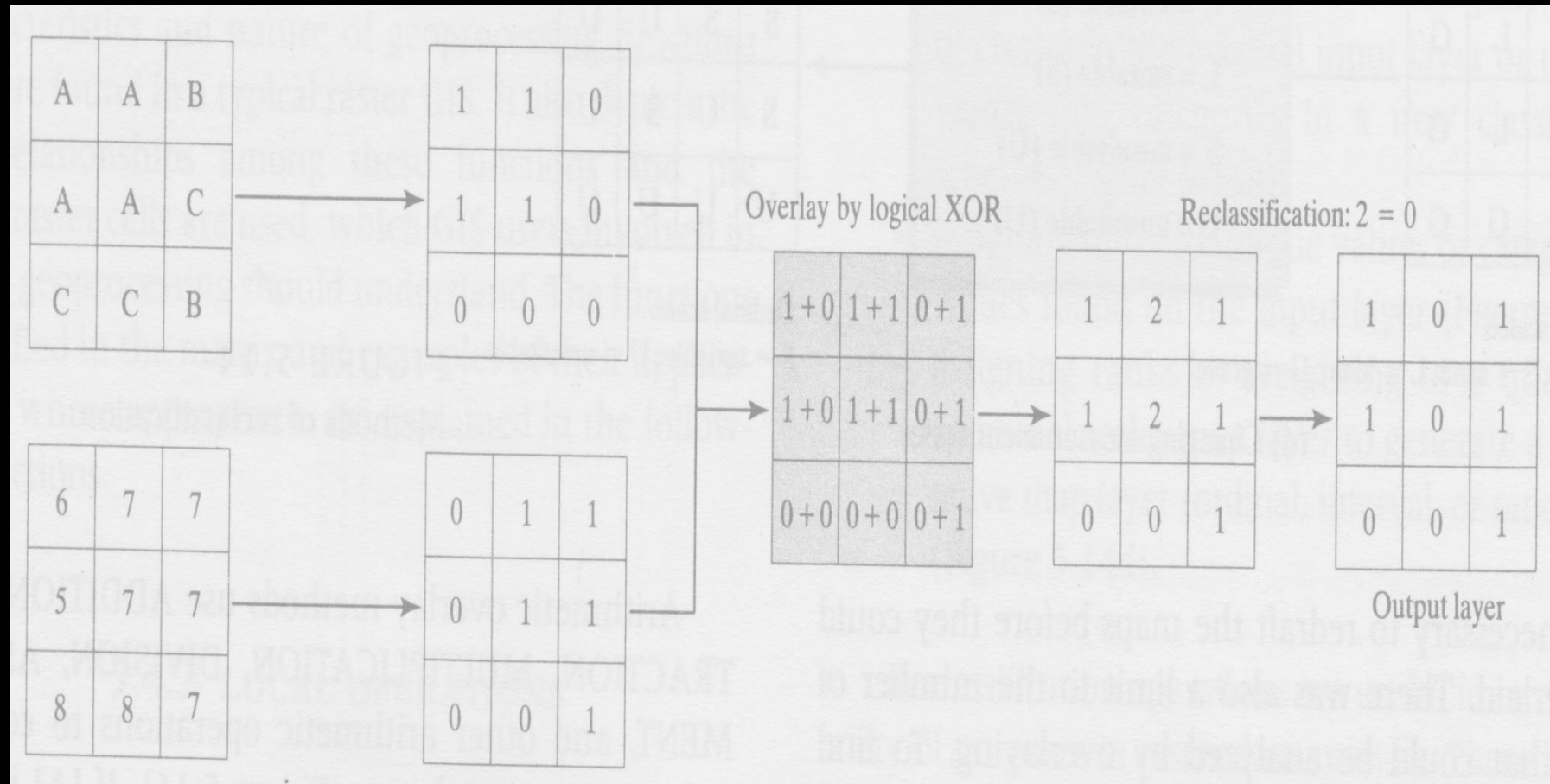
b) Overlay by logical OR to find 'A' and "7" in input raster layer



Logical OR reclassifies output cell values that are equal to 2 from 2 to 1.
It is thought of as **'either or even both'**.

Examples of logical overlaying operations

c) Overlay by logical XOR to find 'A' and "7" in input raster layer



Logical XOR reclassifies these cell values from 2 to 0 (Figure 5). It is thought of as **'one or the other but not both'**.

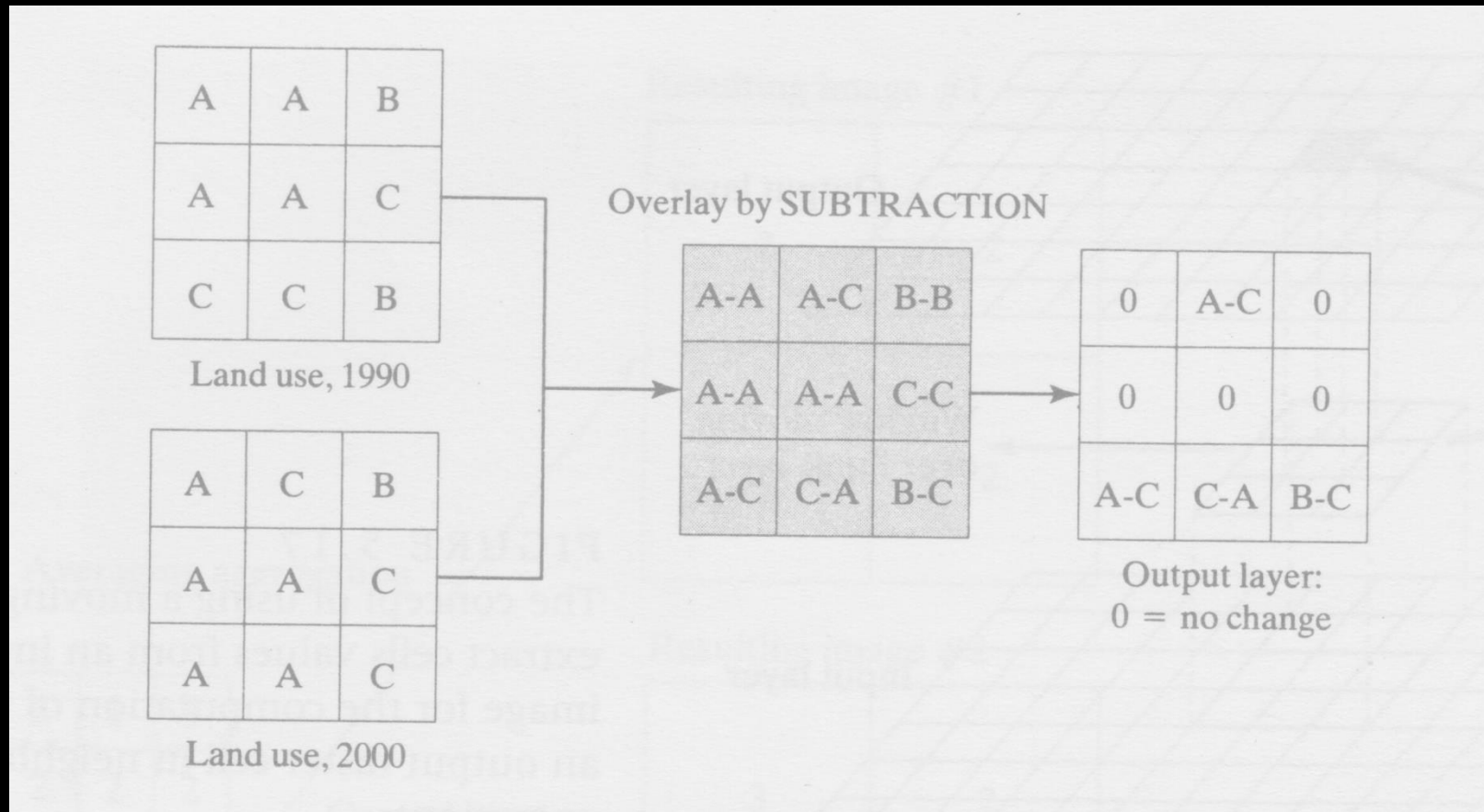
Overlay analysis

- Arithmetic overlay methods use ADDITION, SUBTRACTION, MULTIPLICATION,, DIVISION, ASSIGNMENT, and other arithmetic operations to combine two or more map layers.
- If [A] is input layer 1, [B] is input layer 2, and [O] is the output layer created by overlaying, the following arithmetic operations are possible.

```
[Oa] = [A] + [B]
[Ob] = ([A] + [B])/2
[Oc] = [A] - [B]
[Od] = [A] * [B]
[Oe] = [A]/[B]
[Of] = [A] if [A] > [B]
[Og] = [A] if [A] < [B]
[Oh] = [B] if [B] > 0; otherwise
[Oh] = [A]
```

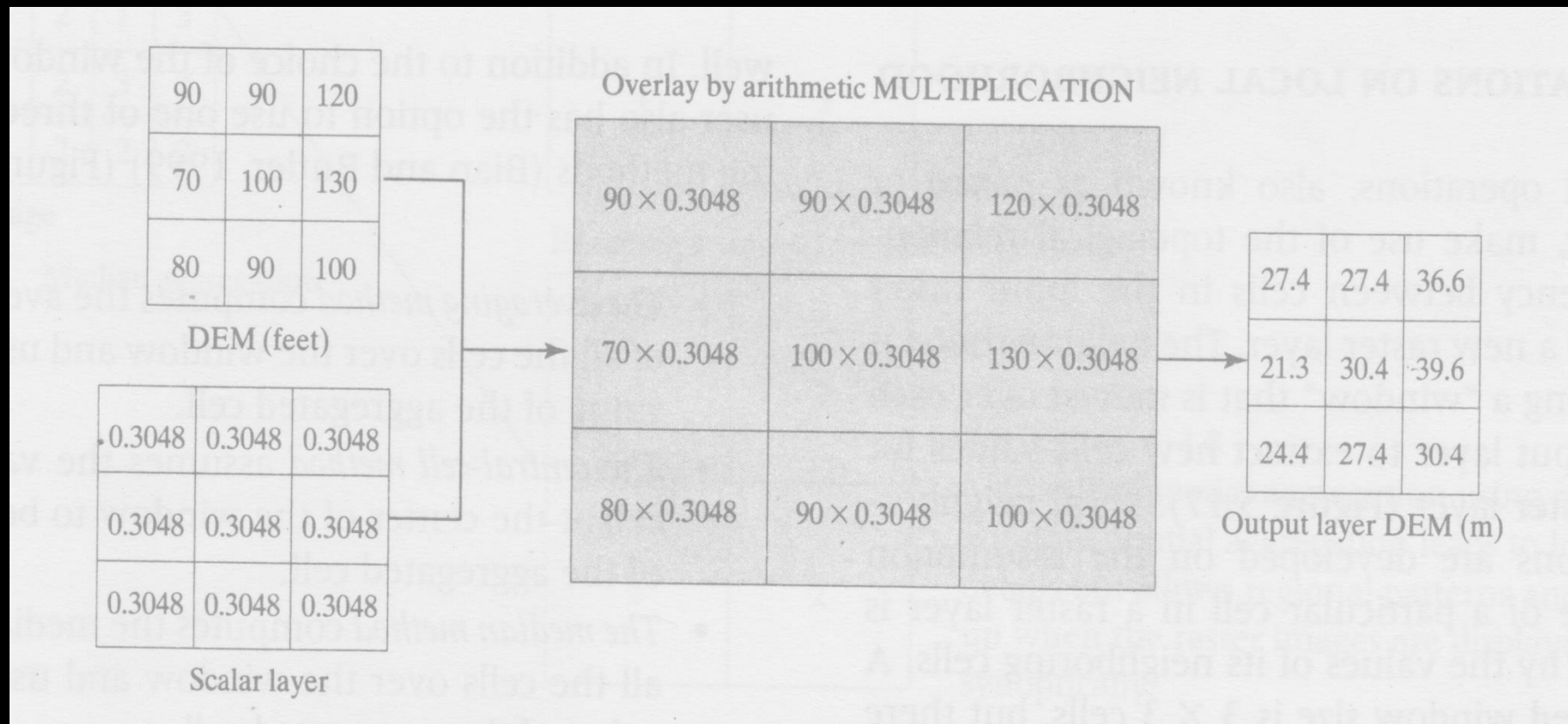

Examples of arithmetic overlaying operations

a) Overlay by arithmetic SUBTRACTION to detect land-use change



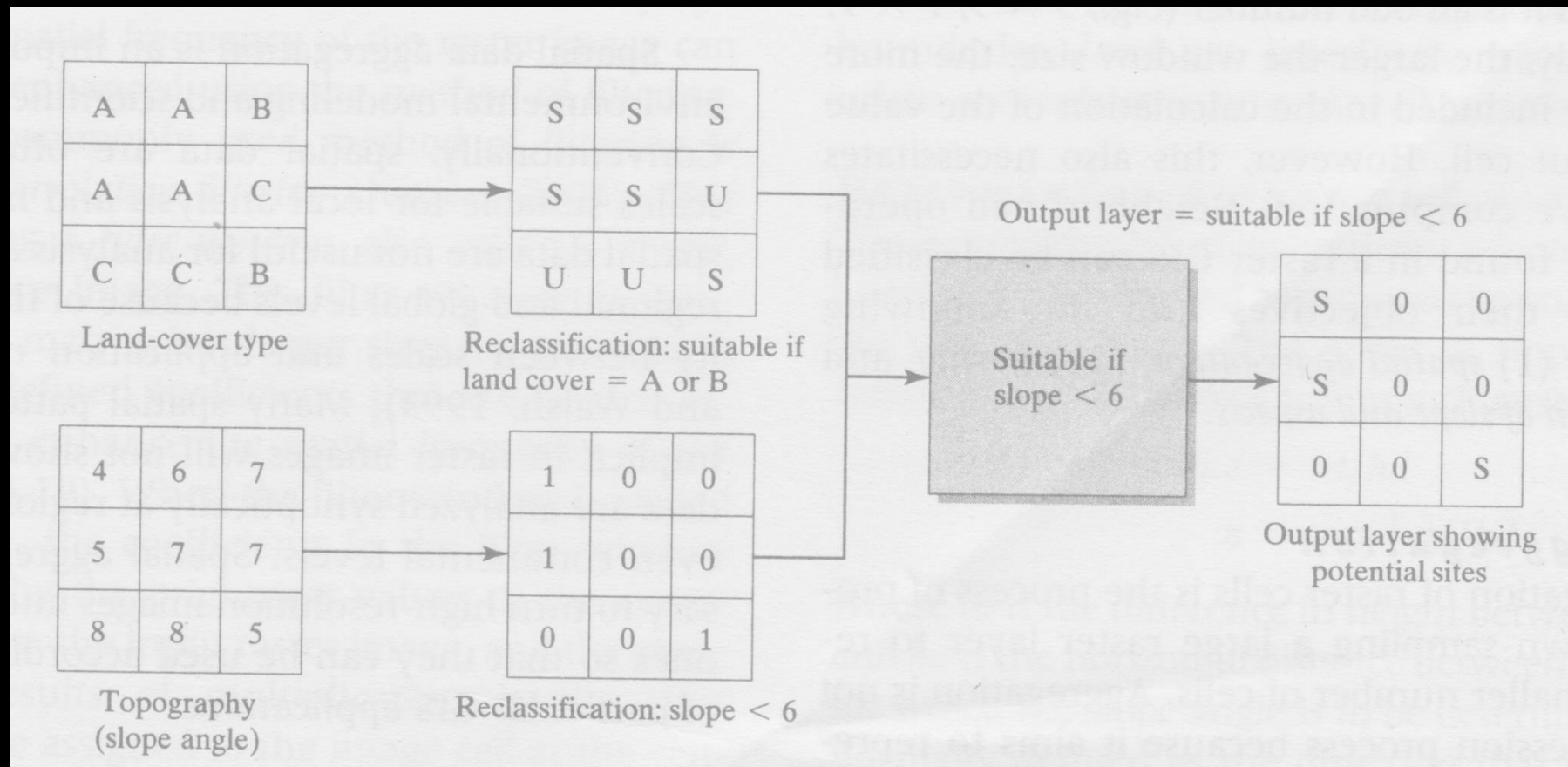
Examples of arithmetic overlaying operations

b) Overlay by arithmetic MULTIPLICATION to convert DEM data from feet to meters



Examples of arithmetic overlaying operations

c) Overlay by arithmetic ASSIGNMENT to select suitable sites

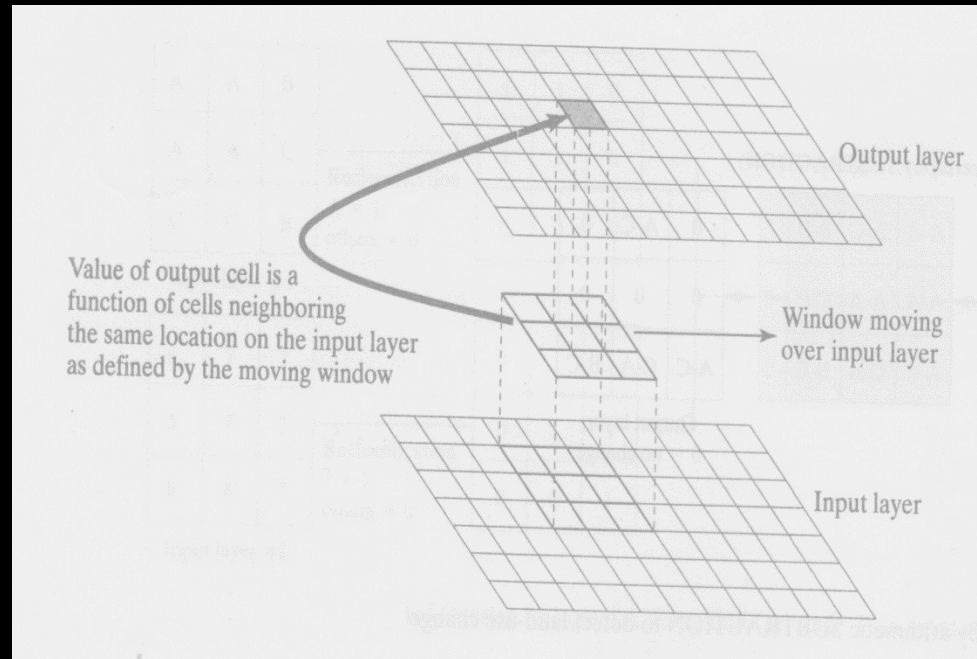


Raster Geoprocessing Techniques

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Operations on Local Neighbourhood

- Neighborhood operations, also known as **context** or **focal** operations, make use of the **topological relationship of adjacency** between cells in the input raster layer to create a new raster layer.
- The neighborhood is defined by using a "**window**" that is moved over each cell on the input layer to extract new cells values for the output raster layer.
- Local neighborhood operations are developed on the assumption that the value of a particular cell in a raster layer is often affected by the values of its neighboring cells.



Operations on Local Neighbourhood

- A commonly used window size is 3 X 3 cells, but there is no theoretical limit to the window size that can be used, provided it is an **odd number** (e.g., 5 X 5, 7 X 7, etc.).
- Obviously, the **larger the window size**, the more input cells are included in the calculation of the value of each output cell. However, this also necessitates **more intensive** computation.
- Neighborhood operations that are found in a raster GIS can be classified according to their objectives into the following three groups:
 - (1) spatial aggregation,
 - (2) filtering, and
 - (3) computation of slope and aspects.

Operations on Local Neighbourhood

1. Spatial aggregation:

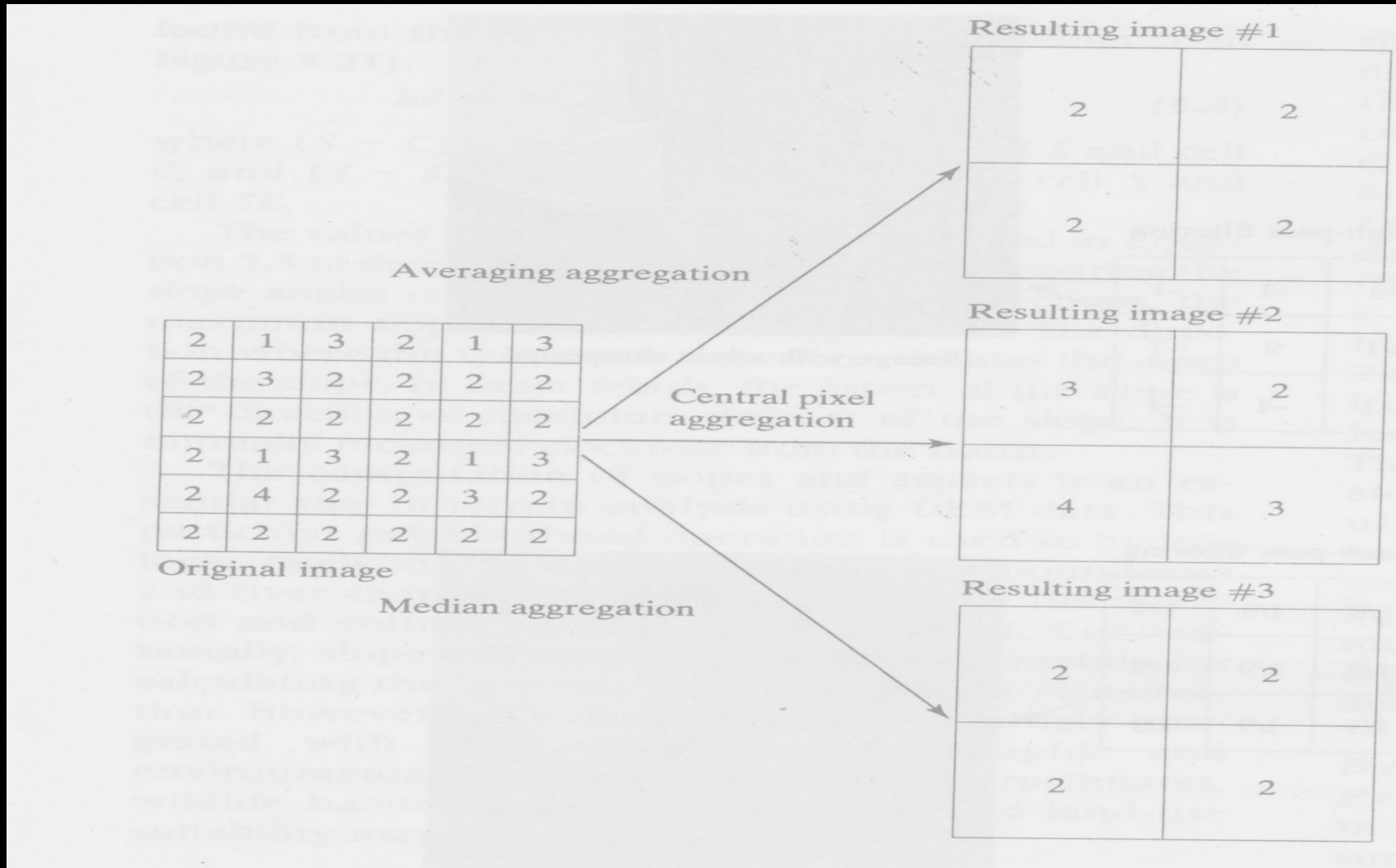
- It is the process of **progressively down-sampling a large raster layer** to reduce it to a smaller number of cells.
- Aggregation → not a data compression process.

It aims to represent the same geographic space using a smaller number of cells by means of a coarser spatial resolution, rather than by compressing the file size mathematically.

- It is basically a **cartographic generalization** process. As a neighborhood operation, spatial aggregation can be carried out using **a window of any size** to compute the values of the cells on the output layer.
- A larger window size will result in a higher aggregation level. A high aggregation level means a smaller aggregated layer, but it implies a greater loss of detail as well.
- In addition to the choice of the window size, one can use three aggregating methods:
 - Averaging method: computes the **average** value of all cells over the window and uses it as the value of the aggregated cell.
 - Central-cell method: assumes the **value of the cell at the center** of the window to be the value of the aggregated cell.
 - Median method: computes the **median** value of all the cells over the window and uses it as the value of the aggregated cell.

Operations on Local Neighbourhood

Methods of spatial aggregation using a 3x3 window



Operations on Local Neighbourhood

1. Spatial aggregation:

- Spatial data aggregation is an important process in environmental modeling and scientific research.
- Conventionally, spatial data are often collected at scales suitable for local analysis and modeling.
- These spatial data are not useful for analysis and modeling at regional and global levels because of the **incompatibility between scales and application objectives**.
- Many spatial patterns and trends implicit in raster images will not show up unless the data are analyzed synoptically at regional, national, or even continental levels.
- Spatial aggregation is necessary to **turn high-resolution images into low-resolution** ones so that they can be used according to the needs of particular GIS applications.

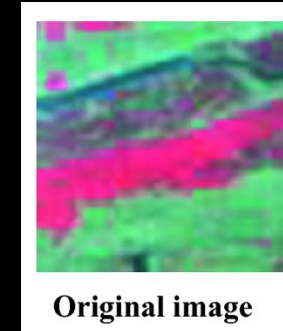
Operations on Local Neighbourhood

2. Filtering:

- Filtering is a digital image processing function for image enhancement.
- It is useful for improving image quality through the removal of noise and other defects in the original data.
- In remote sensing images,
 - If brightness value changes gradually across an area:
→ low-frequency area
 - If brightness value changes rapidly across an area:
→ high-frequency area
- It involves passing a *filter window* across the entire image. The window contains coefficients to either:
 - subdue (*Low pass-filter*), or
 - enhance (*High-pass filter*) the spatial frequency of the image.
- Low-pass → To reduce minor variations and emphasize homogeneous areas.
- High-pass → To sharpen linear features (e.g., drainage networks, roads, land-use interfaces etc.).

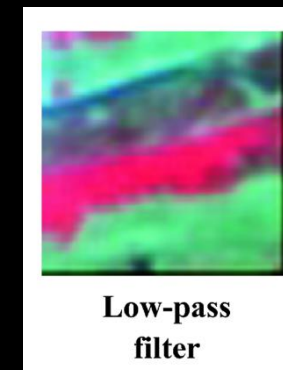
0	0	0
0	0	0
0	0	0

Original image



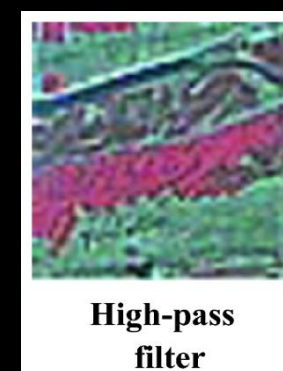
1	1	1
1	1	1
1	1	1

Low-pass filter



-1	-1	-1
-1	+9	-1
-1	-1	-1

High-pass filter



Operations on Local Neighbourhood

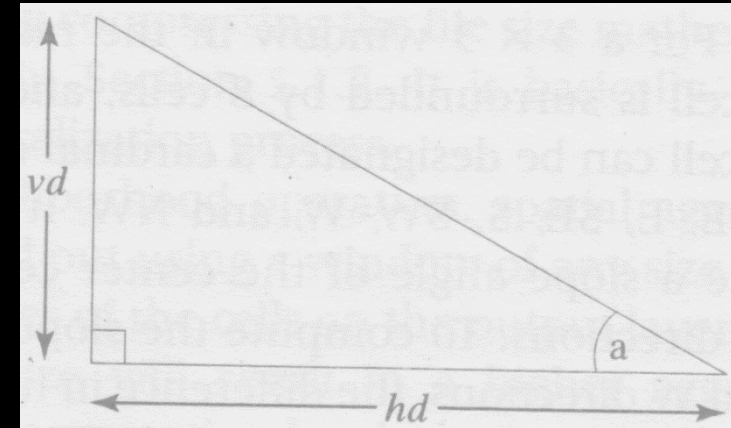
3. Determining Slopes and Aspects:

These neighborhood operations are used for DEMs.

SLOPE:

- The angle of slope (a) is determined by the following formula.

$$\tan a = vd/hd$$
$$a = \tan^{-1} (vd/hd)$$



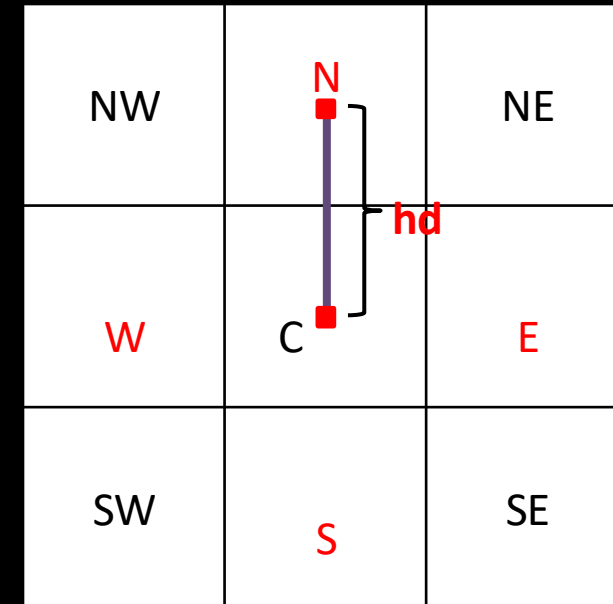
where

- vd is the difference in height between two points.
- hd is the horizontal distance between the two points considered.

Operations on Local Neighbourhood

ASPECT:

- Aspect is normally defined as the **direction** that the **maximum slope** faces.
- For a 3 X 3 window in the raster layer, the center cell is surrounded by 8 cells, and each neighboring cell can be designated a cardinal direction such as N, NE, E, SE, S, SW, W, and NW.
- It is possible to compute a slope angle of the center cell for each of these 8 directions.
- *To compute the slope angles for N, E, S, and W directions,*
 - **Vertical distance (vd)** →
difference in height between the center cell and the cells in these four directions.
 - **Horizontal distance (hd)** →
cell size or resolution of the raster (assuming that square cells are used).



Operations on Local Neighbourhood

- To compute the slope angles for NE, SE, SW, and NW directions,

- **Vertical distance** (vd) →

difference in height between the center cell and the cells in these four directions.

- **Horizontal distance** (hd) →

using Pythagorean Formula

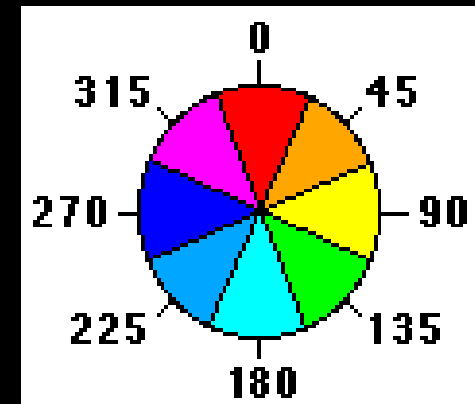
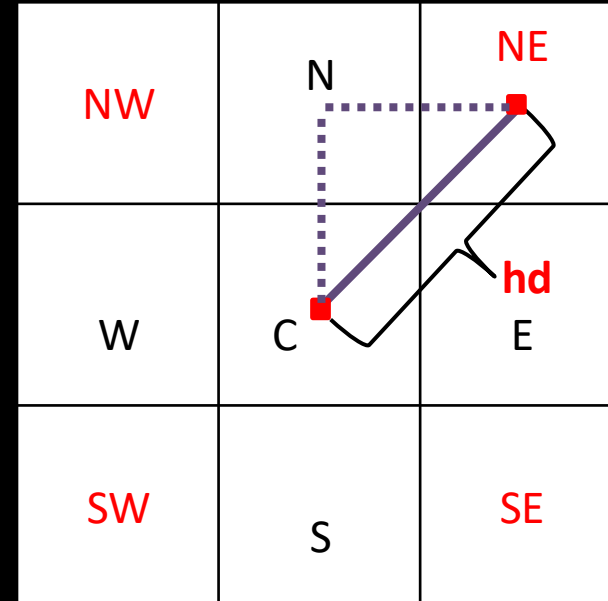
$$\text{hd} = \sqrt{(\text{N-to-C})^2 + (\text{N-to-NE})^2}$$

where

(N-to-C) is the distance b/w cells N and C,

(N-to-NE) is the distance between cells N and NE.

- By comparing the slope angles computed for all the eight directions, the maximum slope angle can be determined.
- The **direction where the maximum slope faces** defines the aspect of the slope. In other words, the aspect of the slope is the direction of maximum descent of the slope.
- It is normally measured **clockwise from the north**.



Operations on Local Neighbourhood

3. Determining Slopes and Aspects:

- The computation of slopes and aspects is an essential step in **terrain analysis** using DEM data.
- This particular neighborhood operation is used in **hydrologic modeling** to delineate watershed boundaries and river channels, as well as to compute the **direction and volume of flow**.
- Conventionally, slope and aspect data were used mainly for **calculating the intensity of light in terrain visualization**.
- However, such data are now increasingly integrated with other sources of topographic and environmental data for studies in microclimates, wildlife habitat analysis, biodiversity, and land-use suitability mapping.

Raster Geoprocessing Techniques

	Local Operations	Neighborhood Operations	Extended Neighborhood Operations	Regional Operations
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Operations on Extended Neighbourhoods

- It process a set of points that extend over a much larger area than the local neighborhoods and hence involve a greater number of cells.
- A new raster layer is produced as a result of an extended neighbourhood operation.
- There are five important groups of operations :
 - (1) statistical analysis of the raster;
 - (2) determination of distance, proximity, and connectivity;
 - (3) geometric transformation of raster layers;
 - (4) buffering, and
 - (5) viewshed analysis
- Since the spatial extent of an extended neighborhood can include the entire raster layer, these operations are sometimes referred to as global operations.

Operations on Extended Neighbourhoods

Proximity Analysis:

Proximity analysis is another of the most basic GIS operations, to answer questions such as:

- How close is this well to a landfill?
- Do any roads pass within 1000 meters of a stream?
- What is the distance between two locations?
- What is the nearest or farthest feature from something?
- What is the distance between each feature in a layer and the features in another layer?
- What is the shortest street network route from some location to another?

Operations on Extended Neighbourhoods

Proximity Tools:

Buffering:

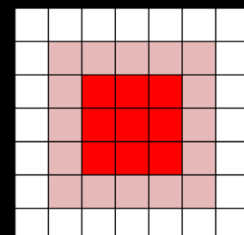
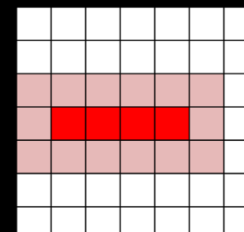
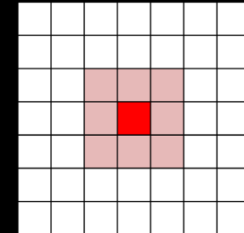
- The process of creating buffers, buffering, is a very important function in GIS applications, useful in many occasions:

For example: → to provide greenways or easements around lakes and water courses

→ to show areas along highways where traffic noise is above a certain level

→ to delineate a safety zone around hazardous facilities.

- In geoprocessing, buffer → area around a particular spatial feature (point / line / area).
- In raster geoprocessing, buffer → cells adjacent to or affected by selected cells having a particular value.
- In the raster data model, the concept of spatial feature does not apply because **individual features are represented not as independently identifiable entities** but as a collection of contiguous raster cells having the same attribute values.
- Hence, a buffer is defined as the **raster cells that are at a specific distance** from a particular cell, a linear array of cells, or a cluster of cells.

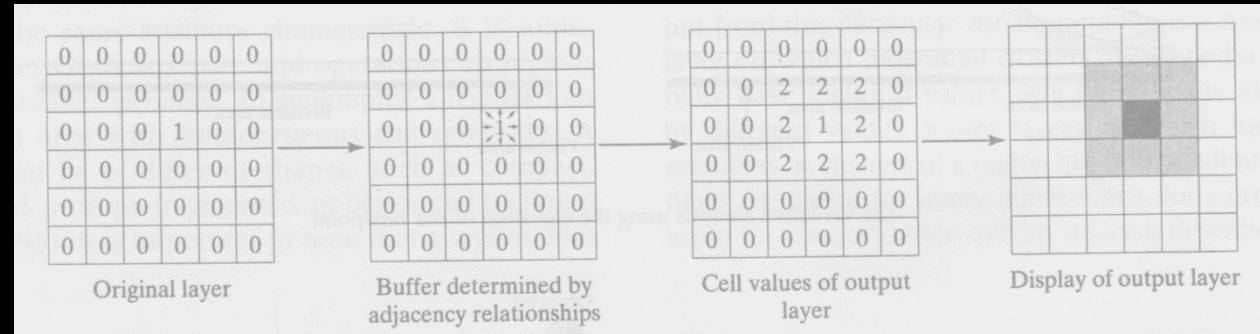


Operations on Extended Neighbourhoods

Buffering:

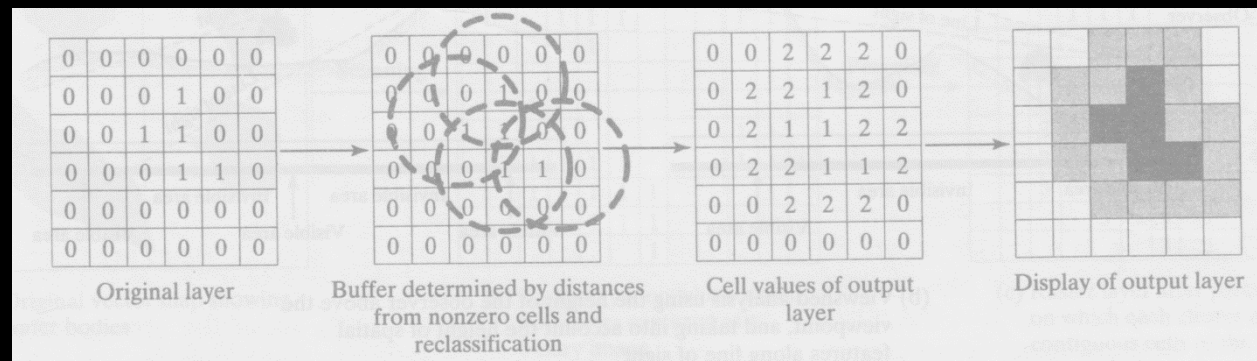
- In raster geoprocessing, buffers can be created in two ways:

First approach: Uses the 'adjacency' topological relationship to spread the value of identified cells to neighboring cells up to a specific buffer width from these cells.



Second approach:

- First the distance operator is used to compute the distances of all the cells in a raster layer from those cells to be buffered.
- Then the cell distances are reclassified into two categories:
 - cells $\leq$ buffer width $\rightarrow 1$, and
 - cells $>$ buffer width $\rightarrow 0$.

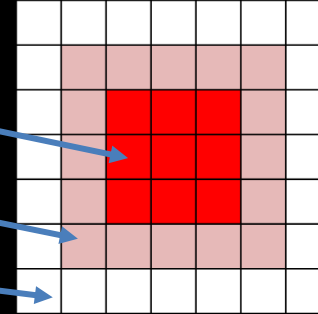


Operations on Extended Neighbourhoods

Buffering:

- The results of these extended neighbourhood operations are new raster layers containing three classes of cells:

- cells representing the original cells,
- cells forming the buffer, and
- cells outside the buffer.



Operations on Extended Neighbourhoods

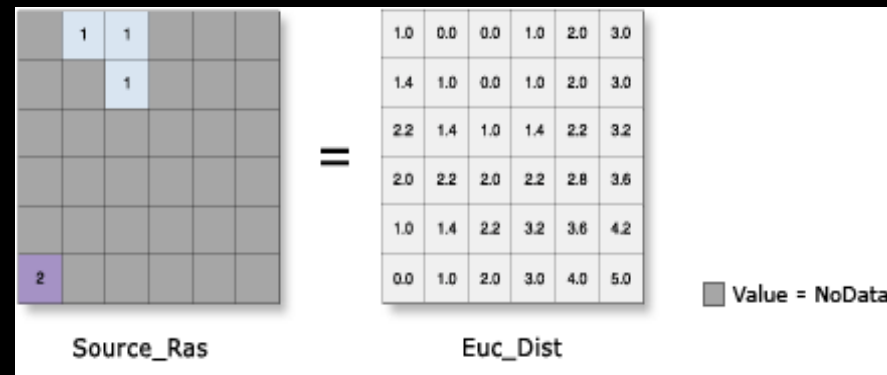
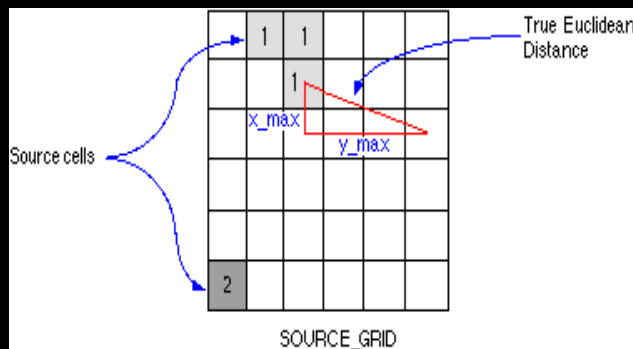
Distance tools:

- To create rasters showing the **shortest distance of each cell from a set of features**, or that allocate each cell to the closest feature.
- They can also calculate the **shortest path** across a surface, or the corridor between two locations that minimizes two sets of costs.
- Distance surfaces are often used as inputs for overlay analyses;

For example: In habitat suitability model, **distance from streams** could be an important factor for water-loving species, or **distance from roads** could be a factor for timid species.

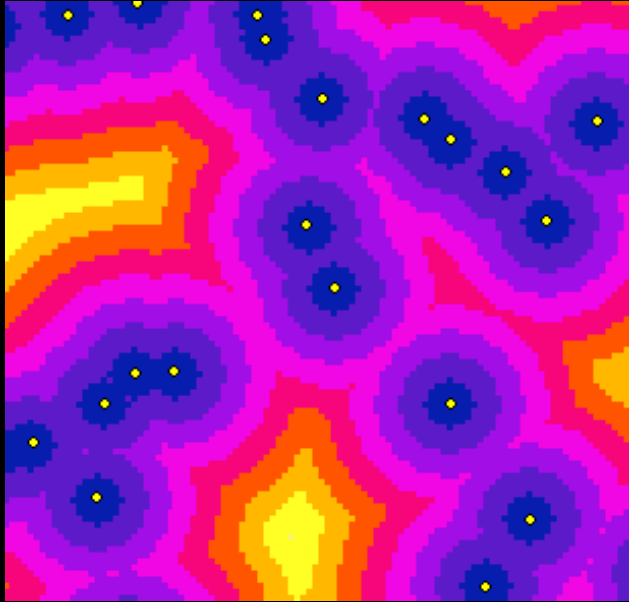
Euclidean distance:

- It is the straight-line distance, or distance measured "**as the crow flies**". For a given set of input features, the minimum distance to a feature is calculated for every cell.

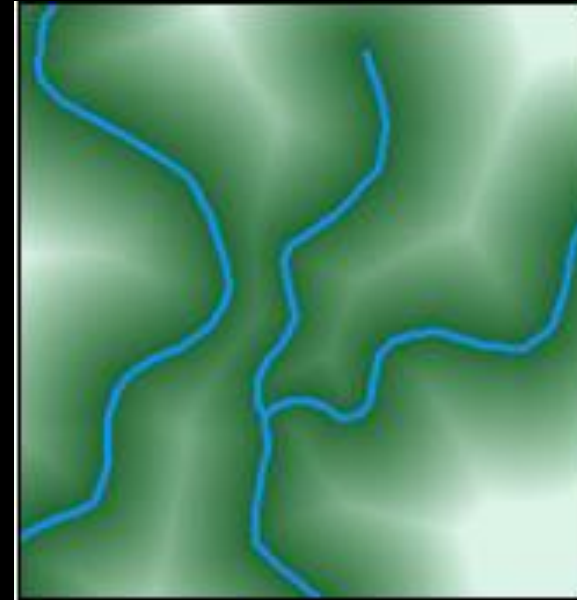


Operations on Extended Neighbourhoods

Illustrations of the Euclidean Distance Output Raster:



● Source points



Distance to the nearest line feature

Euclidean Distance can be used as part an forest fire model, where the probability of a given cell igniting is a function of distance from a currently burning cell.

Raster Geoprocessing Techniques

	Local Operations	Neighborhood Operations	Extended Neighborhood Operations	Regional Operations
Logical Operations	• Reclassification			
Arithmetic Operations	• Reclassification	• Aggregation • Filtering	• Statistical analysis	
Overlay Operations	• Logical • Arithmetic			• Category-wide overlay
Geometric Property Operations		• Slope and aspects	• Distance, proximity, and connectivity	• Area • Perimeter • Shape
Geometric Transformation Operations			• Rotation • Translation • Scaling	
Geometric Derivation Operations			• Buffering • Viewshed analysis	• Identification and reclassification

Operations on Regions

- On a raster layer, a region is a collection of cells that exhibit the same attribute characteristic.
- It is sometimes referred to as a zone, and operations on regions are called zonal operations.
- In geography, a region is an extended area with homogeneous characteristics.
- A region can be of different shapes, such as compact, elongated, prompt, fragmented, or perforated.
- It is important to note that a region does not have to be connected on a raster layer.

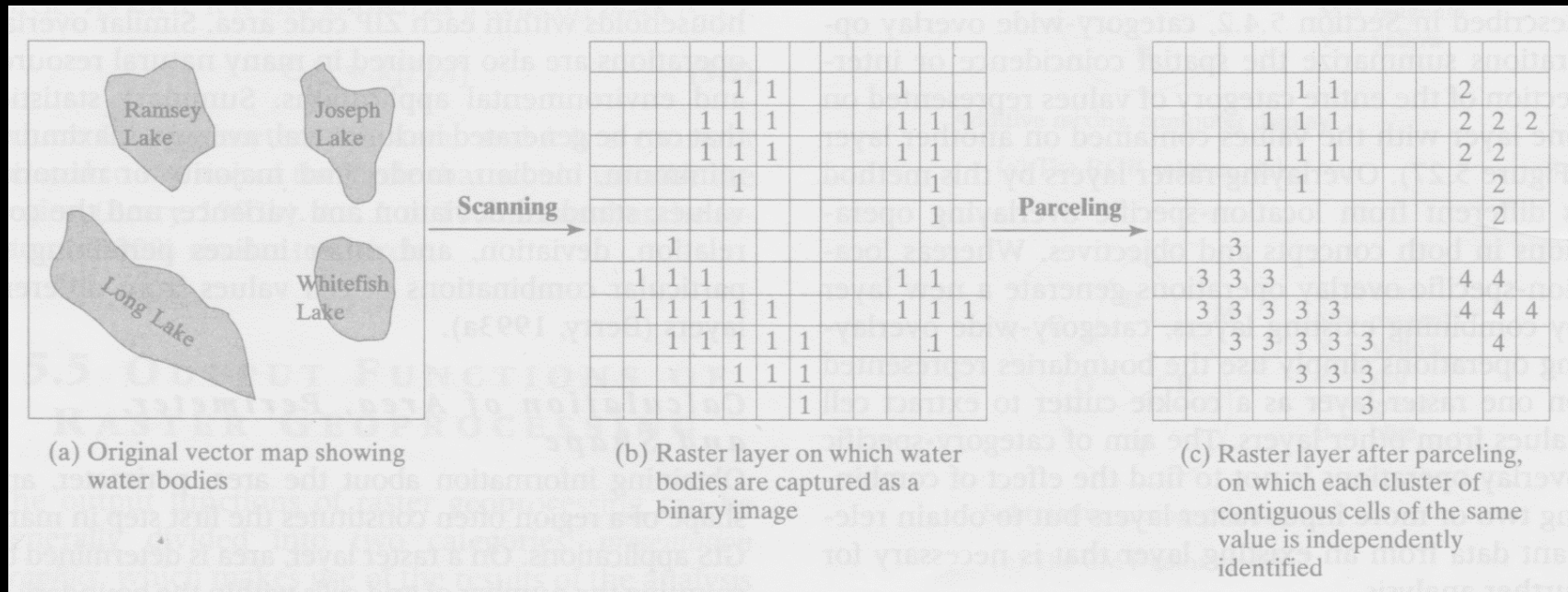
For example: all the wheat-growing areas in the India, which are distributed all over the country as disparate individual clumps, can be defined as one single region since they are all characterized by the same attribute value.

- In raster geoprocessing, there are three major categories of zonal operations:
 - (1) identification of regions and reclassification,
 - (2) category-wide overlay, and
 - (3) calculation of area, perimeter, and shape.

Operations on Regions

Identification of Region and Reclassification:

- On a raster layer, a region is identifiable **visually** by differentiating its cell values, as represented by a unique color than other values.
- In the simplest case, a region is made up of **only one single cluster or clump of cells**, but, as noted, it is common for a region to include many individual scattered clusters.
- In many applications, it is necessary to identify each cluster of a region individually. The zonal operation that carries out this function in raster geoprocessing is known as **parceling**.
- This operation identifies clusters of spatially contiguous cells that have the same values and reclassifies them by giving each identified cluster a unique value.



Operations on Regions

Identification of Region and Reclassification:

- By applying this operation, for example, all wheat-growing areas in India can be individually identified from a generalized raster layer representing **all wheat-growing areas as one single category**.
- The output from this particular zonal operation is a new raster layer on which individual clusters of cells, which originally have **identical values**, will be uniquely identified by **different values**.
- Raster layers on which individual members or clusters of a region are **independently identified** are useful for many applications, for example, as input for category-wide overlay analysis.

Weighted Overlay

- Choosing the a new house for relocation -
 - Proximity to schools,
 - Property price per sq. ft.,
 - Proximity to main roads, etc.

Weighted Overlay

- It is a technique for applying a common measurement scale of values to diverse and dissimilar inputs to create an integrated analysis.
- Geographic problems often require the analysis of many different factors.
E.g.: Choosing the a new house for relocation involves assessing multiple criteria-
Proximity to schools, Property price per sq. ft., Proximity to main roads, etc.

*>> Hence, **SELECTION OF CRITERIA** → First Stage of Weighted Overlay Analysis*

- This information exists in different raster layers with different value scales.
E.g.: Proximity to schools (in minutes),
Property price per sq. ft. (in Rs.),
Proximity to main roads (in m).
- But, we can't simply add a raster of proximity (time/distance) to a raster of insurance cost (Rs.) to obtain a meaningful result.

*>> Hence, **STANDARDIZATION OF CRITERION SCORES** → Second Stage Of Weighted Overlay Analysis*

Weighted Overlay

Stage 1: Selection of Criteria

Proximity to schools
(minutes)

20	15
12	10

Property price/ sq. ft
(Rs..)

100	1000
500	750

Proximity to main roads
(m)

1500	1000
1000	2000

Stage 2: Standardization of Criterion Scores

Proximity to schools
(minutes)

20 10

0 1

0	0.5
0.8	1

Property price/ sq. ft
(Rs..)

1000 100

0 1

1	0
0.5	0.25

Proximity to main roads
(m)

2000 1000

0 1

0.5	1
1	0

Weighted Overlay

- Additionally, the factors in our analysis may not be equally important.
E.g.: Proximity to schools >> Property price >> Proximity to main roads

The importance (weight) of each criterion is for you to decide.

>> Hence, **ALLOCATION OF WEIGHTS** → *Third Stage* of Weighted Overlay Analysis

From Stage 2- Standardized Criterion Scores

Proximity to schools
(minutes)

0	0.5
0.8	1

Property price/ sq. ft
(Rs..)

1	0
0.5	0.25

Proximity to main roads
(m)

0.5	1
1	0

Stage 3: Allocation of Weights

Proximity to schools
(minutes) x 50%

0	0.25
0.4	0.5

Property price/ sq. ft
(Rs..) x 30%

0.3	0
0.15	0.075

Proximity to main roads
(m) x 20%

0.1	0.2
0.2	0

Weighted Overlay

- Finally, we add (overlay) the rasters to compute the final suitability score.
>> Hence, **COMBINING THE RASTERS** → Fourth Stage of Weighted Overlay Analysis

From Stage 3- Allocation of Weights

Proximity to schools
(minutes)

0	0.25
0.4	0.5

Property price/ sq. ft
(Rs..)

0.3	0
0.15	0.075

Proximity to main roads
(m)

0.1	0.2
0.2	0

+

Stage 4: Combining of Rasters

0.4	0.45
0.75	0.575

Stage 1: Selection of Criteria

Proximity to schools
(minutes)

20	15
12	10

Property price/ sq. ft
(Rs..)

100	1000
500	750

Proximity to main roads
(m)

1500	1000
1000	2000

Stage 2: Standardization of Criterion Scores

0	0.5
0.8	1

1	0
0.5	0.25

0.5	1
1	0

Stage 3: Allocation of Weights

0	0.25
0.4	0.5

0.3	0
0.15	0.075

0.1	0.2
0.2	0

Stage 4: Combining of Rasters

0.4	0.45
0.75	0.575

Weighted Overlay- Key points to remember

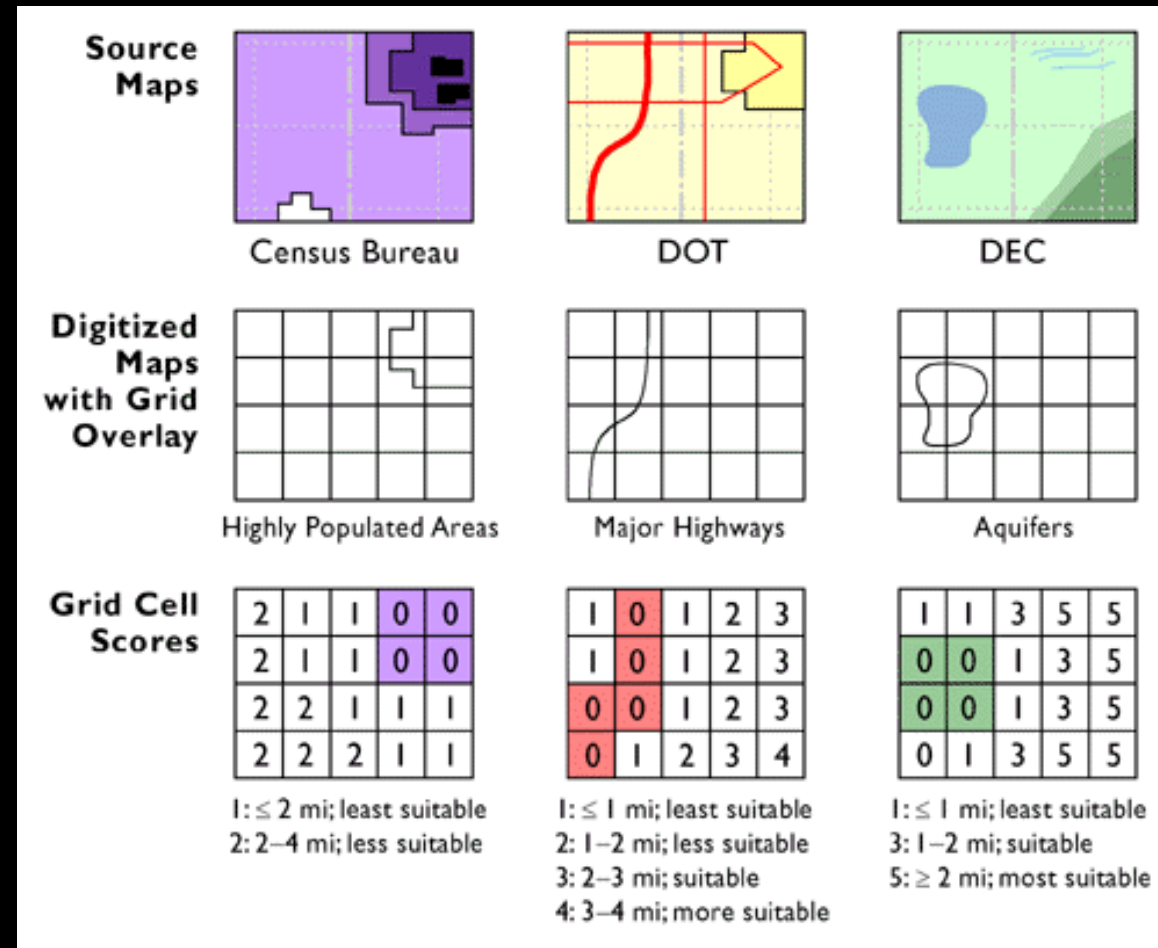
- Define the problem, break the model into sub-models, and identify the input layers.
- Weighted overlay only accepts **integer rasters as input**, such as a raster of land use or soil types.
- Continuous (floating point) rasters must be **reclassified as integer** before they can be used, with the help of Reclassify tool.
- Generally, the values of continuous rasters are **grouped into ranges**, such as for slope, or Euclidean distance outputs. Each range must be assigned a single value before it can be used in the Weighted Overlay tool.

For example: Value = 1, if slope = 0.0 to 5.0 degrees,
 Value = 2, if slope = 5.01 to 10.00 degrees, and
 Value = 3, if slope = 10.01 to 15.00 degrees.

Weighted Overlay- Key points to remember

- Evaluation scale for each inputs should be of **uniform scale**, say 0-5; 0-10 etc.
- An assigned preference on the common scale implies the phenomenon's preference for the criterion. **The preference values are on a relative scale**. That is, a preference of 10 is twice as preferred as a preference of 5.
- The preference values not only should be assigned relative to each other within the layer but should have the **same meaning between the layers**.
E.g.: If a location for one criterion is assigned a preference of 5, it will have the same influence on the phenomenon as a 5 in a second criterion.
- **Understand the scale values applied to input rasters so you know what the values in the output raster mean**. If the analysis is used to generate a cost surface (i.e. cost to travel through the landscape), high values will generally indicate higher travel costs.
- With the correct evaluation scale chosen, the cells in the output raster will already be set according to suitability.

Weighted Overlay Example of land suitability analysis for Solid waste Dumping



Example of land suitability analysis for Solid waste Dumping: (Contd.,)

Weighted Scores

40	20	20	E	E
40	20	20	E	E
40	40	20	20	20
40	40	40	20	20

Factor = 20

30	E	30	60	90
30	E	30	60	90
E	E	30	60	90
E	30	60	90	120

Factor = 30

50	50	150	250	250
E	E	50	150	250
E	E	50	150	250
50	50	150	250	250

Factor = 50

Composite Scores

120	E	200	E	E
E	E	100	E	E
E	E	100	230	360
E	120	250	360	390

Factor = 20

Composite Suitability Map

